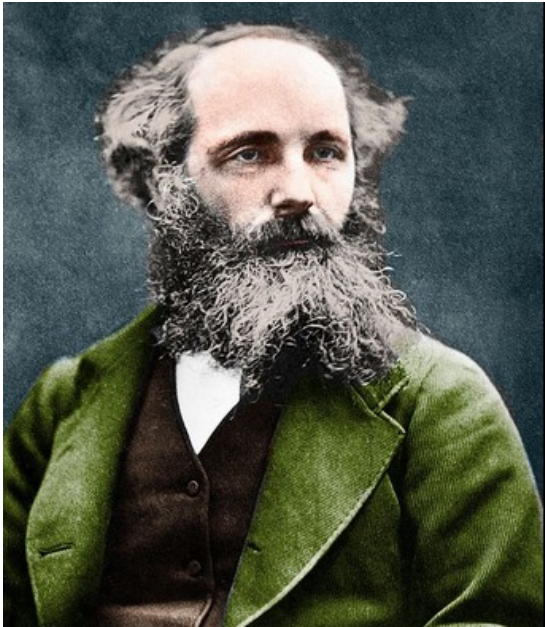


Electromagnetic Waves



- Born when Faraday discovered his law
- died when Einstein was born
- Did work on rings of Saturn
- kinetic theory of gases
- moved to London (from Scotland) in 1860
To King's College. Met with Faraday. Admired him.
- was home schooled. Very good mathematician

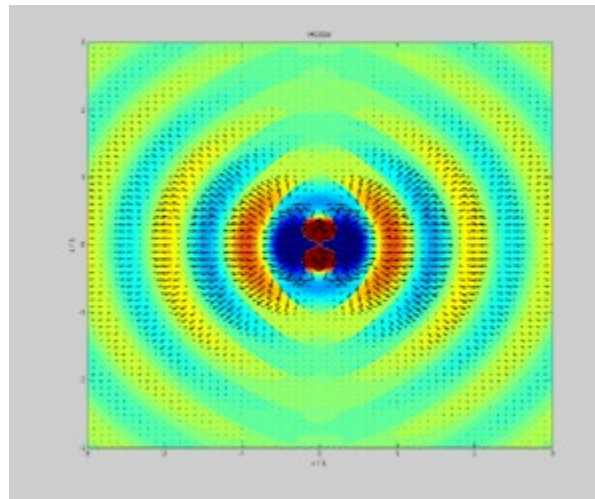
James Clerk Maxwell (13 June 1831 – 5 November 1879) was a Scottish mathematician[1][2] and scientist responsible for the classical theory of electromagnetic radiation, which was the first theory to describe electricity, magnetism and light as different manifestations of the same phenomenon. Maxwell's equations for electromagnetism have been called the "second great unification in physics"[3] where the first one had been realized by Isaac Newton.

EM waves, Maxwell Equations

- Maxwell equations and Amperes law amended by Maxwell
- speed of light, EM waves
- making EM waves, dipole radiation
- Energy in EM waves, Poynting vector
- polarization

SEE simulation
In folder

@veronique lankar 2023



What We Know about EM Fields

Electric fields arise from charges

- Coulomb's law, Gauss's law
- Electric field lines begin and end on charges

Magnetic fields arise from moving charges

- Biot-Savart law, Ampère's law

Magnetic field lines don't begin or end

- No magnetic monopoles

Changing magnetic fields give rise to electric fields

→ Maxwell → you don't need a conductor.

Electric field is induced in space (even without wire)

- Faraday's law

Next step Maxwell → changing electric field → changing magnetic field.
So they induce each other in space if they are changing.

Shake a magnet, shake a charged balloon → EM wave !

Electromagnetism in Math Language

Law	Math statement	English statement
Gauss's law for electricity	$\oint \vec{E} \cdot d\vec{A} = 4\pi kq$	Electric fields arise from electric charges
Gauss's law for magnetism	$\oint \vec{B} \cdot d\vec{A} = 0$	No magnetic charge; field lines don't begin or end
Faraday's law	$\oint \vec{E} \cdot d\vec{r} = -\frac{d\Phi_B}{dt}$	Electric fields arise from changing magnetic fields
Ampère's law (so far)	$\oint \vec{B} \cdot d\vec{r} = \mu_0 I$	Magnetic fields arise from moving electric charges

Video to visualize the equations. Excellent visual.

https://www.youtube.com/watch?v=SS4tcajTsW8&list=PL8_xPU5epJddRABXqJ5h5G0dk-XGtA5cZ&index=39
(15:50 of the video)

If a changing magnetic flux produces a circulation of electric field, then can a

A changing electric field flux produce a circulation of magnetic field? Asked Maxwell?

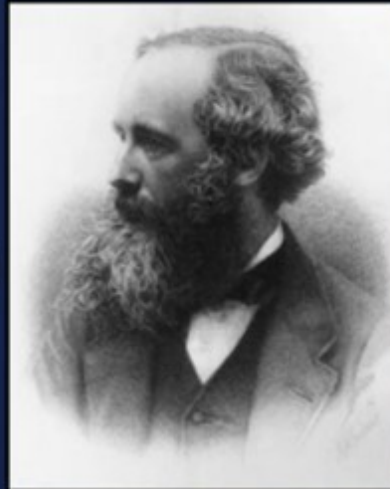
Maxwell's Insight (1864)

Changing electric fields give rise to magnetic fields

- Faraday's law: changing magnetic flux BA gives rise to electric field
- So: add changing electric flux to Ampère's law
New source of magnetic field

Ampère's law:

$$\oint \vec{B} \cdot d\vec{r} = \mu_0 I + \frac{\mu_0}{4\pi k} \frac{d\Phi_E}{dt}$$



James Clerk Maxwell
1831 – 1879

Maxwell added another term. He called this the displacement Current. A changing electric flux creates through a surface (membrane) creates A circulating magnetic field .

Ampere's Law with Displacement Current

Reminder that the magnetic field is a vector

Dot product tells you to find the part of $\vec{B}$ parallel to $d\vec{l}$ (along path C)

An incremental segment of path C

The electric current in amperes

The rate of change with time

The magnetic field in teslas

Reminder that only the enclosed current contributes

The magnetic permeability of free space

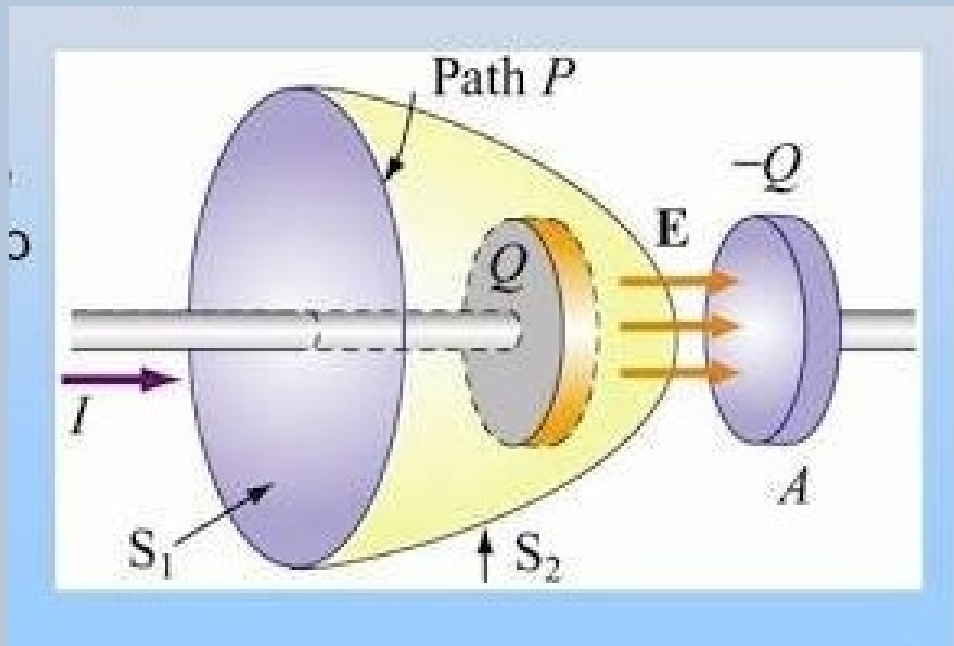
The electric permittivity of free space

The electric flux through a surface bounded by C

Tells you to sum up the contributions from each portion of closed path C in a direction given by the right-hand rule

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 \left(I_{\text{enc}} + \epsilon_0 \frac{d}{dt} \int_S \vec{E} \cdot \hat{n} da \right)$$

https://www.youtube.com/watch?v=SS4tcajTsW8&list=PL8_xPU5epJddRABXqJ5h5G0dk-XGtA5cZ&index=40
Go to 19:35 of the video to see the demonstration



Ampère's law:

$$\oint \vec{B} \cdot d\vec{r} = \mu_0 I + \frac{\mu_0}{4\pi k} \frac{d\Phi_E}{dt}$$

Magnetic field arising from... electric current...

or changing electric flux

Ampere's law says if the blue membrane is poked through by I then there is a circulating B around I . But if we replace the area by the yellow membrane while the capacitor is charging (so while there is a current) then no current is poking through the membrane. And still, here is a circulating magnetic field. So there is a term missing.

The changing electric field poking through the membrane is like a current and generates a circulation of B . This is called the displacement current.

It's proportional to the rate of change of the electric field flux. Video

Maxwell's Equations

Law	Math statement	English statement
Gauss's for E	$\oint \vec{E} \cdot d\vec{A} = 4\pi kq$	Electric fields arise from electric charges
Gauss's for B	$\oint \vec{B} \cdot d\vec{A} = 0$	No magnetic charge; field lines don't begin or end
Faraday	$\oint \vec{E} \cdot d\vec{r} = -\frac{d\Phi_B}{dt}$	Electric fields arise from changing magnetic fields
Ampère	$\oint \vec{B} \cdot d\vec{r} = \mu_0 I + \frac{\mu_0}{4\pi k} \frac{d\Phi_E}{dt}$	Magnetic fields arise from moving electric charges and changing electric fields

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m} \quad K_e = 1/(4\pi \epsilon_0) = 9 \times 10^9 \text{ N} \cdot \text{m}^2 \cdot \text{C}^{-2}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m} \quad K_m = \mu_0/(4\pi) = 10^{-7} \text{ H/m}$$

Recommended videos: <https://www.youtube.com/watch?v=hJD8ywGrXks>
<https://www.youtube.com/watch?v=rB83DpBJQsE>

Maxwell's Equations

Law	Math statement	English statement
Gauss's for E	$\oint \vec{E} \cdot d\vec{A} = 4\pi kq$	Electric fields arise from charges
Gauss's for B	$\oint \vec{B} \cdot d\vec{A} = 0$	No magnetic charge; field lines don't begin or end
Faraday	$\oint \vec{E} \cdot d\vec{r} = -\frac{d\Phi_B}{dt}$?	Electric fields arise from changing magnetic fields
Ampère	$\oint \vec{B} \cdot d\vec{r} = \boxed{\mu_0 I} + \frac{\mu_0}{4\pi k} \frac{d\Phi_E}{dt}$	Magnetic fields arise from moving charges and changing electric fields

Its not totally symmetric. A moving charge creates a magnetic field
But there is no moving monopole.

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m} \quad K_e = 1/(4\pi \epsilon_0) = 9 \times 10^9 \text{ N} \cdot \text{m}^2 \cdot \text{C}^{-2}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m} = K_m = \mu_0/(4\pi) = 10^{-7} \text{ H/m}$$

$$\nabla = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$$

$$\oiint \mathbf{E} \cdot d\mathbf{A} = \frac{q}{\epsilon_0}$$



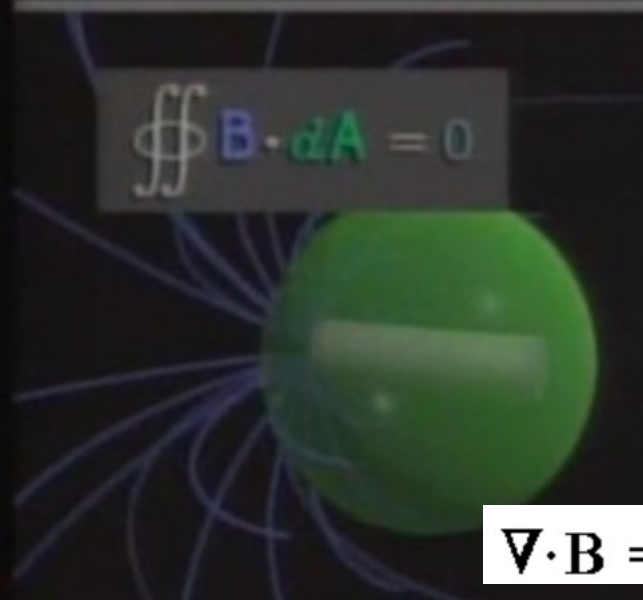
$$\nabla \cdot \mathbf{E} = \rho / \epsilon_0$$

$$\nabla \times \mathbf{E} = - \frac{\partial \mathbf{B}}{\partial t}$$

$$\oint \mathbf{E} \cdot d\mathbf{r} = - \frac{d}{dt} \Phi$$



$$\oiint \mathbf{B} \cdot d\mathbf{A} = 0$$



$$\nabla \cdot \mathbf{B} = 0$$

$$\oint \mathbf{B} \cdot d\mathbf{r} = \mu_0 \left(I + \epsilon_0 \frac{d\Phi}{dt} \right)$$



$$\nabla \times \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} + \mu_0 \mathbf{j}_c$$

Maxwell's Equations in Vacuum

Law	Math statement	English statement
Gauss's for E	$\oint \vec{E} \cdot d\vec{A} = 0$	No electric charges in vacuum
Gauss's for B	$\oint \vec{B} \cdot d\vec{A} = 0$	No magnetic charge (in vacuum or elsewhere)
Faraday	$\oint \vec{E} \cdot d\vec{r} = -\frac{d\Phi_B}{dt}$	Electric fields arise from changing magnetic fields
Ampère	$\oint \vec{B} \cdot d\vec{r} = \frac{\mu_0}{4\pi k} \frac{d\Phi_E}{dt}$	Magnetic fields arise from changing electric fields

Maxwell predicted the EM waves (detected by Heinrich Hertz) using his equations and he predicted

The speed of light (measured by Albert A. Michelson).

Video (mechanical universe again).

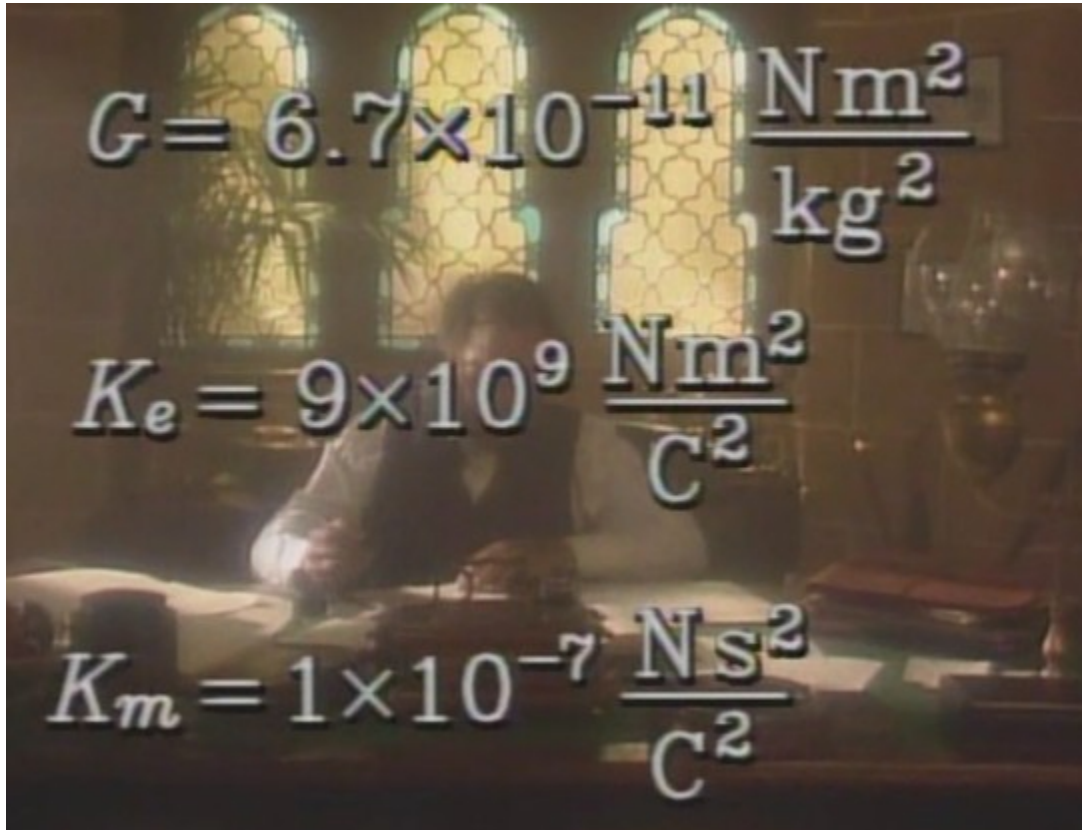
https://www.youtube.com/watch?v=lp_jdcA8fcw&list=PL8_xPU5epJddRABXqJ5h5G0dk-XGtA5cZ&index=41

24.3 The Speed of Light

Maxwell's prediction of the speed of light

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$$

? Compute. What do you get??

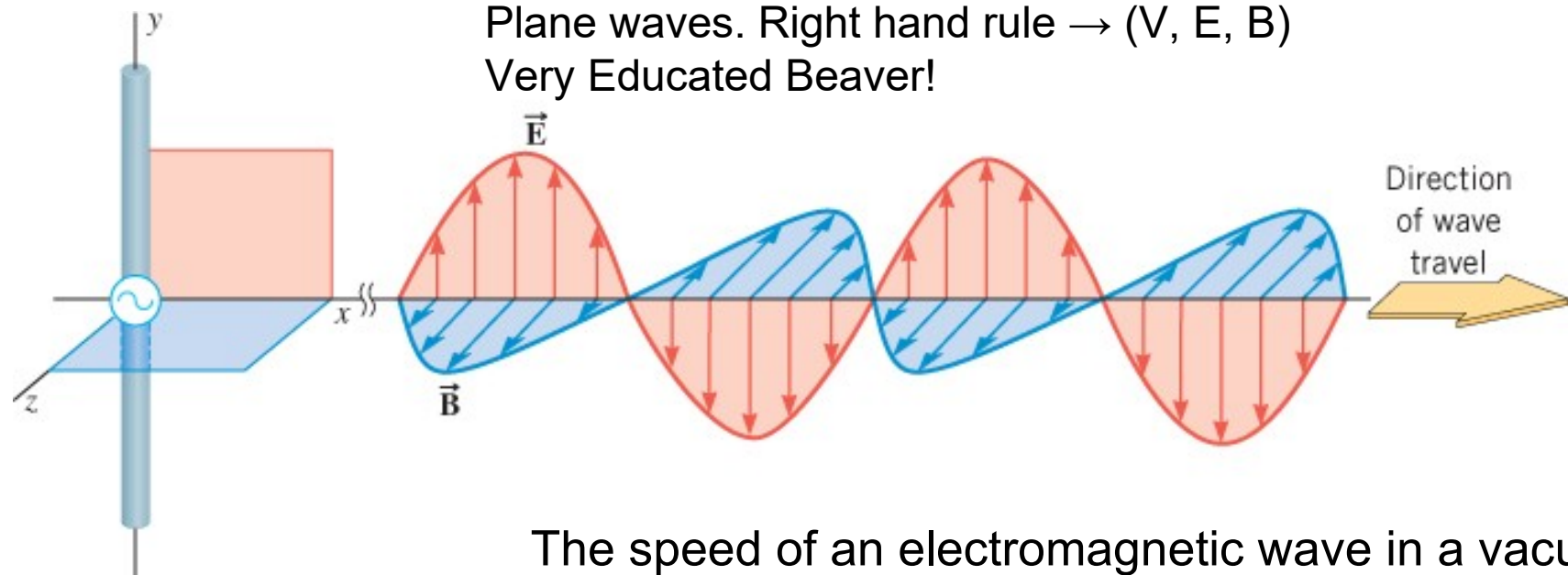


$$\frac{K_e}{K_m} = 9 \times 10^{16} \frac{\text{m}^2}{\text{s}^2}$$

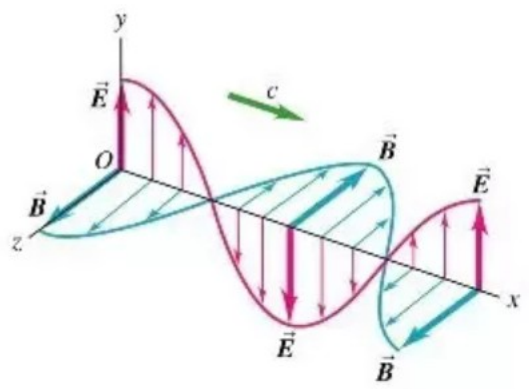
24.1 The Nature of Electromagnetic Waves

This picture shows the wave of the radiation field far from the antenna.

Far away, it is a plane wave. Review waves with Phet sims
Video EM waves from mechanical universe.
Plane waves. Right hand rule → (V, E, B)
Very Educated Beaver!



The speed of an electromagnetic wave in a vacuum is:



$$c = 3.00 \times 10^8 \text{ m/s}$$

Thomas Young
1801: Interference
experiments show
light is a wave



1800

1810

1820

Phet https://phet.colorado.edu/sims/html/wave-interference/latest/wave-interference_en.html
Video veritasium The Original Double Slit Experiment – Explain interference.

James Clerk Maxwell
1860s: EM theory
shows EM waves
propagate at c
(speed of light)



1860

1870

1880

Heinrich Hertz
1888: Creates and
detects EM waves
in lab



1890

1900

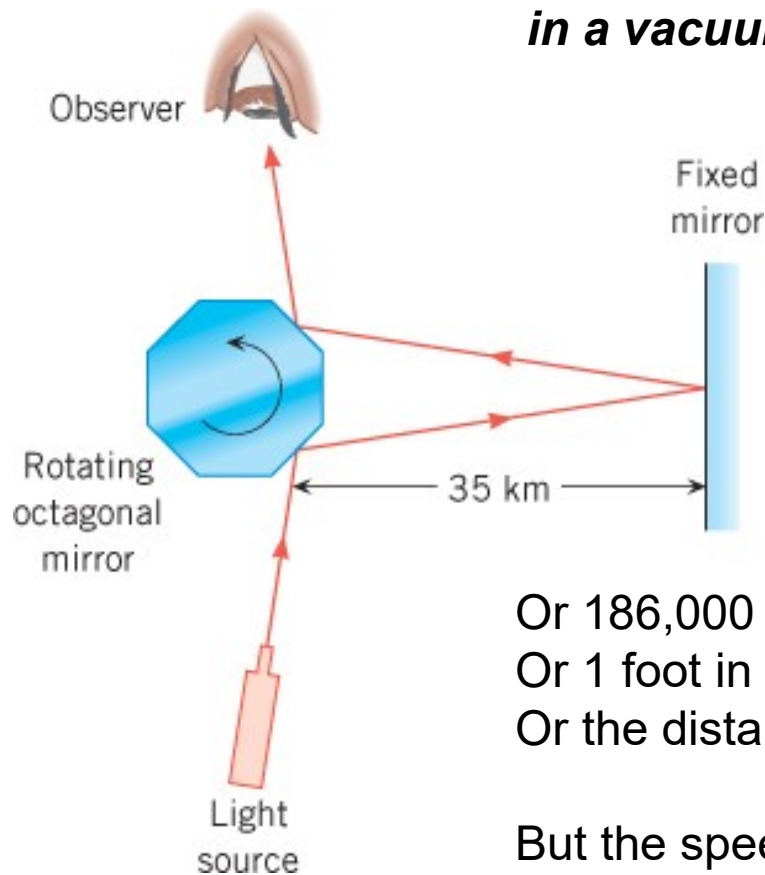
1910



Remember the tank circuit ? (magnetron). With a RLC circuit , the current Oscillates (like a swing) and EM waves are produced. The changing B Induced a induced voltage at a distance, As shown. Video <https://www.youtube.com/watch?v=QZXYFr5YHew>

24.3 The Speed of Light

The speed of light in a vacuum



1879
video



Albert A. Michelson - Wikip...

$$c = 299\,792\,458 \text{ m/s}$$

Or 186,000 miles per second

Or 1 foot in a billionth of a second

Or the distance Earth-Sun (150 million km) in 8 minutes

But the speed of light is finite. And the same in
Any frame of reference (special relativity)

Figure 2.3 The first terrestrial measurement of the speed of light was done by Fizeau in **1849** when he projected a pulsed beam of light onto a distant mirror. Based on the number of teeth and speed of rotation of the toothed wheel, and knowing the distance to the mirror, he was able to calculate a speed of 315,000 km/s.

Electromagnetic waves

A wave is a way to transfer energy. Energy is moving shaking is moving.

Wavelength (m) = size of wave in meter. Frequency (Hz) = number of shaking / second

Speed of wave (m/s) = wavelength x frequency.

High frequency → low wavelength.

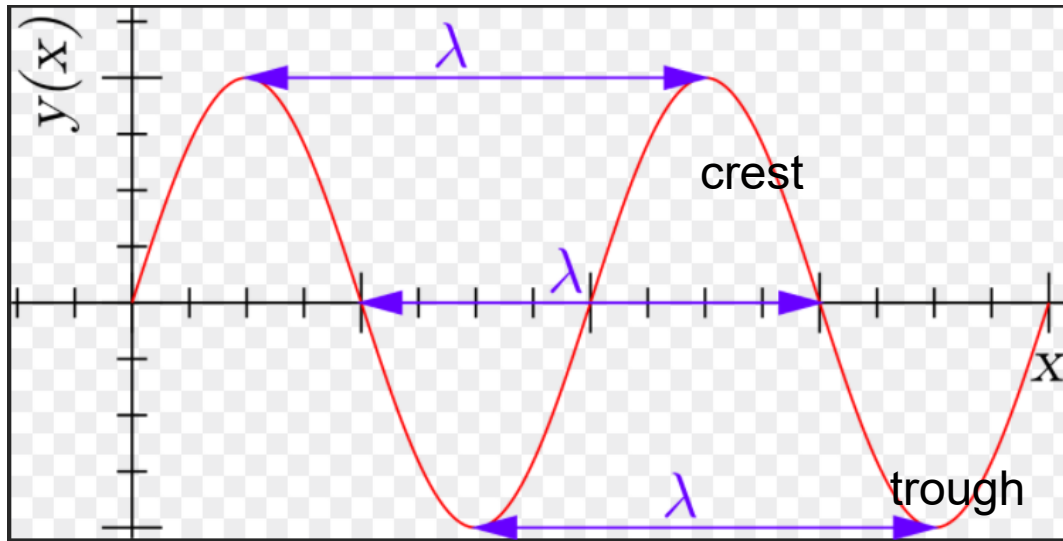
Watch video 3D waves

<http://electron9.phys.utk.edu/optics421/modules/m1/emwaves.htm>

See the equations.

Remember the right hand rule

See the simulation to review waves



A plane EM wave traveling in the x-direction is of the form:
(sine waves. Not all EM waves are sinewaves.

$$E(x,t) = E_{\max} \cos(kx - \omega t + \phi), \quad B(x,t) = B_{\max} \cos(kx - \omega t + \phi).$$

https://phet.colorado.edu/sims/html/wave-on-a-string/latest/wave-on-a-string_en.html

KDKA
1920: Commercial
radio



1920

1930

1940

WNBT
1941: Commercial
television



1940

1950

1960

Raytheon
1947:
Microwave oven



1950

1960

1970

2000s:
Cell phones
everywhere



2000

2010

2020

**2010:
Wireless
everything!**



2010

2020

covid



The Van de Graaf generator gets charged
And creates a static electric field in the space

If you shake the Van de Graaf the electric field
shakes = electromagnetic wave.

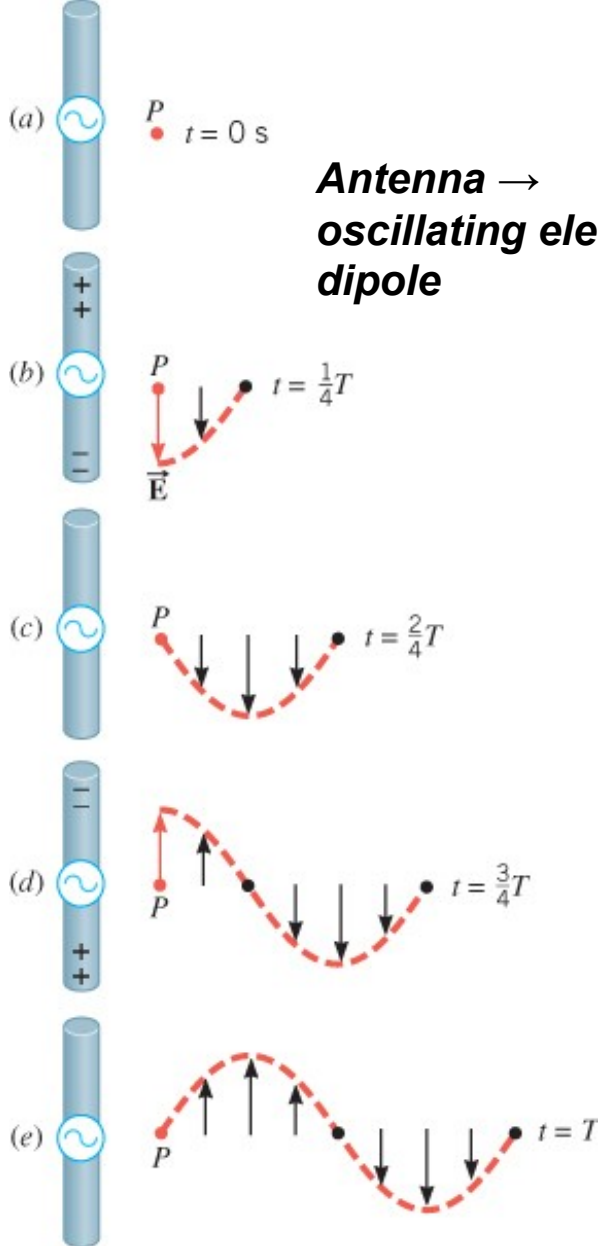
The frequency is small so a radio wave is created.
The wave can be detected by an antenna.

<https://phet.colorado.edu/en/simulation/radio-waves>

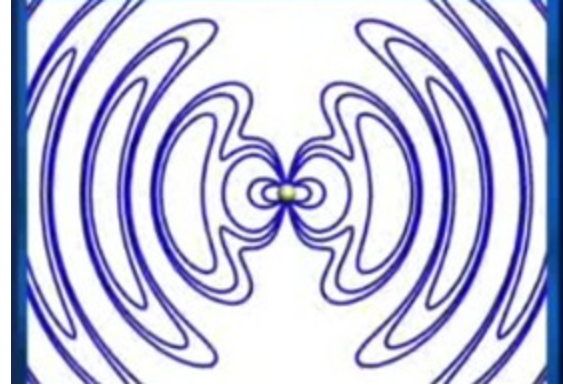
With the publication of "A Dynamical Theory of the Electromagnetic Field" in **1865**, Maxwell demonstrated that electric and magnetic fields travel through space as waves moving at the speed of light. He proposed that light is an undulation in the same medium that is the cause of electric and magnetic phenomena.

Known for: Statistical mechanics; Maxwell's eq





Antenna →
**oscillating electric
dipole**



Two straight wires connected to the terminals of an AC generator can create an **electromagnetic wave**.

Only the electric wave traveling to the right is shown here.

Video oscillating dipole

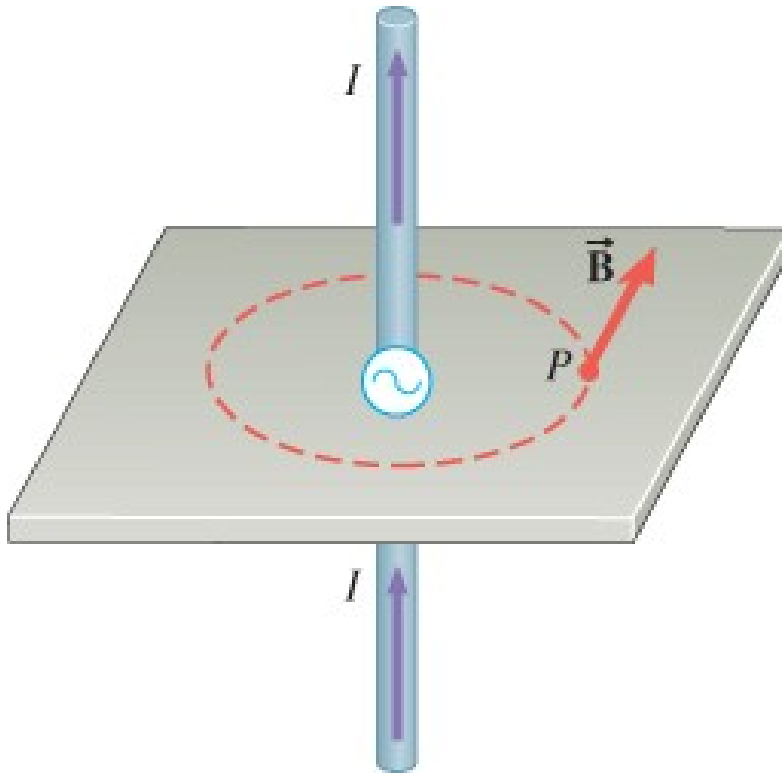
An accelerating charge → EM waves
A disturbance in the electric field

A wave is a traveling disturbance. There is
Something shaking. Here its the electric field.

Simulation Sims

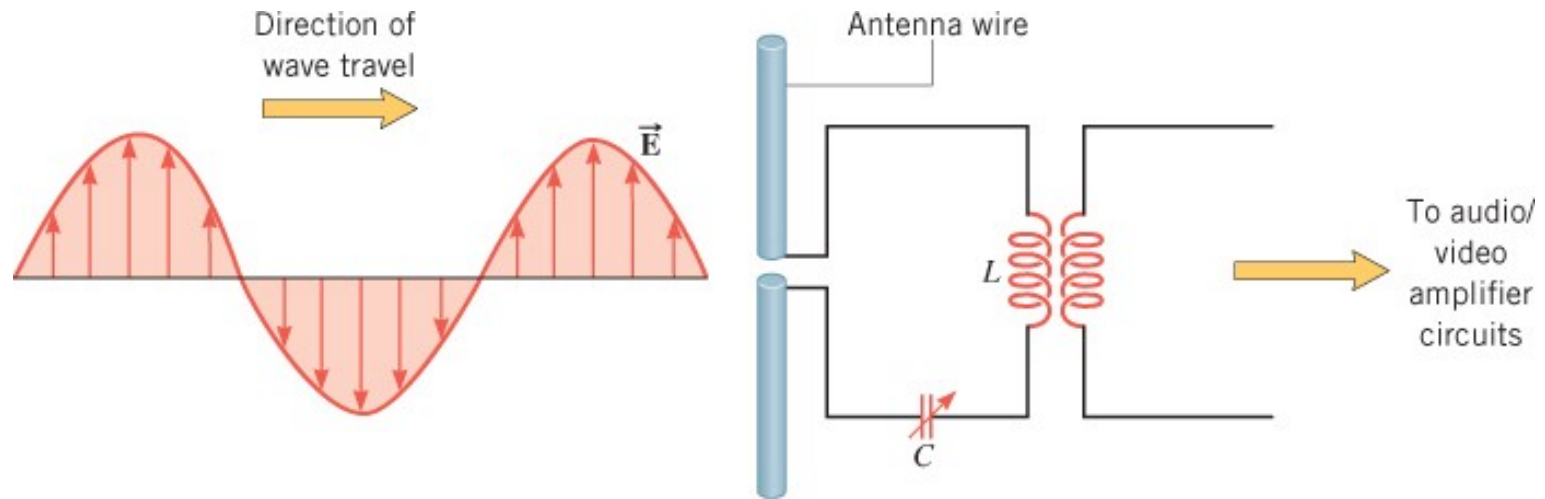
<https://phet.colorado.edu/sims/cheerpj/radio-waves/latest/radio-waves.html?simulation=radio-waves>

24.1 The Nature of Electromagnetic Waves



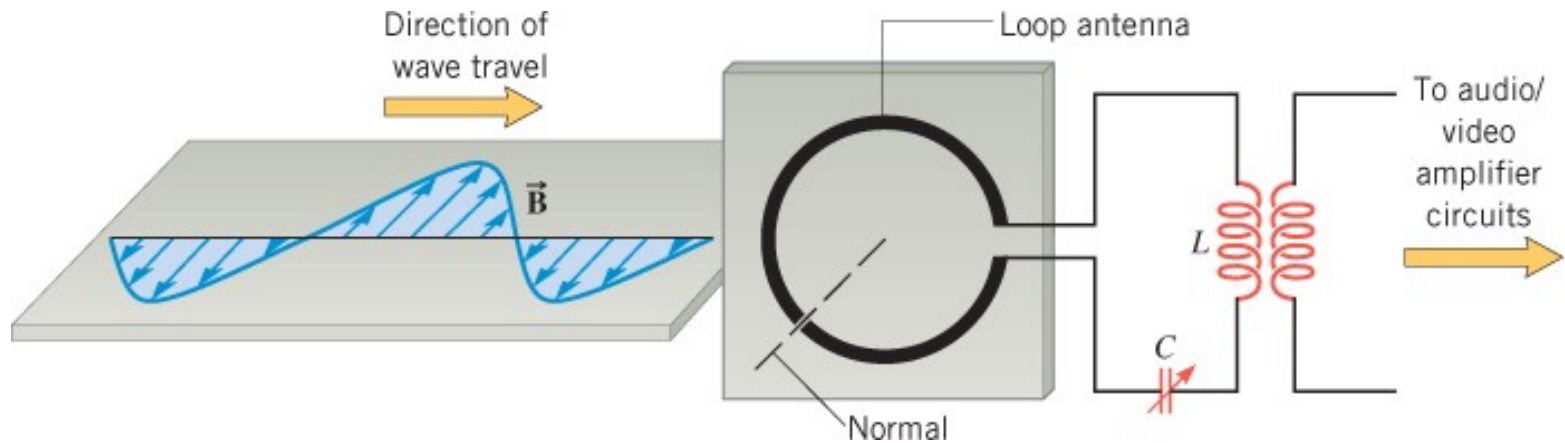
The current used to generate the electric wave creates a magnetic field.

24.1 The Nature of Electromagnetic Waves



A radio wave can be detected with a receiving antenna wire that is parallel to the electric field.

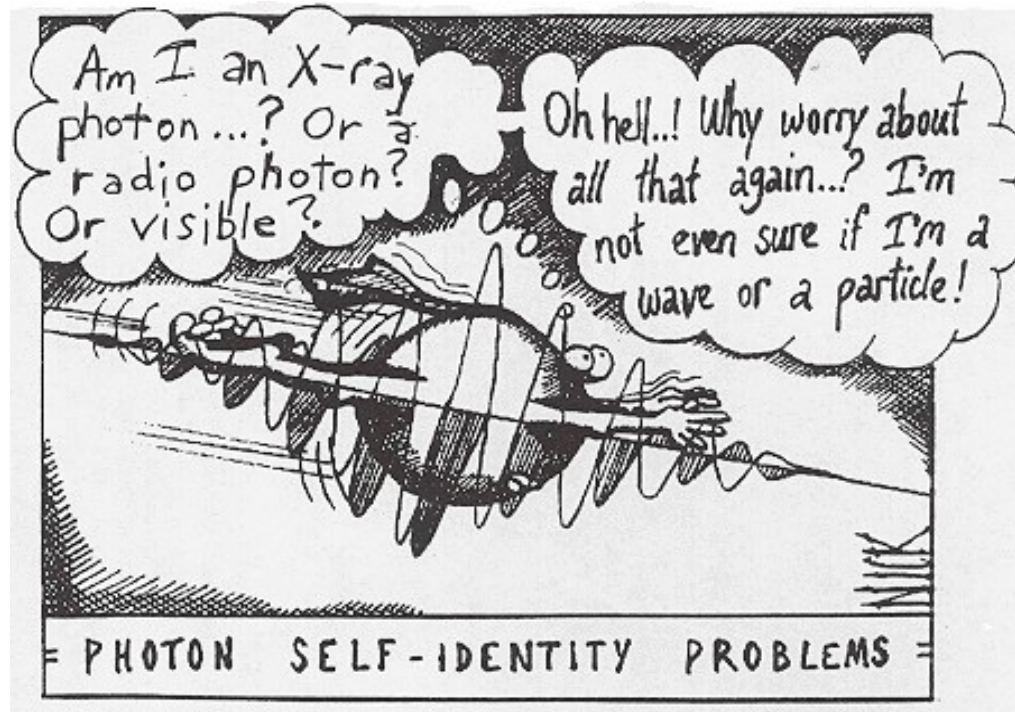
24.1 The Nature of Electromagnetic Waves



With a receiving antenna in the form of a loop, the magnetic field of a radio wave can be detected.

Nature of light :

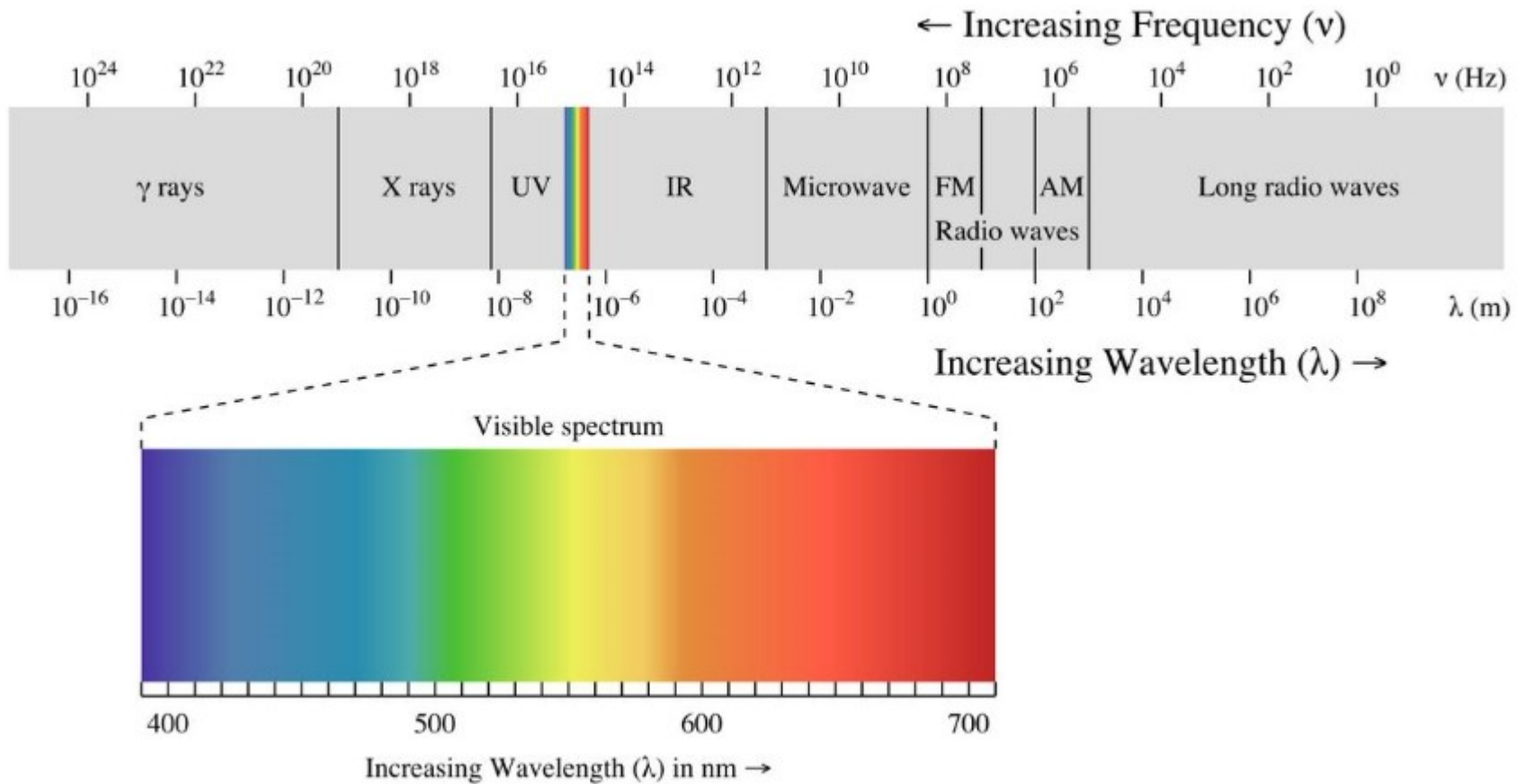
What is light ? a wave or a particle ? Or both ?



Show the experiment of Thomas Young (1801). Interference of light. Light is a wave. Or is it? See Utube video Veritasium

LIGHT for medical imaging / electromagnetic spectrum

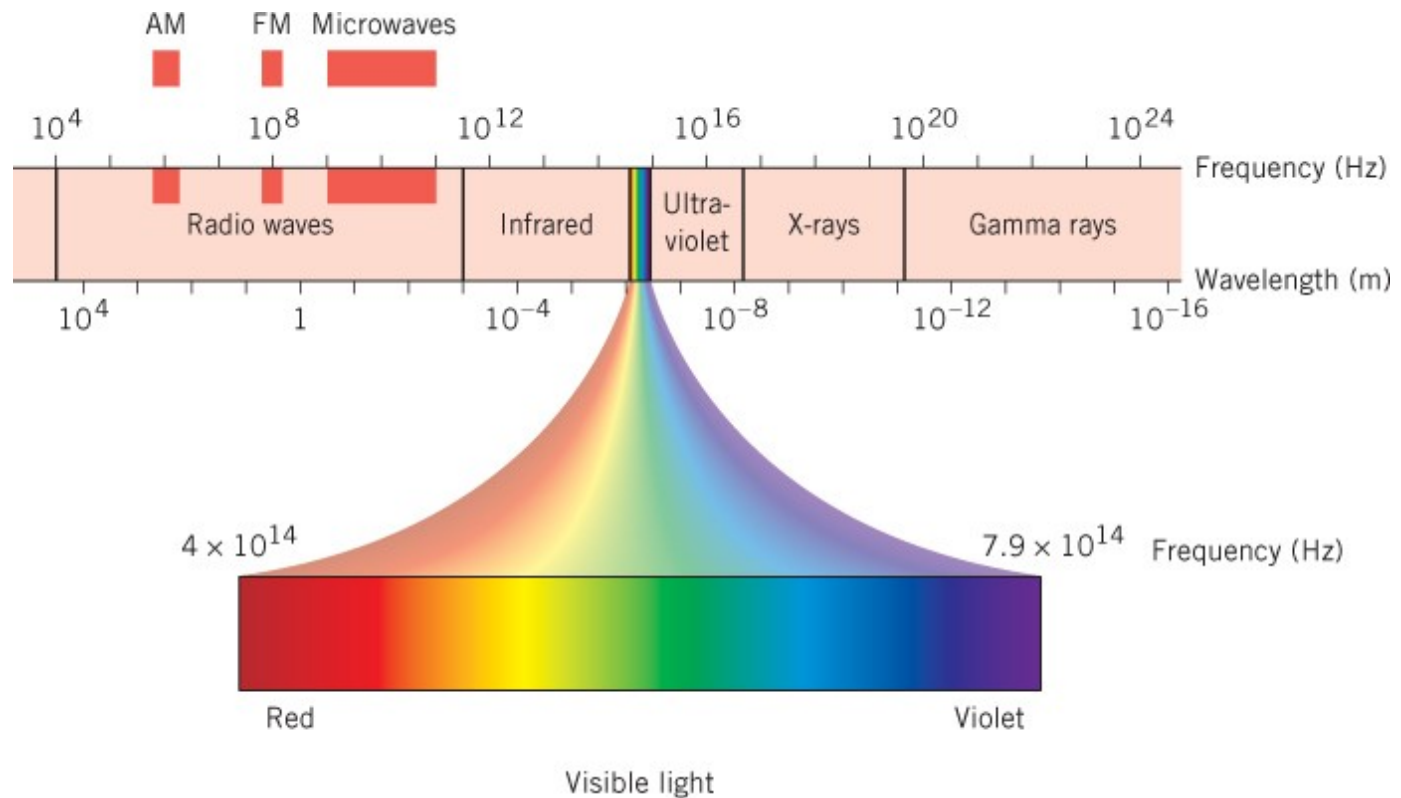
Light can be visible/invisible



Accelerated electric charge is the source of EM waves

- Radio transmitters
- Atomic & molecular dipoles
- X-ray tubes
- Electrons spiraling in magnetic fields
- Nuclear transitions

24.2 The Electromagnetic Spectrum



Like all waves, electromagnetic waves have a wavelength and frequency, related by:

$$c = f\lambda$$

Amplitude, Frequency, Wavelength

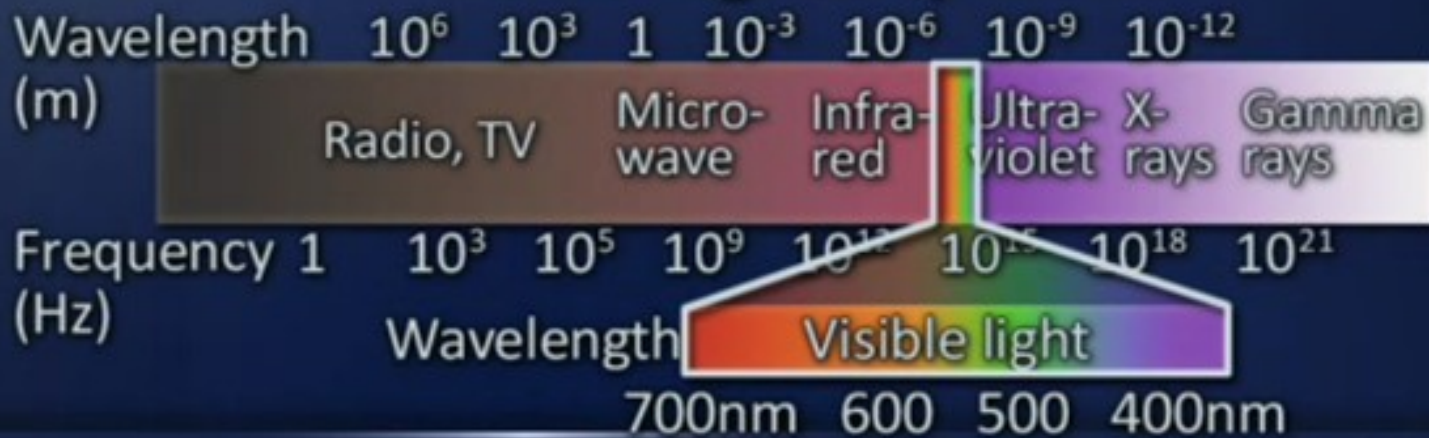
Any amplitude allowed

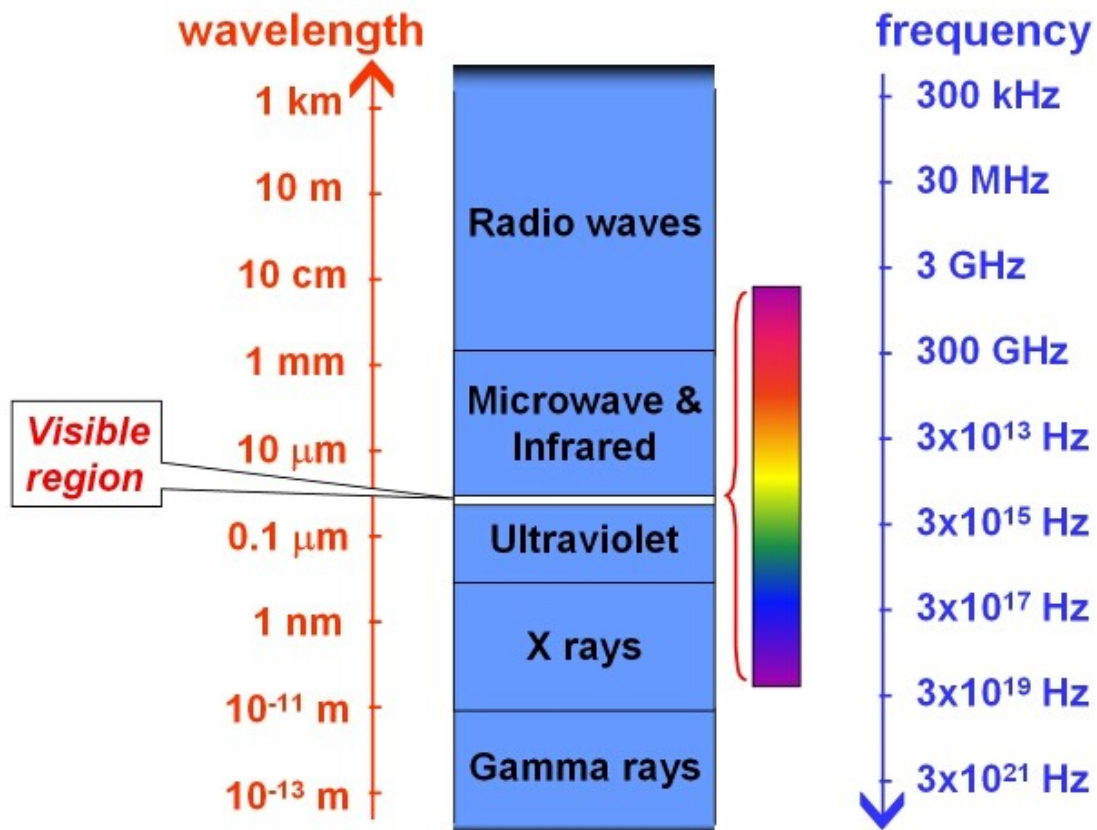
- but E, B related by $E = cB$

Any frequency or wavelength allowed

- but related by $f\lambda = c$

Electromagnetic Spectrum





A light year is the distance traveled in one year by light. It is a unit of distance.

How much is a light year in km? Miles? (1.6km=1mile)

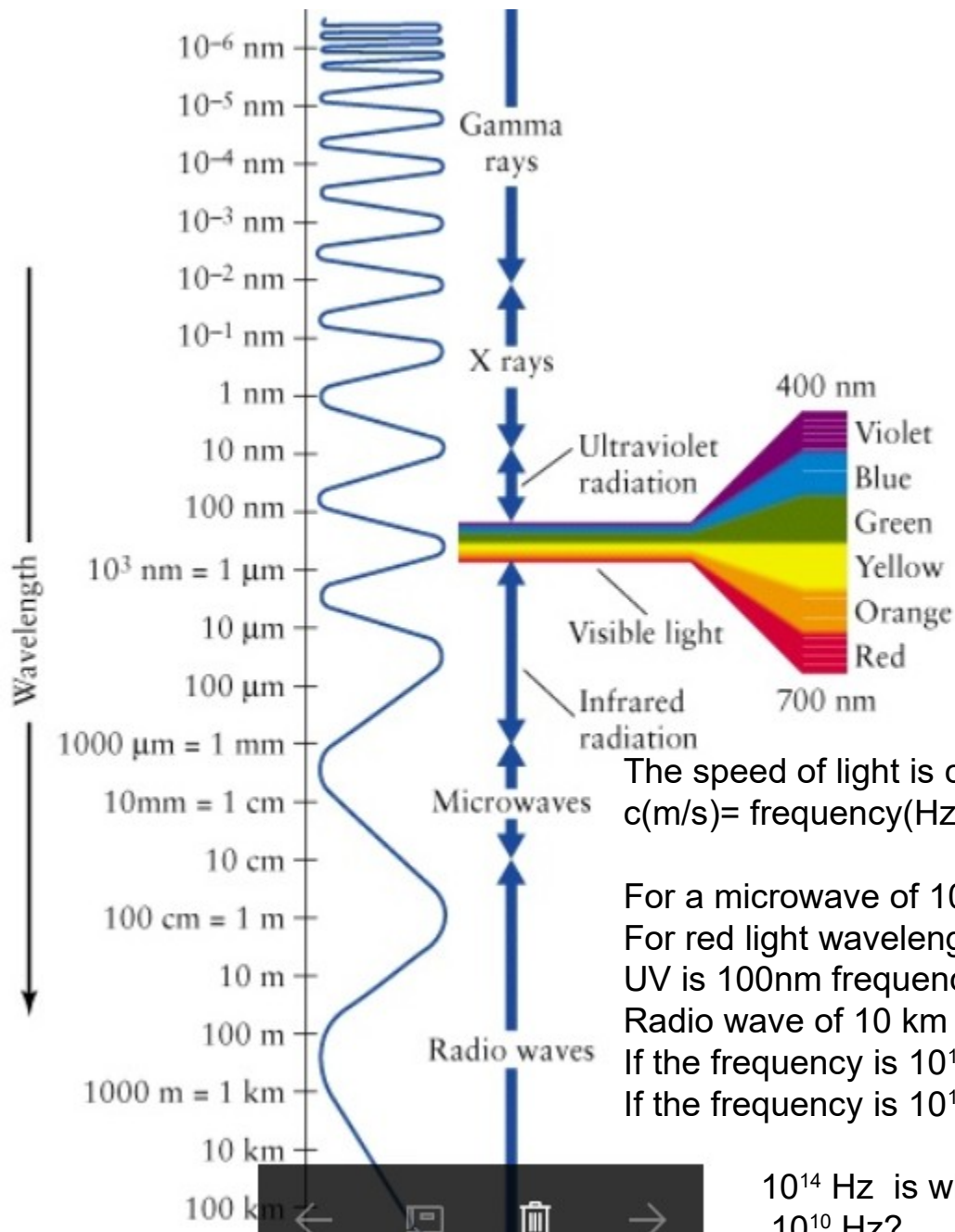
Andromeda galaxy is 2.5 million l.y away.

Are you seeing it as it is now?



**Take AST 1002
To learn more!**





Convert :

10^{-6} nm in meters

10^{-1} nm in meters

100 μ m (micro m) in meters

10km in meters

10^3 cm in meters

The speed of light is $c = 3 \times 10^8$ m/s
 $c(\text{m/s}) = \text{frequency}(\text{Hz}) \times \text{wavelength}(\text{m})$

For a microwave of 10cm what is the frequency?

For red light wavelength = 650nm what is the frequency

UV is 100nm frequency is ?

Radio wave of 10 km ?

If the frequency is 10^{12} Hz what is the wavelength ? What is it?

If the frequency is 10^{18} Hz what is the wavelength ? What is it?

10^{14} Hz is what wavelength ? What is this wave ?

10^{10} Hz?

Example 1 The Wavelength of Visible Light

Find the range in wavelengths for visible light in the frequency range between $4.0 \times 10^{14} \text{ Hz}$ and $7.9 \times 10^{14} \text{ Hz}$.



Example 1 The Wavelength of Visible Light

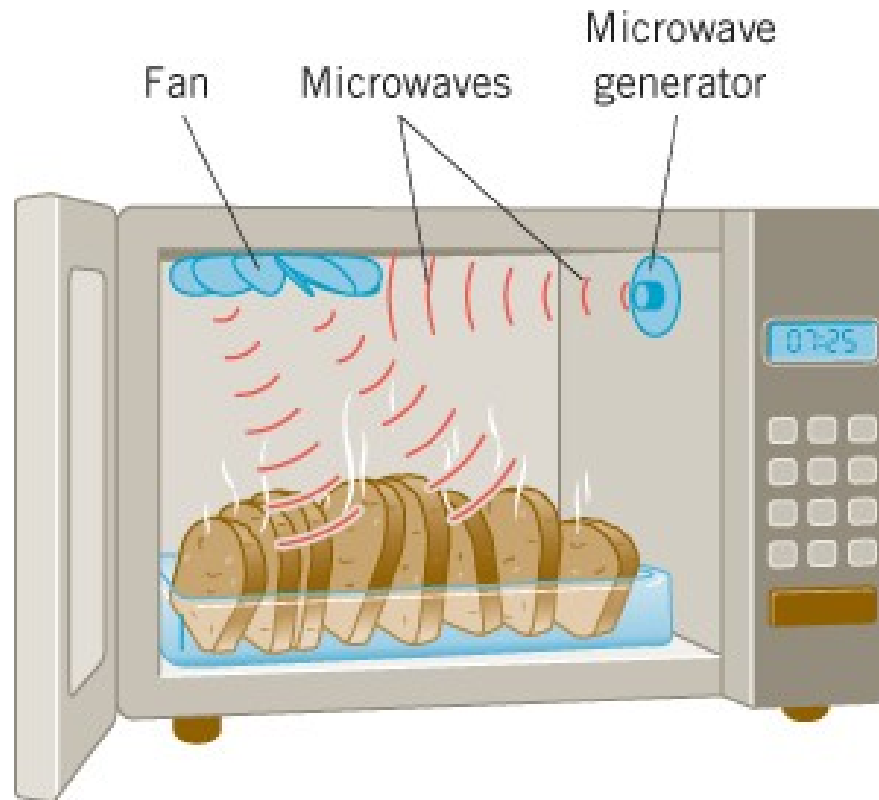
Find the range in wavelengths for visible light in the frequency range between $4.0 \times 10^{14} \text{ Hz}$ and $7.9 \times 10^{14} \text{ Hz}$.

$$\lambda = \frac{c}{f} = \frac{3.00 \times 10^8 \text{ m/s}}{4.0 \times 10^{14} \text{ Hz}} = 7.5 \times 10^{-7} \text{ m} = 750 \text{ nm}$$

$$\lambda = \frac{c}{f} = \frac{3.00 \times 10^8 \text{ m/s}}{7.9 \times 10^{14} \text{ Hz}} = 3.8 \times 10^{-7} \text{ m} = 380 \text{ nm}$$

24.4 *The Energy Carried by Electromagnetic Waves*

Electromagnetic waves, like water waves, carry energy.



24.4 The Energy Carried by Electromagnetic Waves

The total energy density carried by an electromagnetic wave

$$u = \frac{\text{Total energy}}{\text{Volume}} = \frac{1}{2} \epsilon_o E^2 + \frac{1}{2 \mu_o} B^2$$

1- Those terms are equal

2- write term1 = term2 and replace $(\epsilon_o \mu_o)^2 = 1/c^2$ ← $c = \frac{1}{\sqrt{\epsilon_o \mu_o}}$

3 – you get: $E = cB$ So E is much larger than B
But 1/μo is larger than εo

If the EM are sine waves (see picture) ,we can define the average energy density.

Uaverage → use Erms and Brms instead of Eo and Bo

For AC circuit, the average power was Vrms x I rms.

$$U_{\text{average}} = 0.5 \epsilon_o E_{\text{rms}}^2 + 0.5/\mu_o B_{\text{rms}}^2 \qquad E_{\text{rms}} = \frac{1}{\sqrt{2}} E_0 \quad \text{and} \quad B_{\text{rms}} = \frac{1}{\sqrt{2}} B_0$$

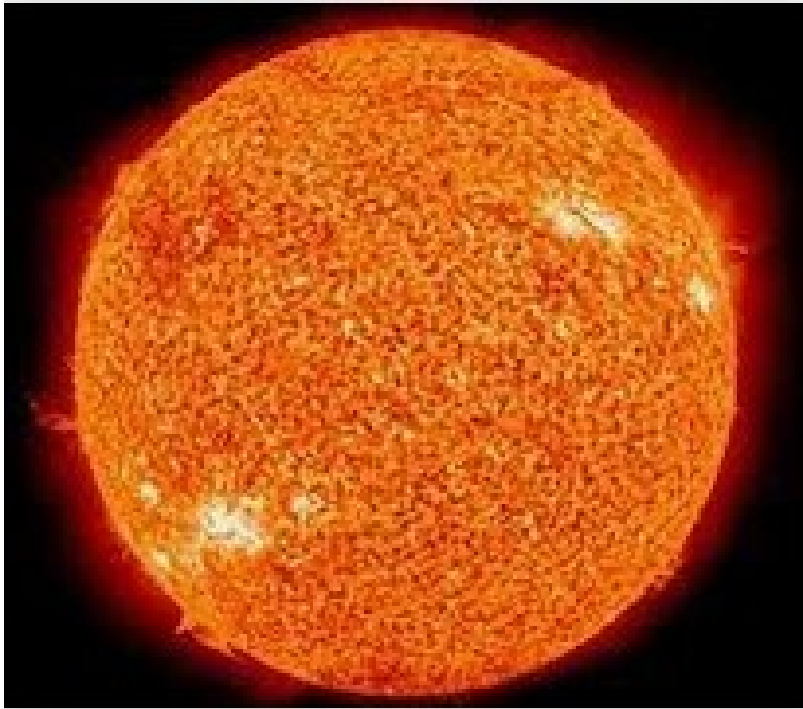
$$U_{\text{average}} = \epsilon_o E_{\text{rms}}^2$$



$$U_{\text{average}} = \epsilon_0 E_{\text{rms}}^2 = (1/\mu_0) B_{\text{rms}}^2$$

Example 4 The Average Energy Density of Sunlight

Sunlight enters the top of the earth's atmosphere with an electric field whose rms value is $E_{\text{rms}} = 720 \text{ N/C}$. Find (a) the average total energy density of this electromagnetic wave and (b) the rms value of the sunlight's magnetic field.



$$U_{\text{average}} = 0.5 \epsilon_0 E_{\text{rms}}^2 + 0.5/\mu_0 B_{\text{rms}}^2$$

$$U_{\text{average}} = \epsilon_0 E_{\text{rms}}^2 = (1/\mu_0) B_{\text{rms}}^2$$

$$\text{With } E_{\text{rms}} = c B_{\text{rms}}$$

Solution (a) According to Equation 24.2b, the average total energy density is

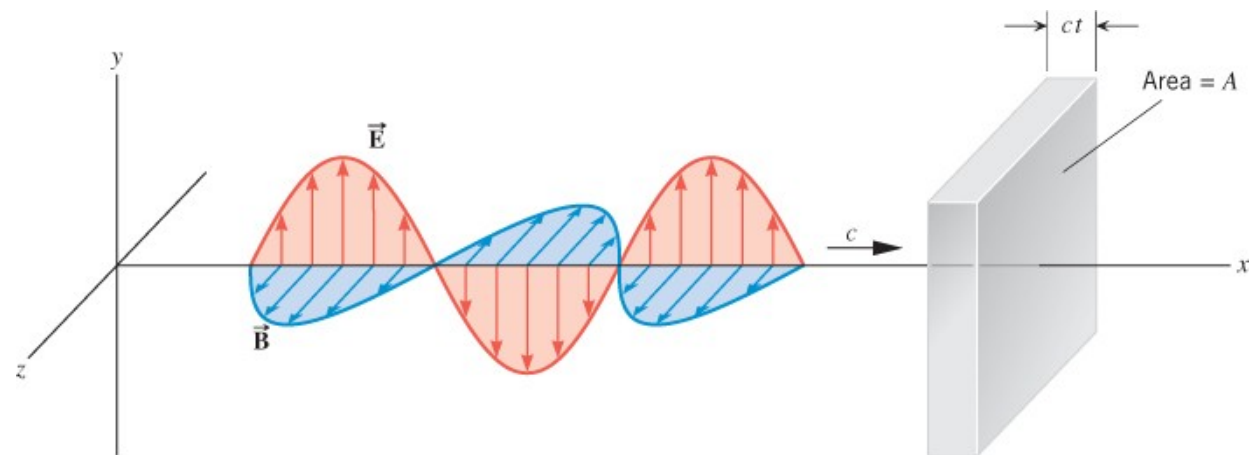
$$\bar{u} = \epsilon_0 E_{\text{rms}}^2 = [8.85 \times 10^{-12} \text{ C}^2/(\text{N} \cdot \text{m}^2)](720 \text{ N/C})^2 = \boxed{4.6 \times 10^{-6} \text{ J/m}^3}$$

(b) Using Equation 24.3, we find that the rms magnetic field is

$$B_{\text{rms}} = \frac{E_{\text{rms}}}{c} = \frac{720 \text{ N/C}}{3.0 \times 10^8 \text{ m/s}} = \boxed{2.4 \times 10^{-6} \text{ T}}$$

Ref: Cutnell & Johnson Physics - 9th edition

24.4 The Energy Carried by Electromagnetic Waves



$$S = \frac{P}{A} = \frac{\text{Total energy}}{t A} = \frac{u c t A}{t A} = c u$$

Poynting vector, a quantity describing the magnitude and direction of the flow of energy in electromagnetic waves. It is named after English physicist John Henry Poynting, who introduced it in 1884. 6 days ago



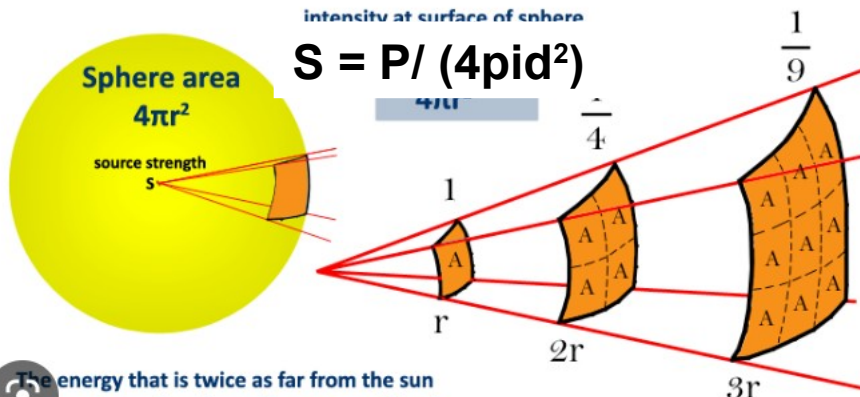
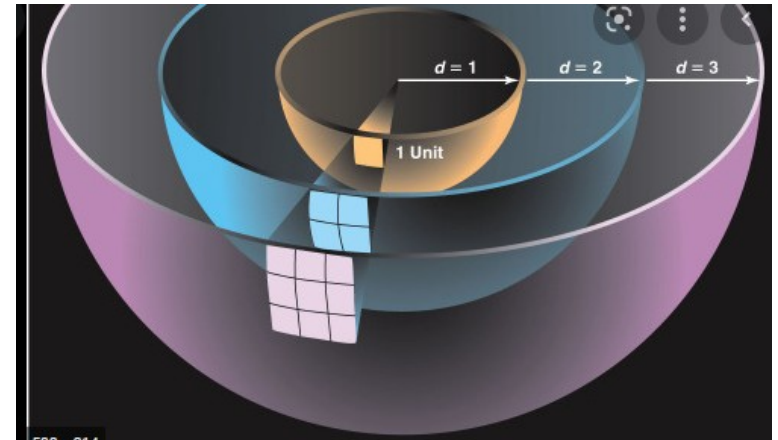
Show that S is the amount of radiation flowing through a unit area A . S depends on the distance from the source. For the same size area, you get less radiation on the skin than you get close to source.

$$S = c\epsilon_0 E^2$$
$$S = \frac{c}{\mu_0} B^2$$

Show video infrared. If radiation is guided, there is no loss in energy crossing a given Area (constant). With a lamp, energy is diluted

Energy flows from the source is $3D$, all direction. Apply Inverse square law.
 Energy output of the source is P (watts or joule/s).
 Radiation through a unit area area is S . (show). Unit is watts/m².

$S = P / (4\pi d^2)$ d is the distance from the source. (draw)
 Same equation for sound. Light.



The energy that is twice as far from the sun is spread over four times the area, making it one-fourth the intensity.

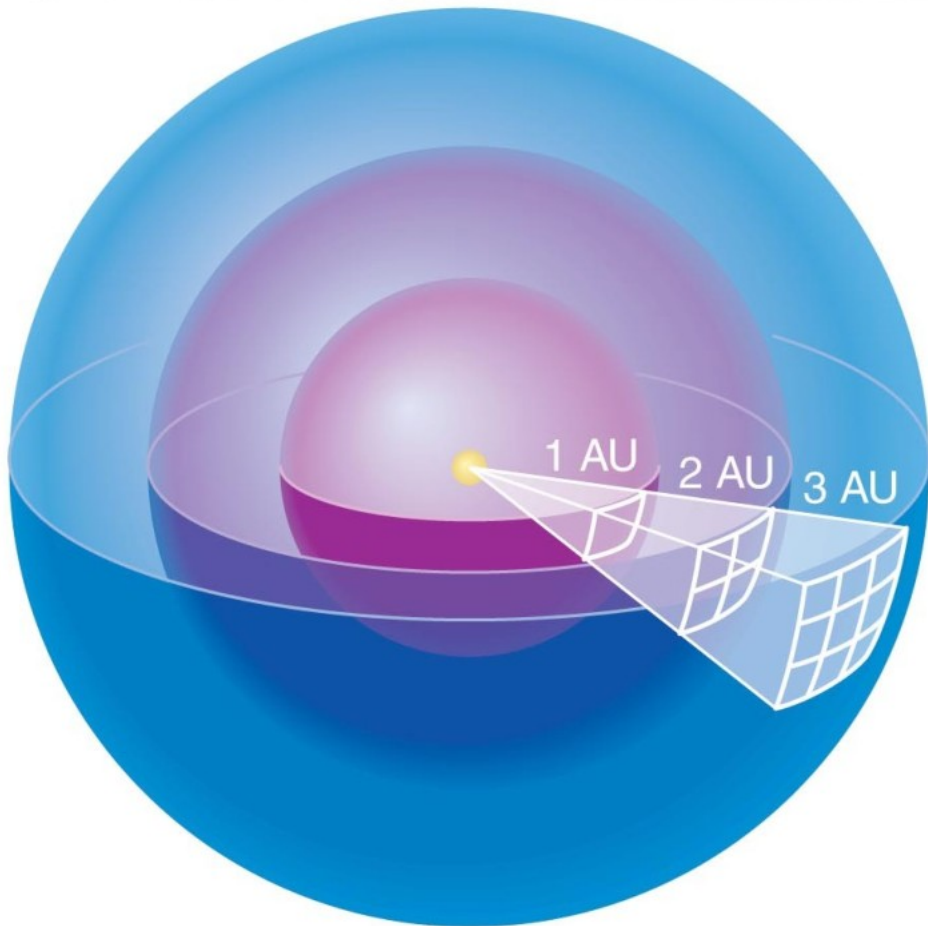
A little square is a unit area.
 $D = 3$. the radiation S is divided by 9

$$S = c\epsilon_0 E^2$$

$$S = \frac{c}{\mu_0} B^2$$

In Astro the power (or intensity) of light is called the luminosity (Watts)
And S (Watts/m²) is called the brightness. If you know the luminosity
And brightness you can get the distance. In astro we use standard candles.

The relationship between luminosity and brightness is simple.



Luminosity passing
through each sphere
is the same.

Area of sphere:

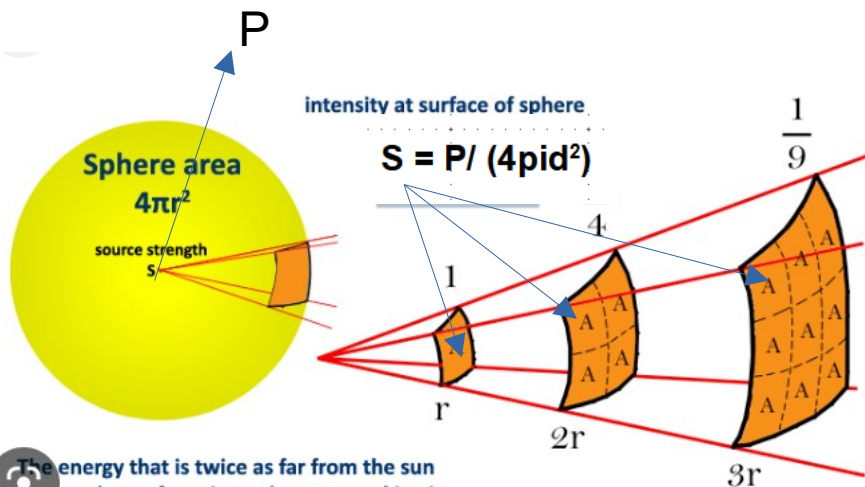
$$4\pi (\text{radius})^2$$

Divide luminosity by
area to get brightness.

$$S = c\epsilon_0 E^2$$

$$S = \frac{c}{\mu_0} B^2$$

$$S = P / (4\pi r^2)$$



The energy that is twice as far from the sun is spread over four times the area, making it one-fourth the intensity.

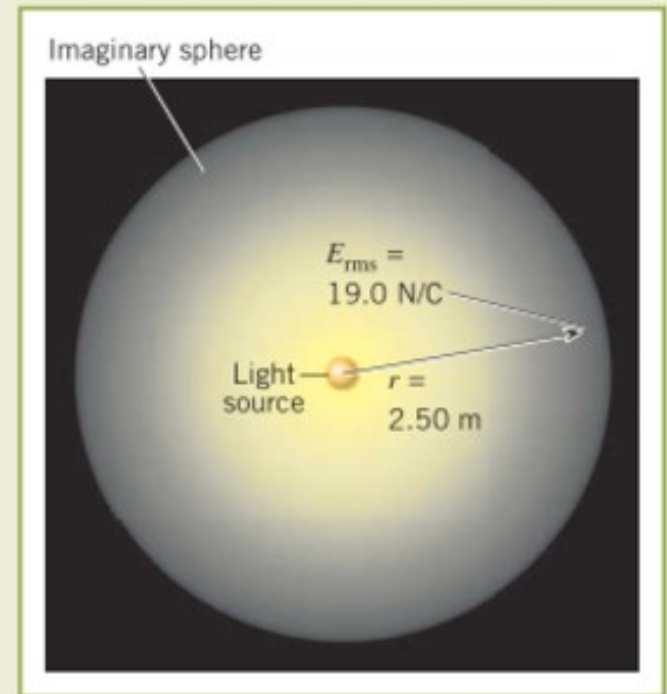


Figure 24.16 At a distance of 2.50 m from the light source, the rms electric field strength of the light has a value of 19.0 N/C.

Example 5 Power and Intensity

Figure 24.16 shows a tiny source that is emitting light uniformly in all directions. At a distance of 2.50 m from the source, the rms electric field strength of the light is 19.0 N/C. Assuming that the light does not reflect from anything in the environment, determine the average power of the light emitted by the source.

Modeling the Problem

STEP 1 Average Intensity According to the discussion in Section 16.7, the average power $\bar{P}$ that passes perpendicularly through the imaginary spherical surface is equal to the average intensity $\bar{S}$ times the area A of the surface, or $\bar{P} = \bar{S}A$. The area of a spherical surface is $A = 4\pi r^2$, where r is the radius of the sphere. Thus, the average power can be written as in Equation 1 at the right. The radius is known, but the average light intensity is not, so we turn to Step 2 to evaluate it.

$$\bar{P} = \bar{S}(4\pi r^2) \quad (1)$$

↑
?

STEP 2 Average Intensity and Electric Field The average intensity $\bar{S}$ of the light passing through the imaginary spherical surface is related to the known rms electric field strength E_{rms} at the surface by Equation 24.5b:

$$\bar{S} = c\epsilon_0 E_{\text{rms}}^2$$

where c is the speed of light in a vacuum and ϵ_0 is the permittivity of free space. We can substitute this expression into Equation 1, as indicated at the right.

$$\bar{P} = \bar{S}(4\pi r^2) \quad (1)$$

↑
 $\bar{S} = c\epsilon_0 E_{\text{rms}}^2$

Solution Algebraically combining the results of each step, we have

$$\begin{array}{cc} \text{STEP 1} & \text{STEP 2} \\ \downarrow & \downarrow \\ \bar{P} = \bar{S}(4\pi r^2) & = c\epsilon_0 E_{\text{rms}}^2(4\pi r^2) \end{array}$$

The average power emitted by the light source is

$$\begin{aligned} \bar{P} &= c\epsilon_0 E_{\text{rms}}^2(4\pi r^2) \\ &= (3.00 \times 10^8 \text{ m/s})[8.85 \times 10^{-12} \text{ C}^2/(\text{N} \cdot \text{m}^2)](19.0 \text{ N/C})^2 4\pi(2.50 \text{ m})^2 = \boxed{75.3 \text{ W}} \end{aligned}$$

$$L_{\text{sun}} = 3.8 \times 10^{26} \text{ W}$$

How much radiation do we get on Earth per m²?

Radio signal, 5 miles from 10 kW transmitter:
 $12 \mu\text{W}/\text{m}^2$

Direct sunlight, Earth's surface: $\sim 1000 \text{ W}/\text{m}^2$

NIF laser fusion experiment: $10^{20} \text{ W}/\text{m}^2$

Radio signal, 5 miles from 10 kW transmitter:
 $12 \mu\text{W}/\text{m}^2$

Laser pointer: $100 \text{ W}/\text{m}^2$

Starlight, bright star: $100 \text{ nW}/\text{m}^2$

Microwave intensity, 2" from cell phone: $10 \text{ W}/\text{m}^2$

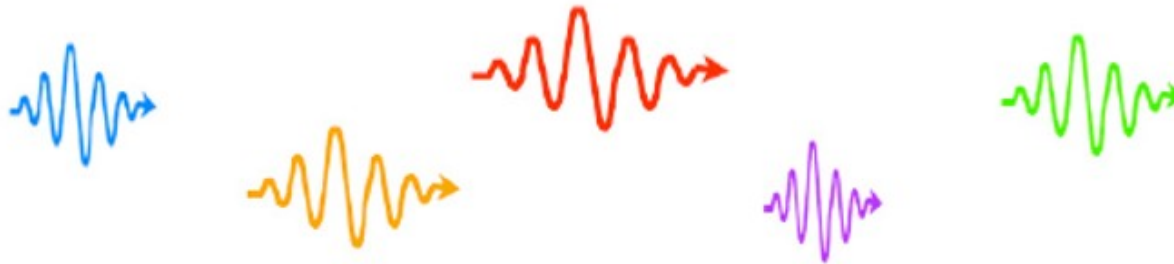
Photons (wave or energy packets)

Electromagnetic waves are composed of particle-like entities called *photons*.

Light also behaves like discrete particles called “photons”



“White light” (sunlight) contains all of the colors



Collectively, lots of photons with the same color (same wavelength) make an electromagnetic wave

$$E=hf \quad c=\lambda f \quad p=h/\lambda$$

Energy (E) = frequency x h = frequency x 4.13×10^{-15} in eV
 And frequency x wavelength = speed of light = 300,000,000
 Frequency in Hz wavelength in meters speed of light in m/s

$6.626070040(81) \times 10^{-34}$

J·s

h for energy in joules

Electronvolt

1.60218e-19

Joule

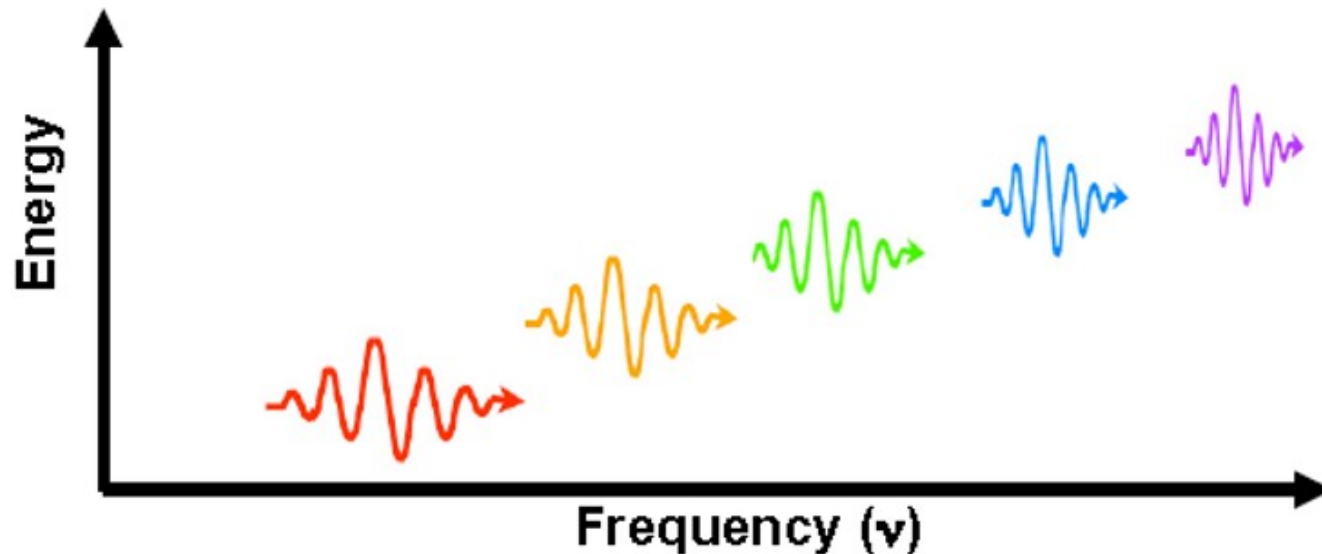
$4.135667662(25) \times 10^{-15}$

eV·s

h for energy in electron-volts

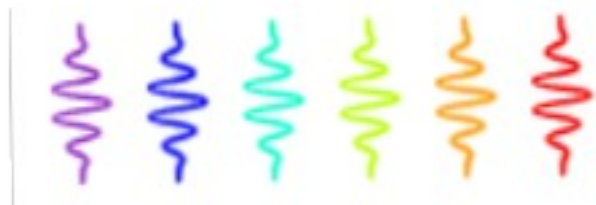
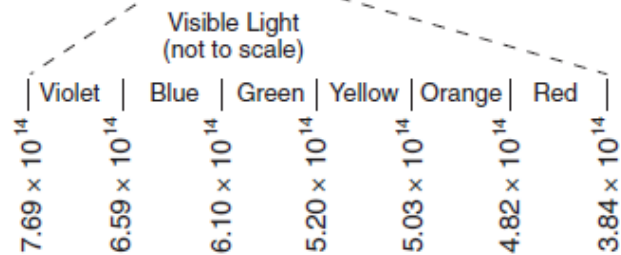
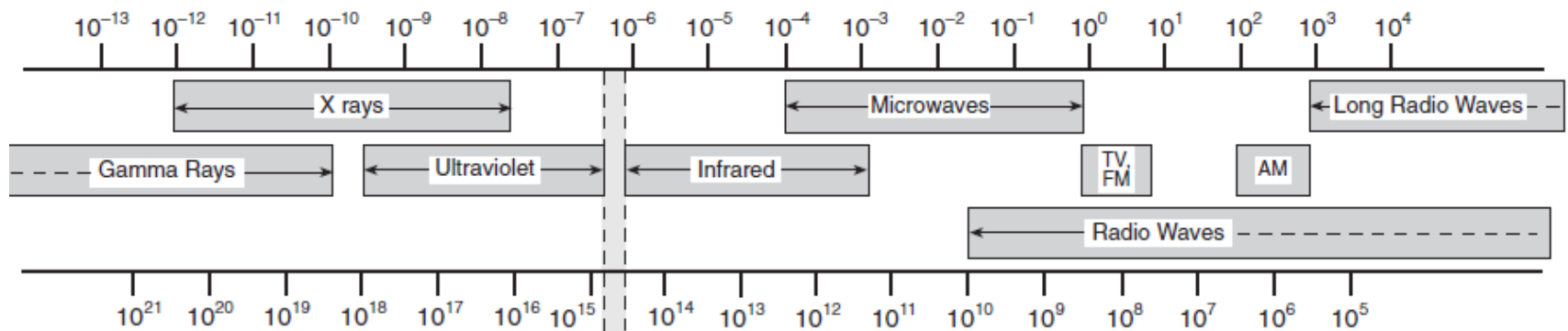
A photon has no mass, but its energy is $E = h\nu$

h = "Planck's constant," which is a VERY small number (6.63×10^{-34} in our units).



Photons of higher energy (E) have a higher frequency (ν) and shorter wavelength (λ)

Energy (E) = frequency $\times$ h = frequency $\times$ 4.13×10^{-15} in eV
And frequency $\times$ wavelength = speed of light = 300,000,000
Frequency in Hz wavelength in meters speed of light in m/s



Region	Wavelength
Hard gamma	1×10^{-9} nm
Gamma	1×10^{-6} nm
Gamma/X-ray	0.001 nm
X-ray	1 nm
X-ray/ Ultraviolet	10 nm
Ultraviolet	100 nm
Visible (blue)	400 nm
Visible (red)	700 nm
Infrared	10000 nm
Microwave	1 cm
Microwave/Radio	10 cm
Radio	100 m
Radio	100 km

Find the frequency and energy in eV

For each region

$1\text{nm} = 10^{-9}\text{m}$

$1\text{cm} = 0.01\text{m}$

$1\text{km} = 1000\text{m}$

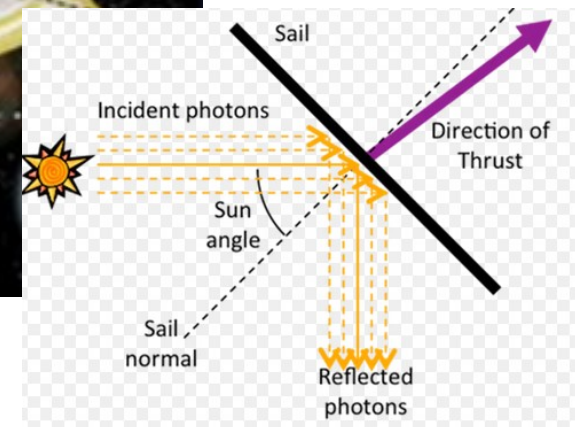
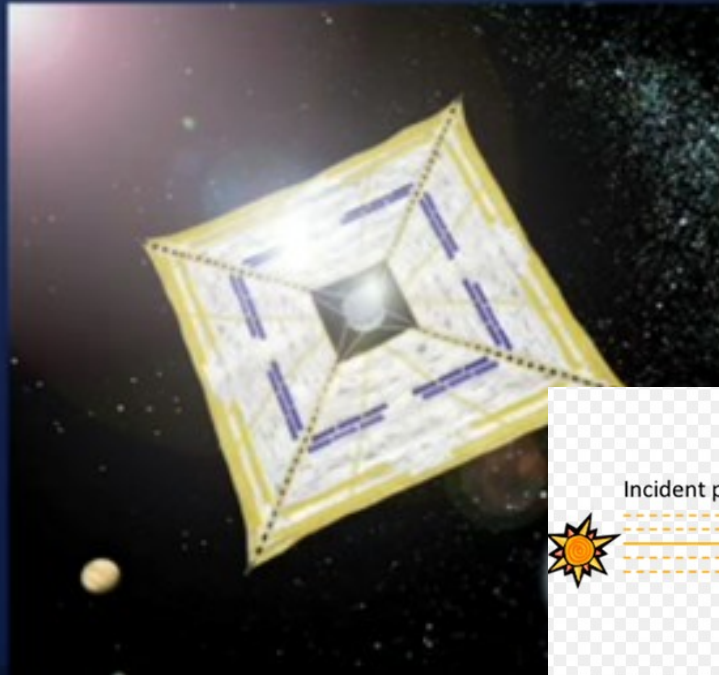
Energy (E) = frequency $\times$ h = frequency $\times$ 4.13×10^{-15} in eV
 And frequency $\times$ wavelength = speed of light = 300,000,000
 Frequency in Hz wavelength in meters speed of light in m/s

Where There's Energy There's Momentum

Momentum transfer results in
radiation pressure:

$$P_{\text{rad}} = \frac{\bar{S}}{c} \text{ Unit: N/m}^2$$

Solar sailing!



Einstein → momentum of a particle (photon) = energy / c
This is called the radiation pressure. More pressure if the photons bounce backs.
Explains the tails of the comets. It pushes the dust away from the sun

Summary

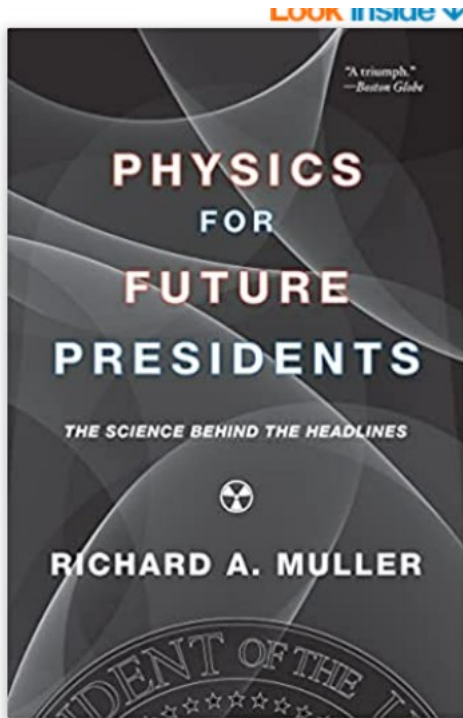
Maxwell's equations describe fully the behavior of electric and magnetic fields

In a vacuum there's complete symmetry between electric and magnetic fields

- Changing magnetic fields give rise to electric fields
- Changing electric fields give rise to magnetic fields

Changing fields result in *electromagnetic waves*

- Propagate at the speed of light, c
- Transverse waves
- Spectrum of possible wavelengths/frequencies
- Produced by *accelerated charge*
- Carry energy and momentum



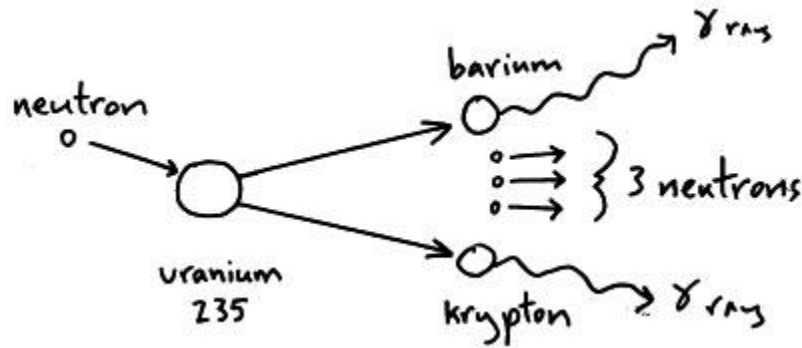
Reference for my slides.
Highly recommended book to read.

Physics for Future Presidents: The Science Behind the Headlines Reprint Edition

by [Richard A. Muller](#) (Author)

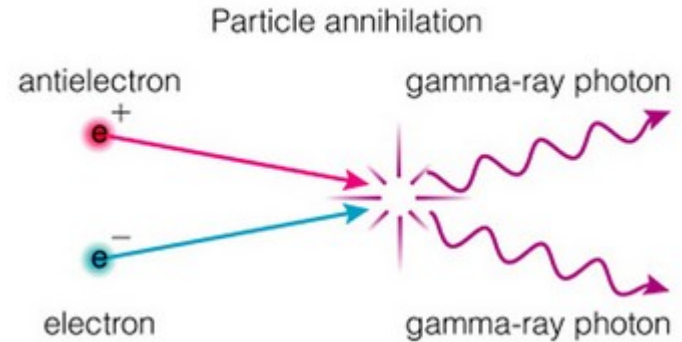
GAMMA RAYS:

Highest frequency range from 3×10^{19} Hz to 10^{23} Hertz. Wavelength same size of nuclei (10^{-12} m and smaller) Emitted in nuclear processes; radioactive decay, Fission, fusion .. damaging. Ionizing radiation. Can cause radiation sickness and Death or induce cancer. But used for imaging.



Gamma rays can be emitted when a nucleus decays. Here is reaction of fission used for the atomic bomb

Explosive and violent event in space
Can produce gamma ray bursts.
It involves neutron stars and black holes.



Gamma rays can be produced when an Electron and an anti-electron (positron) collide.

Matter+ anti_matter = energy

PET SCAN

Using gamma rays

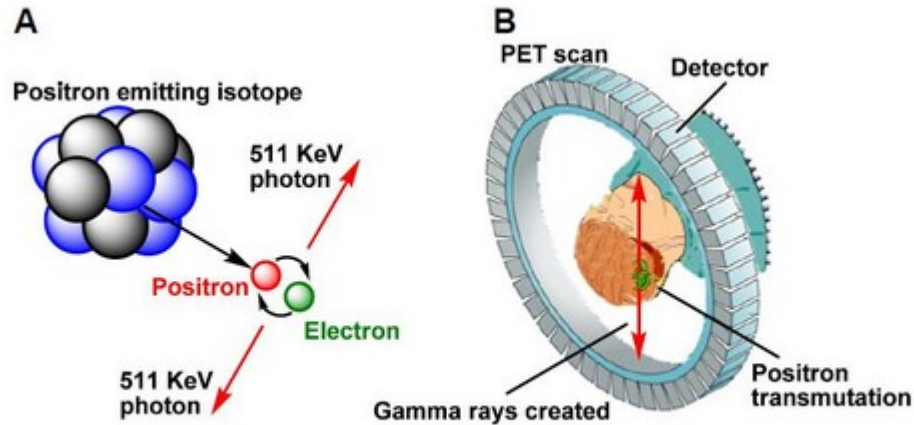
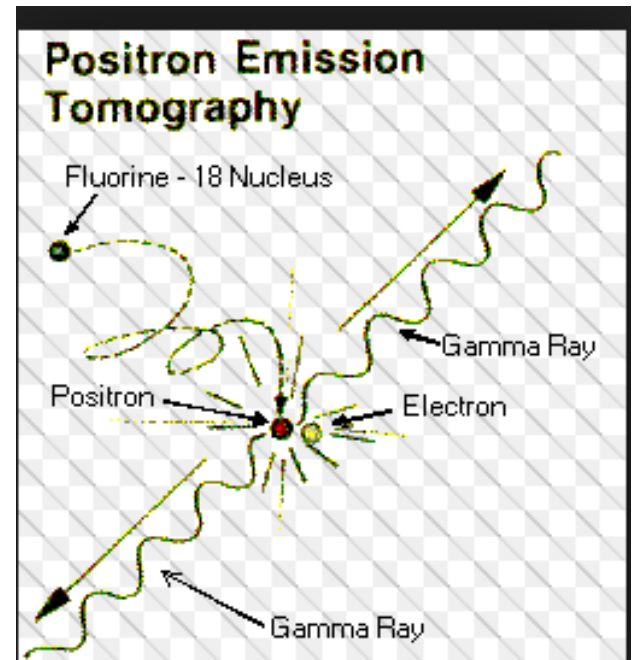
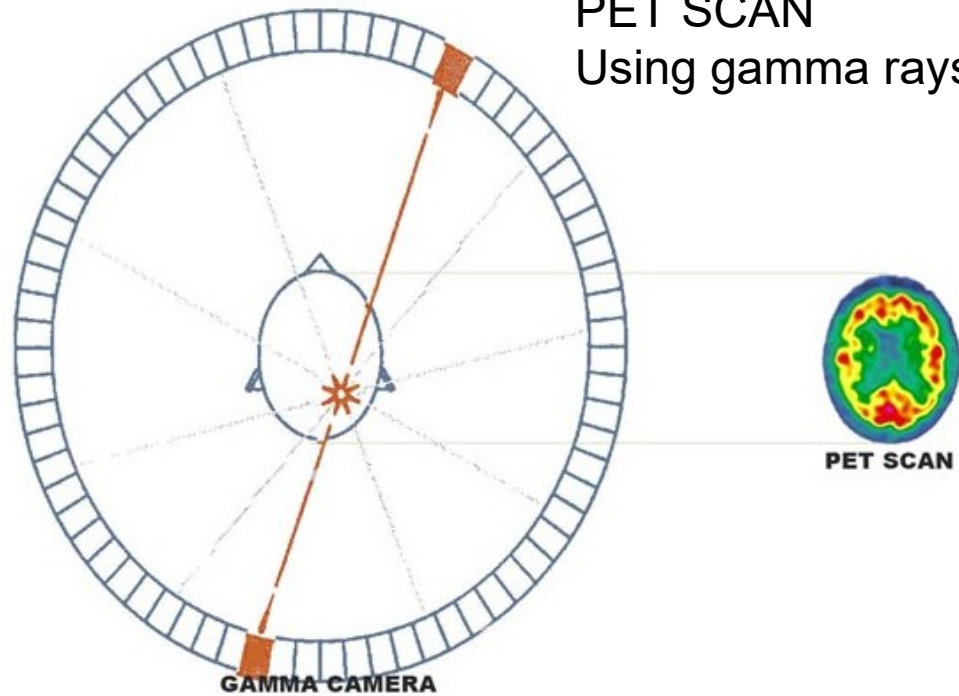
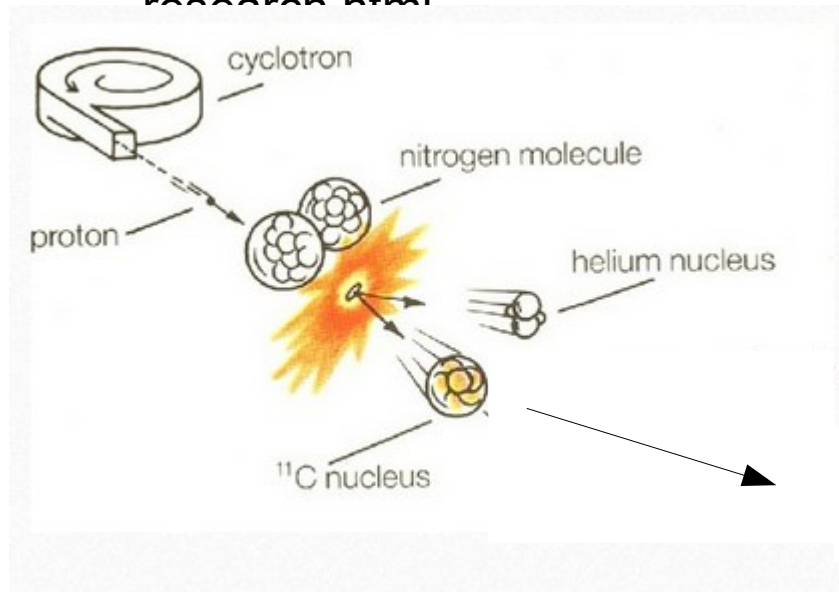


Fig. 1. (A) A positron and an electron annihilate producing two 511 keV photons travelling in opposite directions. (B) The 511 keV photon was registered by the circular gamma ray detector array in the PET camera.

Radionuclide	Half-life	Chemical Form	Nuclear Reaction(s)
^{11}C	20 min	$^{11}\text{CO}_2$ / $^{11}\text{CH}_4$	$^{14}\text{N}(\text{p},\alpha)^{11}\text{C}$
^{13}N	10 min	$^{13}\text{NH}_4^+$ / $^{13}\text{NO}_x$	$^{16}\text{O}(\text{p},\alpha)^{13}\text{N}$
^{15}O	2 min	$^{15}\text{O}_2$	$^{15}\text{N}(\text{p},\text{n})^{15}\text{O}$ $^{14}\text{N}(\text{d},\text{n})^{15}\text{O}$
^{18}F	110 min	$^{18}\text{F}^-$ or $^{18}\text{F}_2$	$^{18}\text{O}(\text{p},\text{n})^{18}\text{F}$ $^{20}\text{Ne}(\text{d},\alpha)^{18}\text{F}$

<http://www.imperial.ac.uk/people/a.gee/research.html>



The radioisotopes used for PET scan have to be made in the hospitals using a cyclotron. (remember a magnetic field is used to trap protons around a circular path. An electric field is used to speed them up.)



Here is a list of some positron emitters :

Carbon-11 (20 min)

Nitrogen -13 (9 min)

Oxygen-15 (2min)

Fluorine-18 (110 min)

Iodine-124 (4.2 days)



The radioisotopes have a very short life. They have to be “ made” in the hospital Before the session. This is expensive because it requires accelerators.

In beta plus decay, a proton is converted, via the weak force, to a neutron, a positron (also known as the "beta plus particle", the antimatter counterpart of an electron), and a neutrino.

Iodine-124 for example, goes to the thyroid. Doctors can detect if there is blank spots or hyperactivity by detecting gamma radiation.

O-15 can flow to the brain for probing the brain.

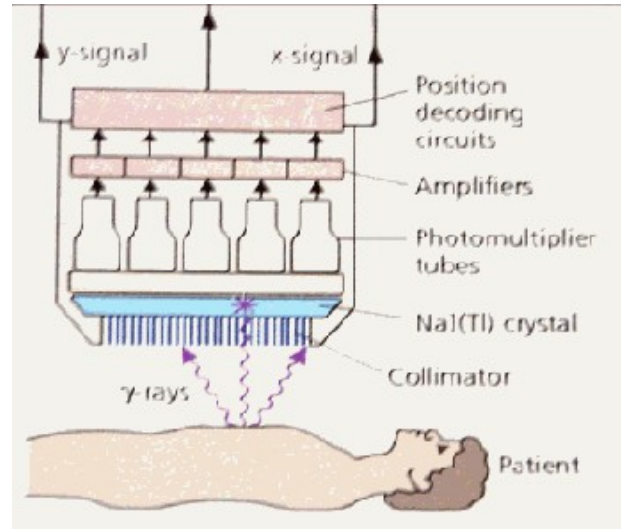
C-11 can flow to biological compounds.

http://www.isotopeworld.com/filestore/EIR_Medical%20Isotopes%20in%20the%2021st%20Century.pdf

http://www.search.com/reference/Positron_emission

SPECT (single photon emission tomography)

Like PET but cheaper:



SPECT is similar to PET in its use of radioactive tracer material and detection of gamma rays. In contrast with PET, however, the tracer used in SPECT emits gamma radiation that is measured directly, whereas PET tracer emits positrons that annihilate with electrons up to a few millimeters away, causing two gamma photons to be emitted in opposite directions. A PET scanner detects these emissions "coincident" in time, which provides more radiation event localization information and, thus, higher resolution images than SPECT (which has about 1 cm resolution). SPECT scans, however, are significantly less expensive than PET scans, in part because they are able to use longer-lived more easily-obtained radioisotopes than PET

http://www.fas.org/irp/imint/docs/rst/Intro/Part2_26d.html

<http://www.slideshare.net/Panduekoyudho/spect-medicine>

Radioisotopes for this are imported from Canada.

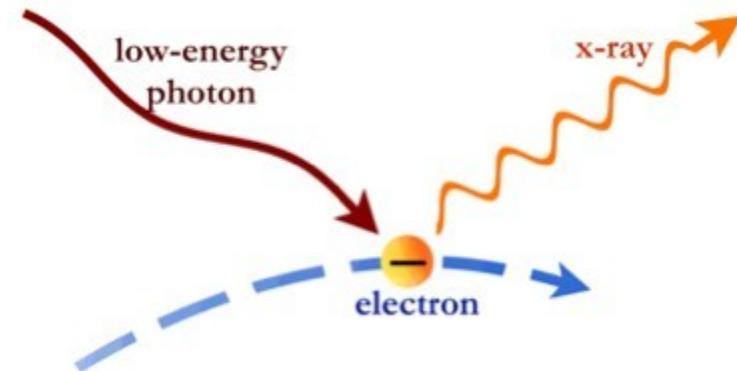
http://en.wikipedia.org/wiki/Single-photon_emission_computed_tomography

http://www.umsl.edu/~tsytsarev/tsytsarev_files/Lecture18.htm



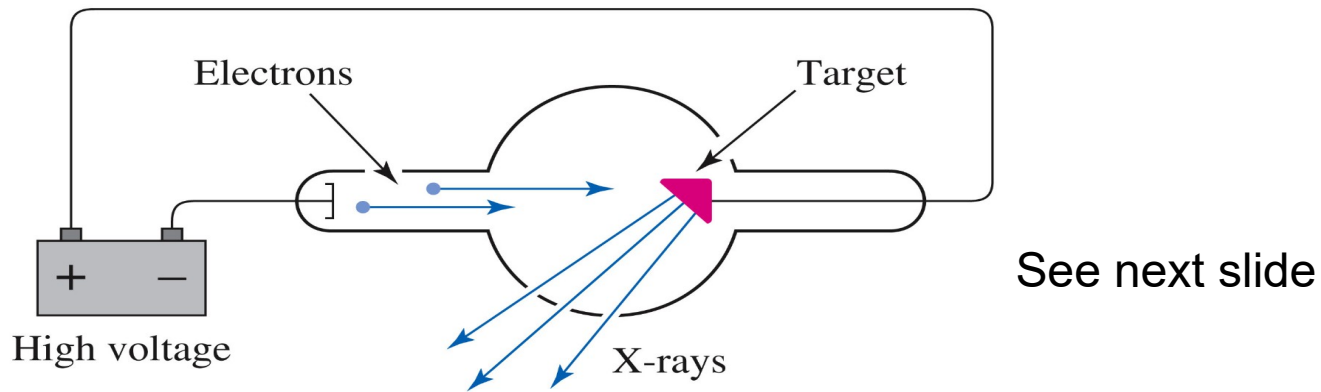
Other ways gamma
Rays are emitted.

Crab nebula, electrons decelerate so much,
they emit flares of X-rays and
Gamma rays.



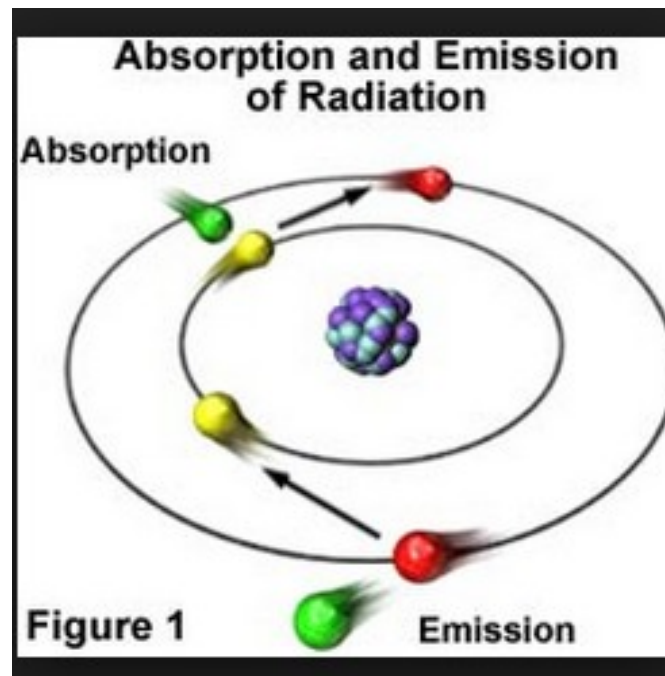
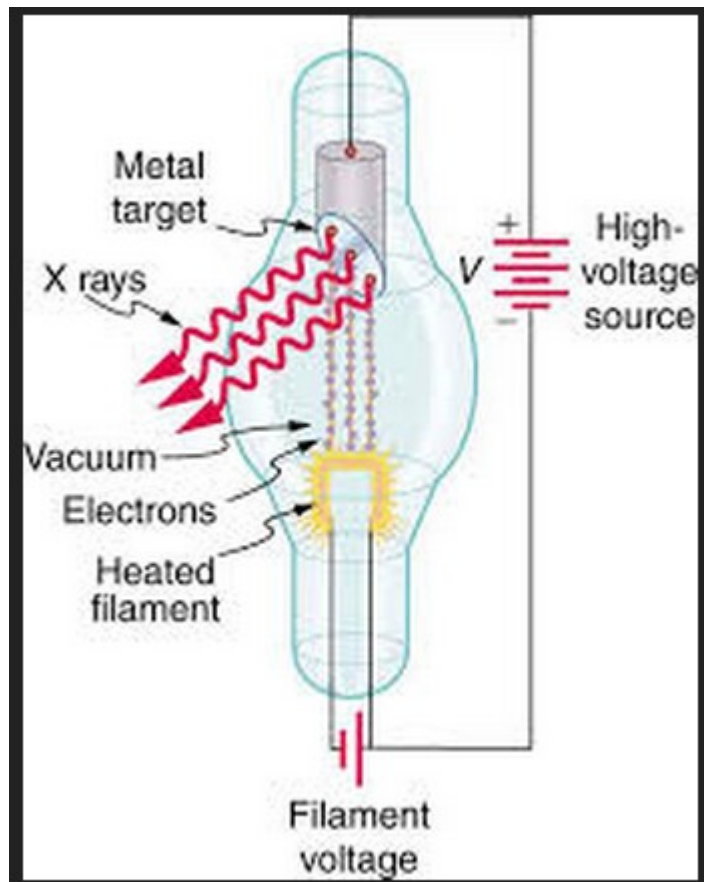
X-RAYS

- penetrating and ionizing too. Can cause damage to cell too.
Size of atoms and molecules (10^{-10}m). Can be used to probe crystals and molecules.
X-rays are produced by smashing high-speed electrons into a “target” made of Tungsten or other metal. This is how x-rays are produced in hospitals.



The electron lose their kinetic energy when decelerating as radiation.
Plus they kick out some electrons in the metals.
Other electrons (from higher energy Levels) take their Place
Emitting X-rays.

<http://www.youtube.com/watch?v=IRBKN4h7u80>



X-RAYS

X-rays pass through the body but they are absorbed by large atoms like calcium. Carbon and hydrogen let them go through. Lead atoms stop x-rays.

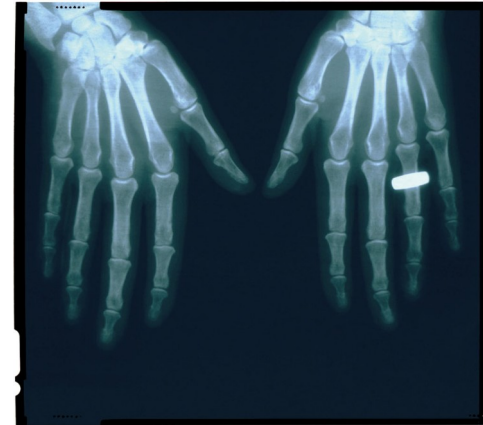
Discovered by 1895 by Wilhelm Conrad Röntgen, who received the first Nobel Prize in Physics in 1901.

He used a cathode ray tube and didn't know what these radiations were. He called them X-rays !!

Very quickly used in the medical field.

First, damaging. Now very safe.

<http://onlinephys.com/giants2.html>



X-RAY EXPOSURE
GIVE ME A DOSE
OH MY POOR HEAD
I CAN'T STAND IT!
IT'S QUICK IT'S SAFE IT'S SURE
OF KOHLER'S ANTIDOTE

KOHLER'S ANTIDOTE

FOR HEADACHE HAS RELIEVED THOUSANDS

WHY NOT YOU? IT WILL CURE THE WORST KIND OF HEADACHE. WHETHER CAUSED BY Sick Stomach, Excess of Spirituous Liquors or Neuralgia

GIVES RELIEF IN 15 MINUTES

8 DOSES Mailed to any address in U. S. 25 CENTS. Post paid, on receipt of price.

KOHLER MFG. CO., BALTIMORE, MD.

When you write, please mention "The Cosmopolitan."



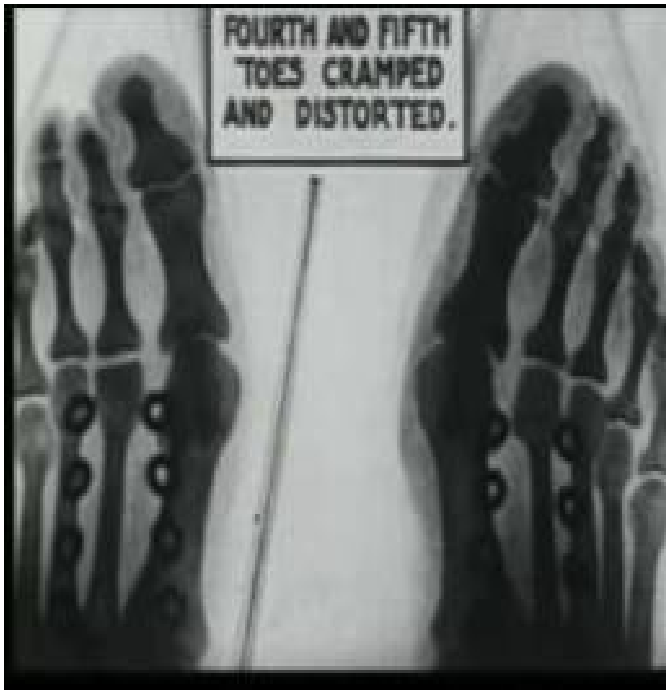
The 8-year-old from Ivory Coast was inside this suitcase when authorities discovered him. (AP)

In the 50s, they used Xrays in shoe stores.

1/5 → cancer

Remember app

See links on website.



Radiation was 30 rem. Find the probability to get cancer in %
1 rem u have a change of 1/2500 to get cancer
If 100,000 people use the machine. How many cancer ? (1200)

CT SCAN or CAT SCAN



The disadvantage is that it uses x-rays.
(you increase the probability of cancer).

CT scans are made by taking x-rays from
Different directions. That's the tomography part.



Less expensive then PET but more invasive
The radiation received is more important.
Not efficient at probing soft tissue.

<http://health.learninginfo.org/pneumothorax.htm>

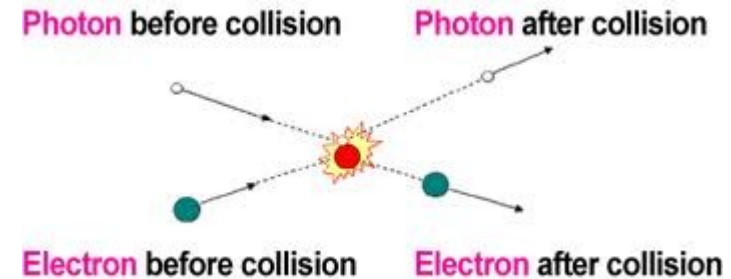
<http://www.ctlab.geo.utexas.edu/overview/index.html>

<http://www.physicscentral.com/explore/action/scans-1.cfm>



XRAY backscatter

Remember : X-ray behaves more like particle than wave because of The high energy of the photons (frequency). The X-ray can either go through The object but they can also bounce off electrons. The amount of back scatter Is roughly determined by the density of the material.



http://articles.cnn.com/2010-05-06/travel/tsa.scanner.assault_1_full-body-scanning-tsa-screener?_s=PM:TRAVEL

**Illegal immigrants
Attempting to enter
Southern Mexico from
Guatemala in truckload of
Bananas !**



UV radiation

Just above the violet. Still energy high enough to induce skin cancer. Trigger a Chemical process responsible for tanning. Not penetrating like x-rays and gamma rays. Emitted by very hot objects. The UV radiation from the Sun is blocked by the Ozone layer. Application : Indian-born physicist Dr. Ashok Gadgil developed a cheap Way to kill the germ in water using UV radiation. Can treat 15 liters per minute.

<http://www.wipo.int/ipadvantage/en/details.jsp?id=2564>

Used to sterilize tools and workplaces. UV laser are used in dermatology and metallurgy(engraving), forensic (to detect body fluid).

Also in fluorescent lamps.

Atoms of mercury are excited by electric discharge.

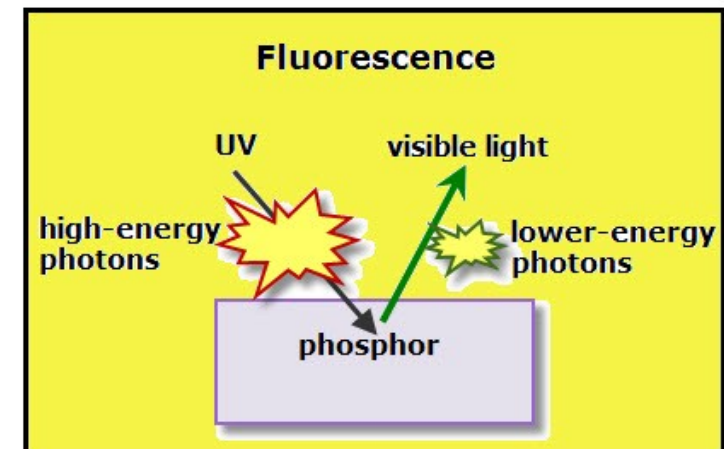
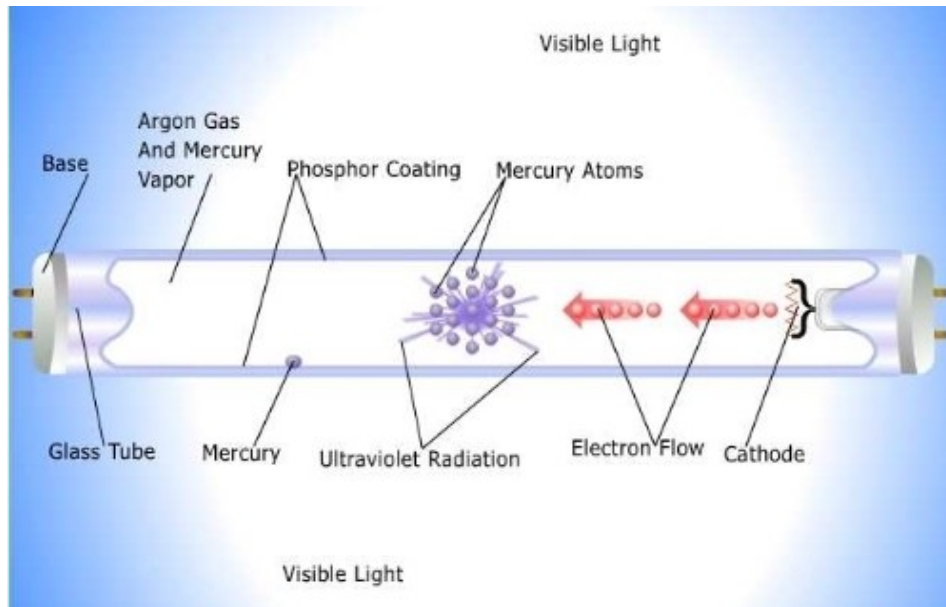
Emit UV. UV absorbed by a powder coating the lamp.

Fluorescent materials. Re emit visible light.



Also in **fluorescent lamps**.

Atoms of mercury are excited by electric discharge. Mercury changes from liquid to gas. The electrons of mercury are bumped to higher levels of energy by collisions (with the Argon gas). When the electrons go back to lower energy level they emit UV photons. The UV photons excite in turn the electrons of the phosphor coating the inside of the lamp. When they go back, visible light is emitted.



<http://www.myclimatechange.net/default.aspx?cat=3&subjectId=51>

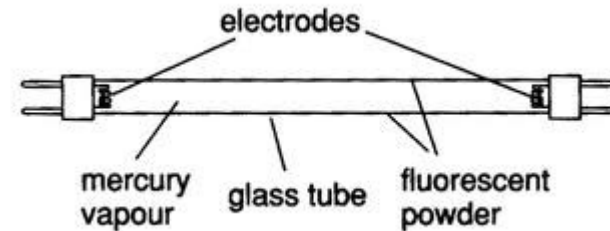
WHITER THAN WHITE

UV light is absorbed by phosphor or fluorescent chemical (for making the jump between energy gaps) and re emit light.



We use this idea in fluorescent light.

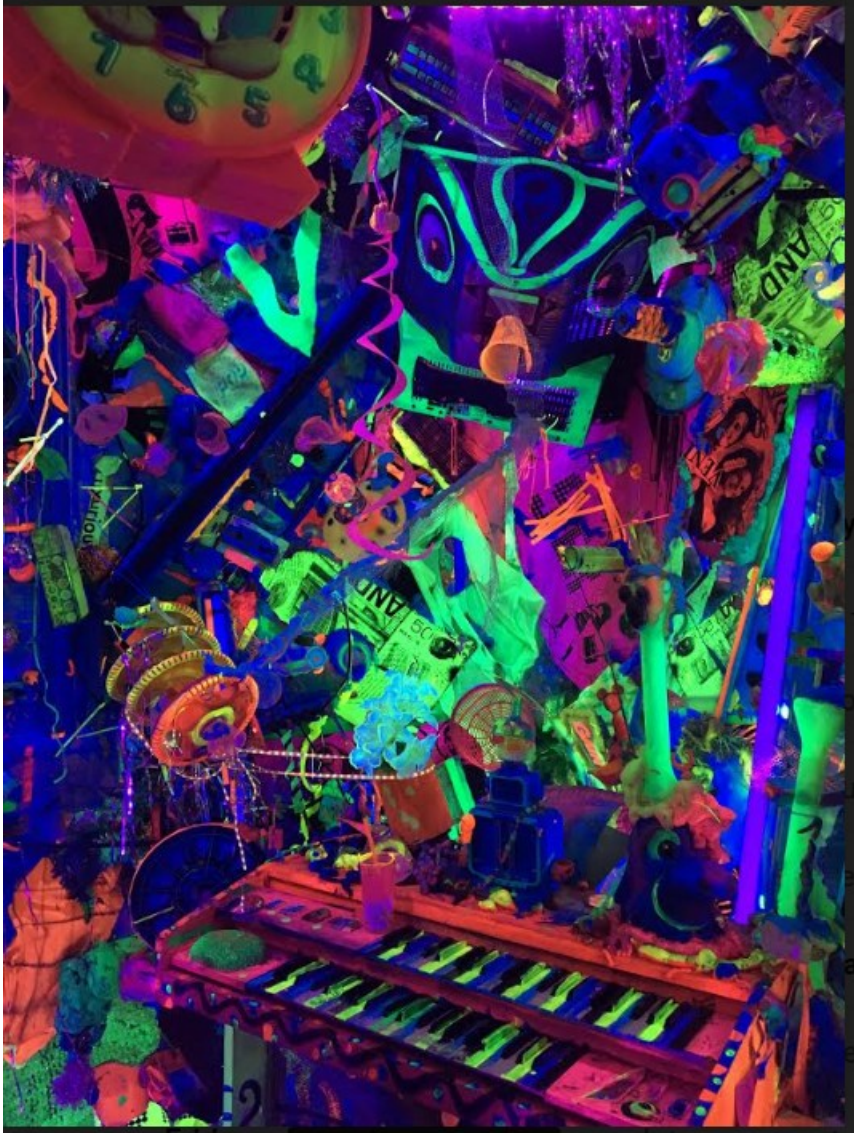
Fluorescent minerals and **laundry soap**.



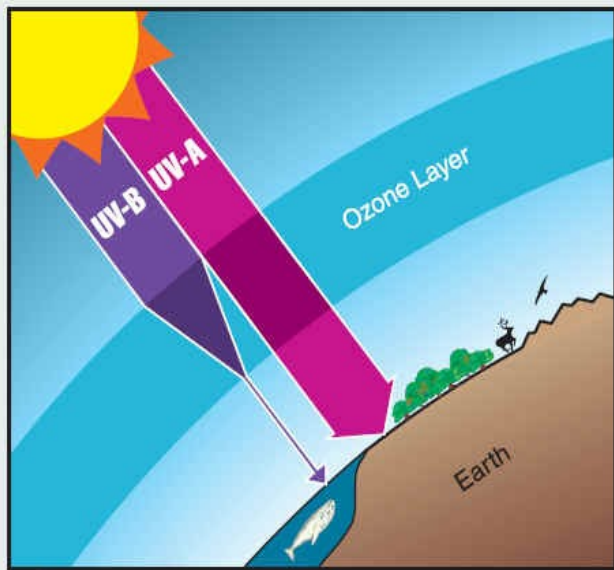
cloth washed with fluorescent chemicals will appear Whiter under the Sun.

UV radiations are absorbed and visible Light emitted Making white brighter.

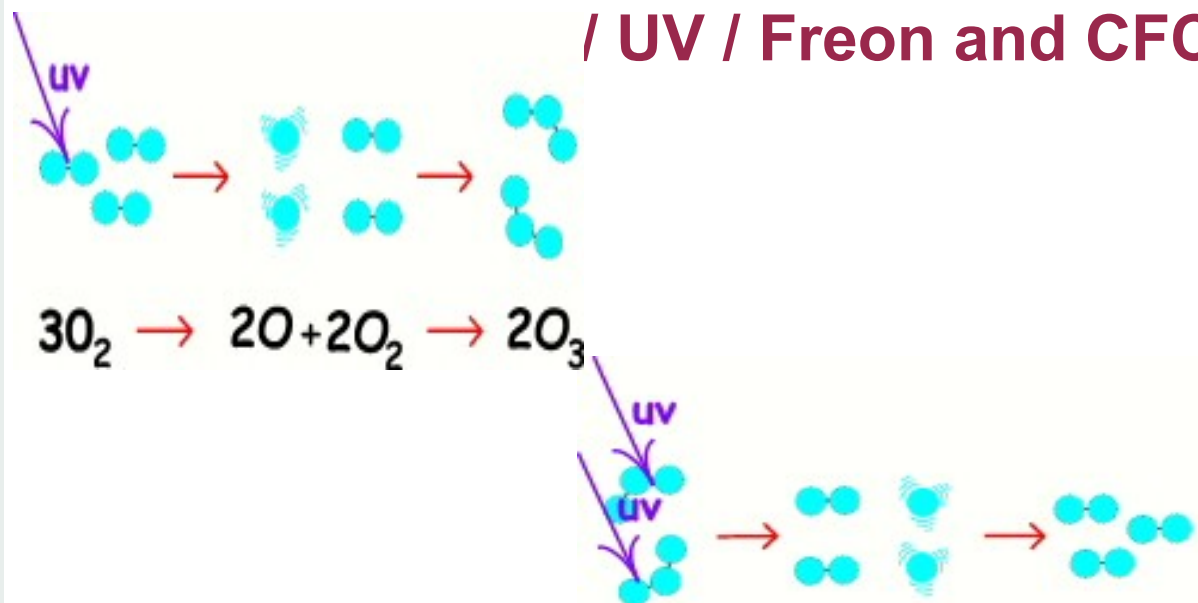
<http://blogs.discovermagazine.com/notrocketscience/2011/12/23/why-do-scorpions-glow-in-the-dark-and-could-their-whole-bodies-be-one-big-eye/#.WsfpMpW5vIV>



UV Protection by the Ozone Layer



/ UV / Freon and CFCs



<https://www.youtube.com/watch?v=IniJx-vRHG0>

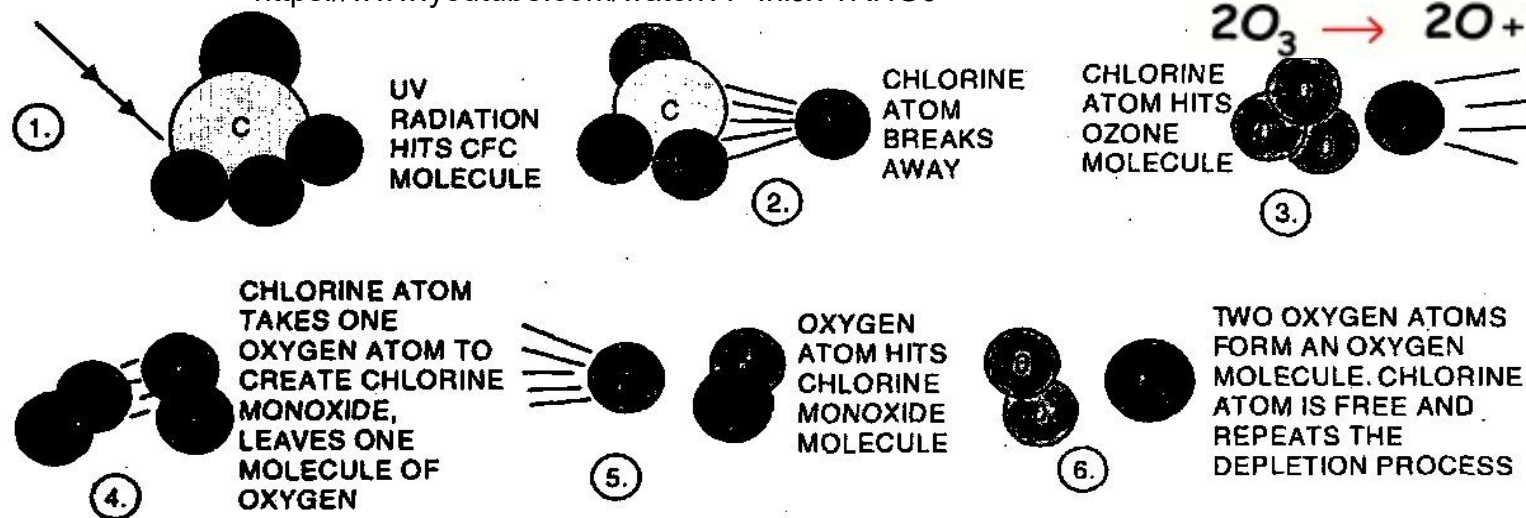
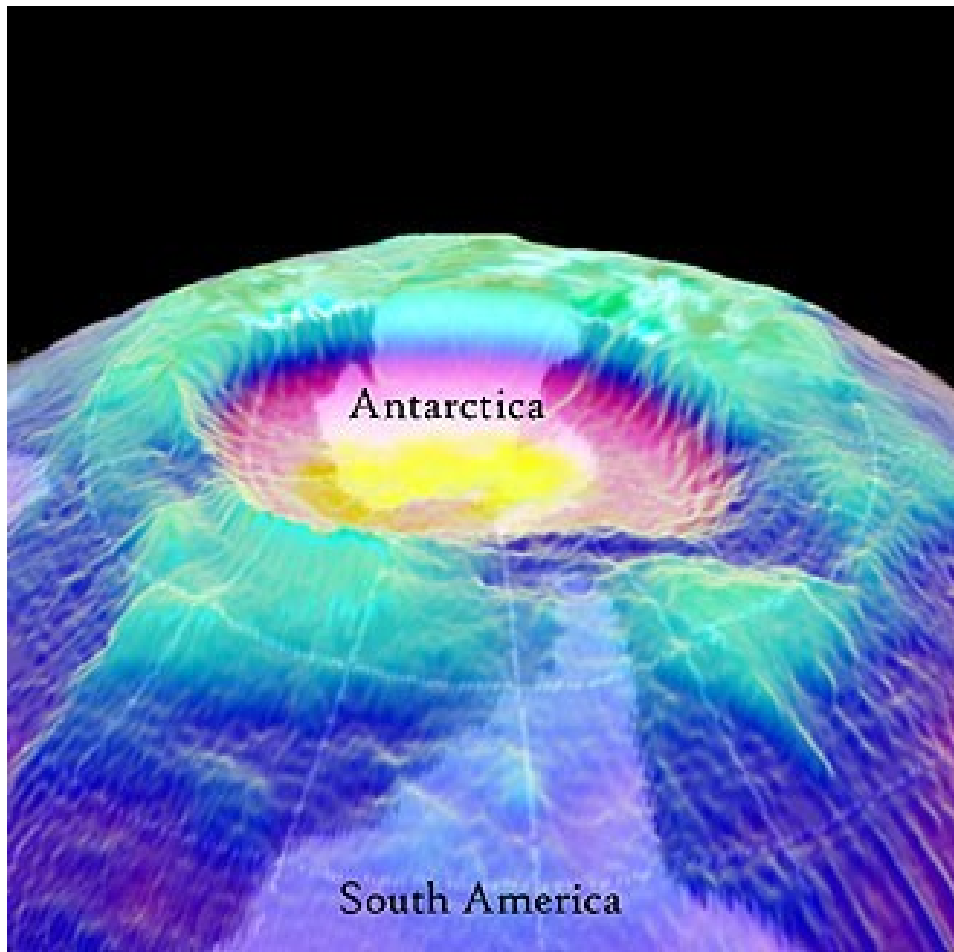


Fig. 1.9. Interaction of CFCs and UV radiation damaging the ozone layer.

<http://resources.yesican-science.ca/trek/scisat/final/grade9/ozone1.html>



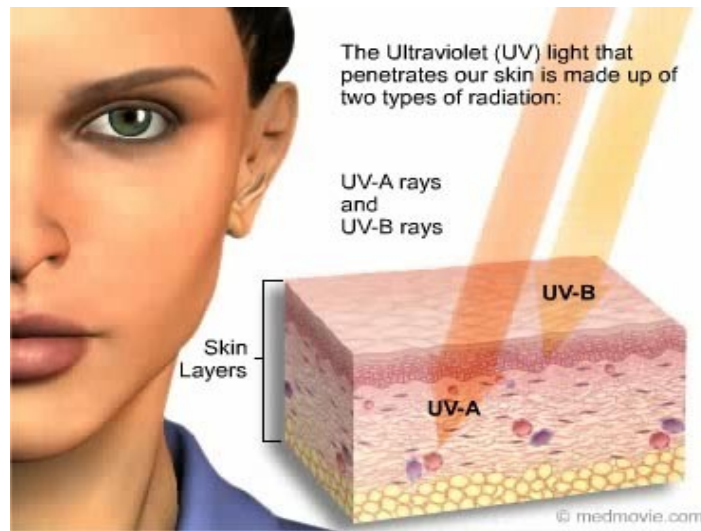
Ozone layer between 40,000 ft to 60,000 ft

Nobody knew why the hole
Was above Antarctica and not
Some where else. Scientists
Realized that certain crystals
Of nitric acid formed there
In the early spring, and on the
Surface of those crystals,
The chlorine and fluorine
Was far more effective at
Destroying the ozone.

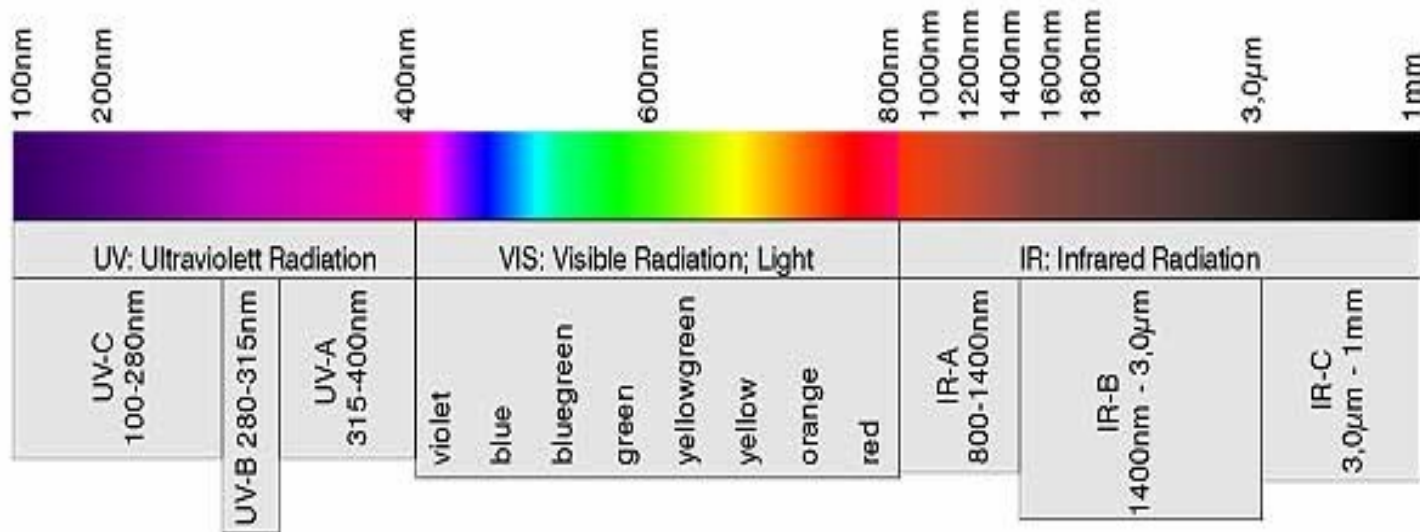
Freon was used in AC,
Refrigerator and propellants. Now,
It's illegal but the CFCs stick for
A long time.

<http://www.freedomspheonix.com/News/049633-2009-05-01-ozone-hole-purportedly-modifies-antarctic-winds.htm?EdNo=001&From=>

Today, nitrous oxide is used as a propellant



Notice:
You have near IR and
Far IR

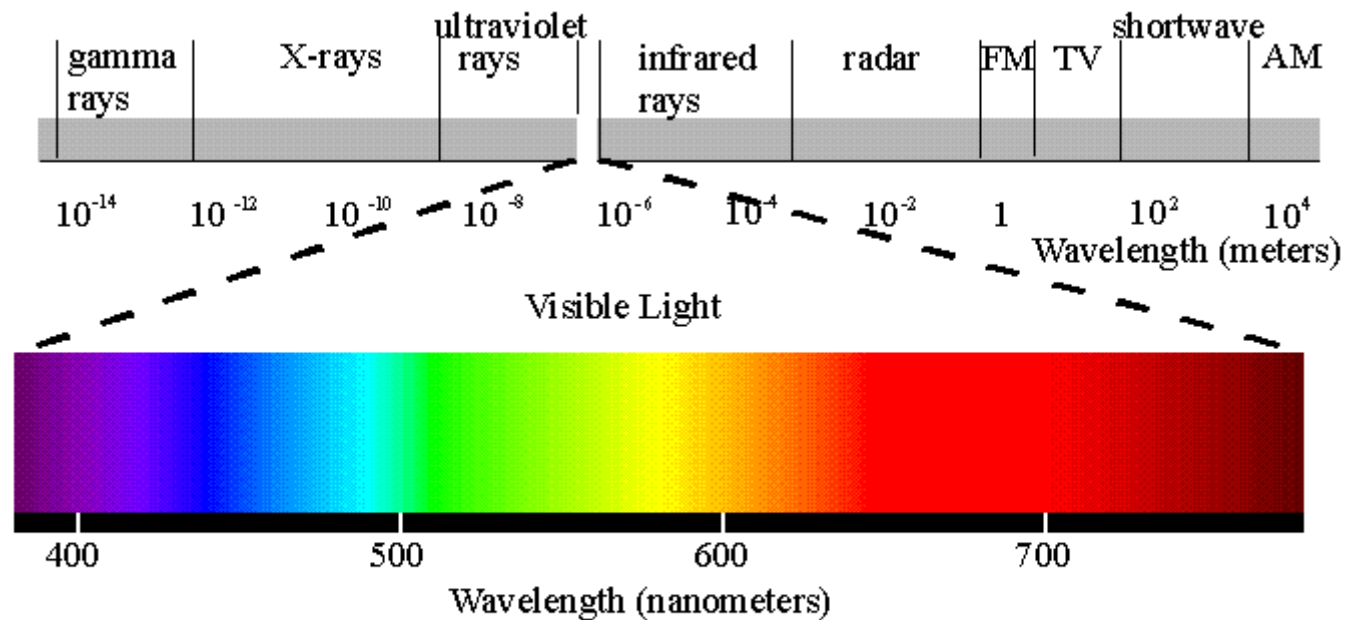


The form of UV that is most potent for burning and cancer has a wavelength Of about 300nm. This kind of radiation is absorbed by ordinary window glass.

Normal glass blocks all of UVB but allows UVA to come through.

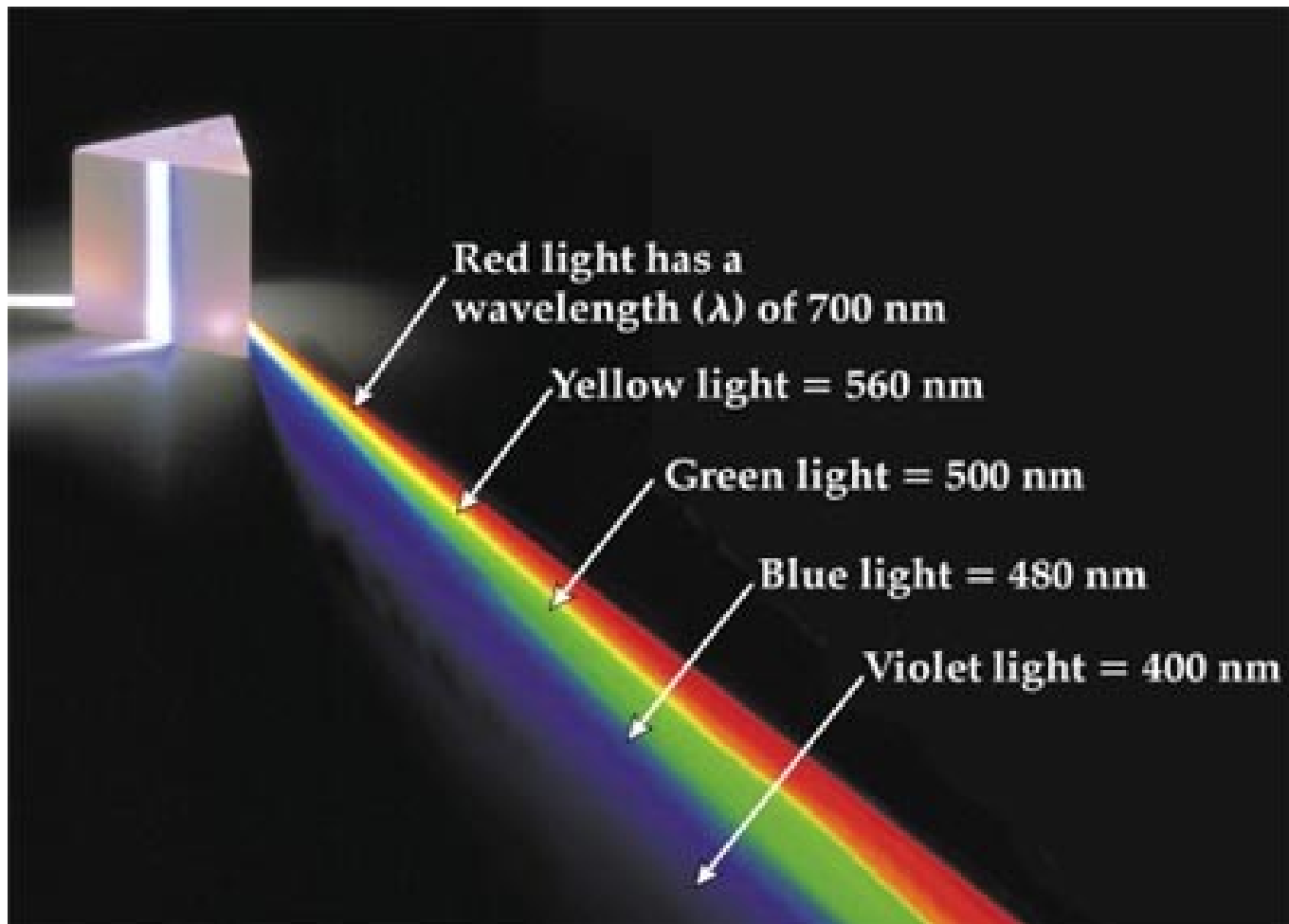
Example 1 The Wavelength of Visible Light

Find the range in wavelengths for visible light in the frequency range between $4.0 \times 10^{14} \text{ Hz}$ and $7.9 \times 10^{14} \text{ Hz}$.



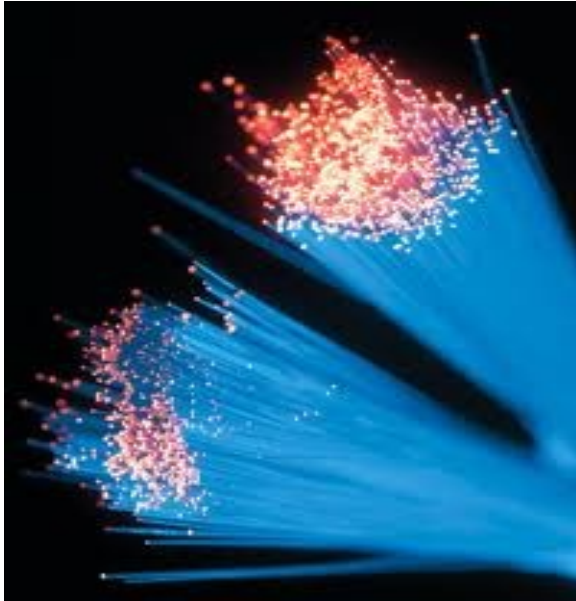
[Table of Contents](#)

[Visual Stimulus](#)



fiber optics: visible light can transport more data

The optical fibers have become very efficient to transport information.



Because of the high frequency of visible light,
You can transport more data per second.
The bits per second is called the baud.
The number of bits of second you can transport
Is the frequency of the wave. (about).

Indeed, information is coded are series of 0 and 1.
So you turn off and on the wave carrying the
Information. You can not send signal faster than the
Frequency. The number of on or off per second =
Frequency of the wave at most.

So you can transport about 10^{14} bits per second. Higher is the frequency, more information can Be transmitted per second.

So visible light can transport more info than Radio wave (smaller frequency).

This rule was found in 1940 by Claude Shannon,

INFRA RED radiations

Heat waves. Produced by any warm body.
(any thing with temperature).

Emitted by the molecules jiggling around.

Used in low energy lasers. Remote

Control. Objects in space emit in the IR.

Used by military to detect tank, planes ...

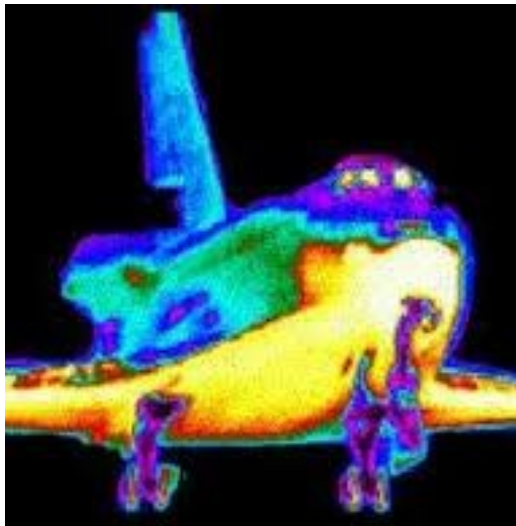
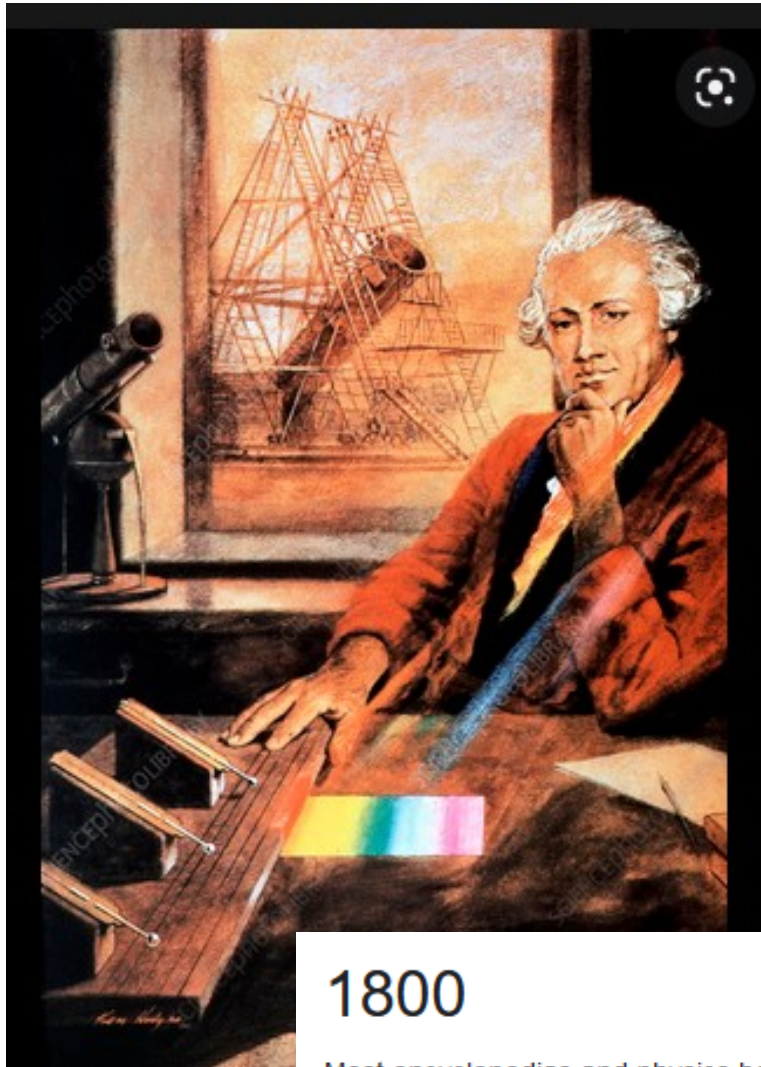


Table 8.1 Approximate Frequencies and Wavelengths of Different Colors

Color	Frequency Range ($\times 10^{14}$ Hz)	Wavelength Range ($\times 10^{-7}$ m)
Red	4.0–4.8	7.5–6.3
Orange	4.8–5.1	6.3–5.9
Yellow	5.1–5.4	5.9–5.6
Green	5.4–6.1	5.6–4.9
Blue	6.1–6.7	4.9–4.5
Violet	6.7–7.5	4.5–4.0



1800

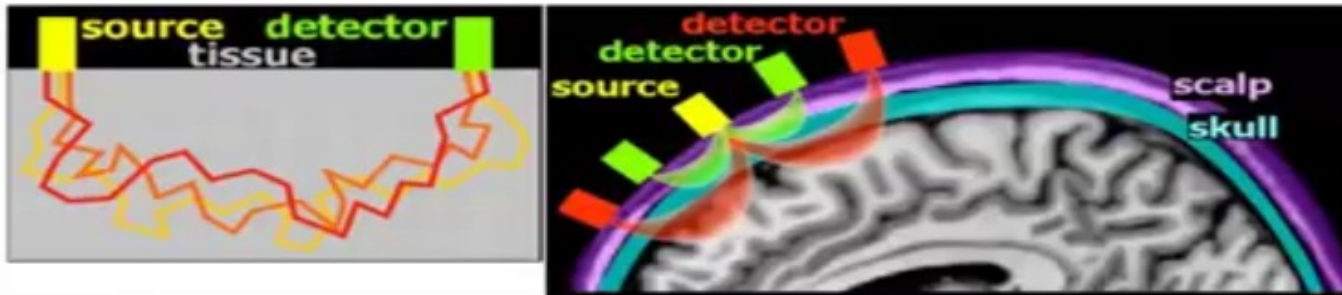
Most encyclopedias and physics books credit the great British astronomer Sir William Herschel with the discovery of infrared radiation **in 1800**.

He discovered Uranus

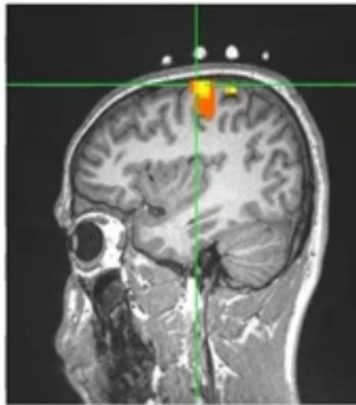


Functional Near-Infrared Spectroscopy (fNIRS)

- Each source-detector pair probes a 'banana-shaped' region
- fNIRS can only image the surface of the brain (cerebral cortex)
- Multiple source-detector pairs are used simultaneously to map neuronal activity on the brain's surface



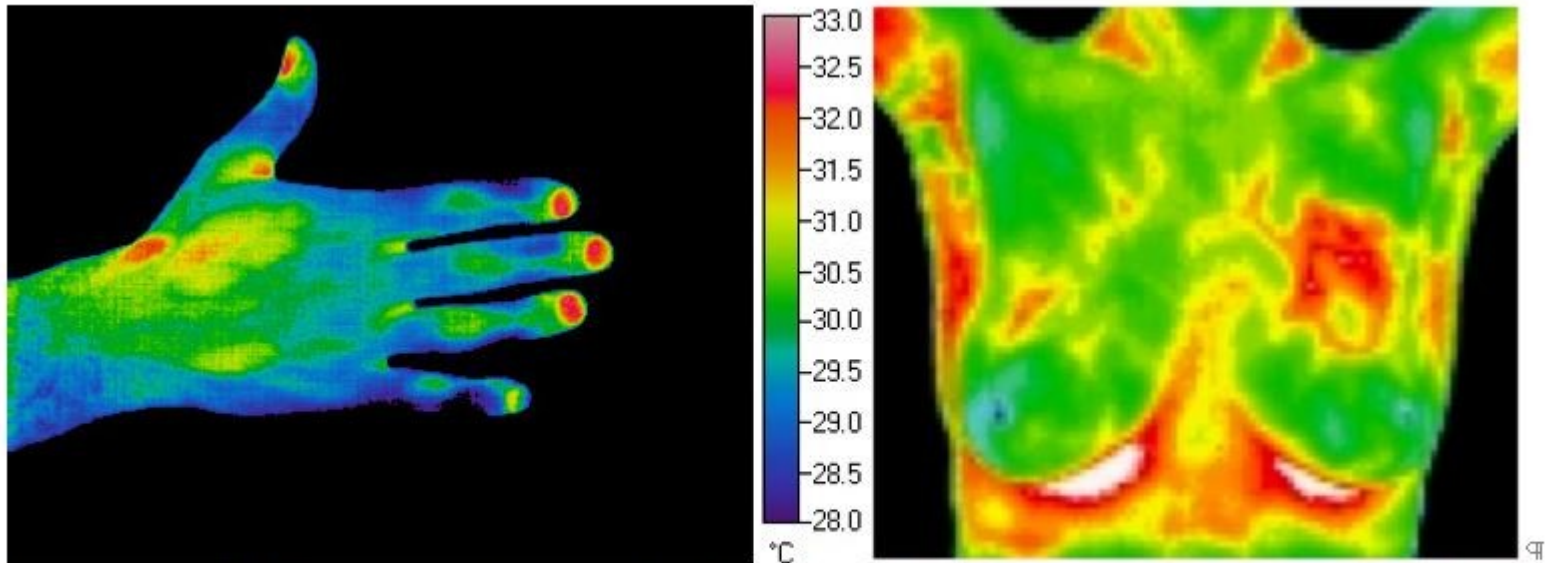
Using NIR to probe the surface of the brain.



https://en.wikipedia.org/wiki/Near-infrared_spectroscopy

A thermogram can also depict abnormal heating indicating infection

In the image below, the right foot of the patient is noticeably cooler, suggesting reduced circulation related to nerve damage



Thermograph image of a human hand, with an infection on the wrist (near the thumb), and thermograph showing breast cancer (on upper breast on right).
(Left photo from NASA; right photo copyright Meditherm.)

http://www.fas.org/irp/imint/docs/rst/Intro/Part2_26d.html

Note: thermography can also be done by painting the skin with a chemical (known as a liquid Crystal) that shows visible color change with temperature

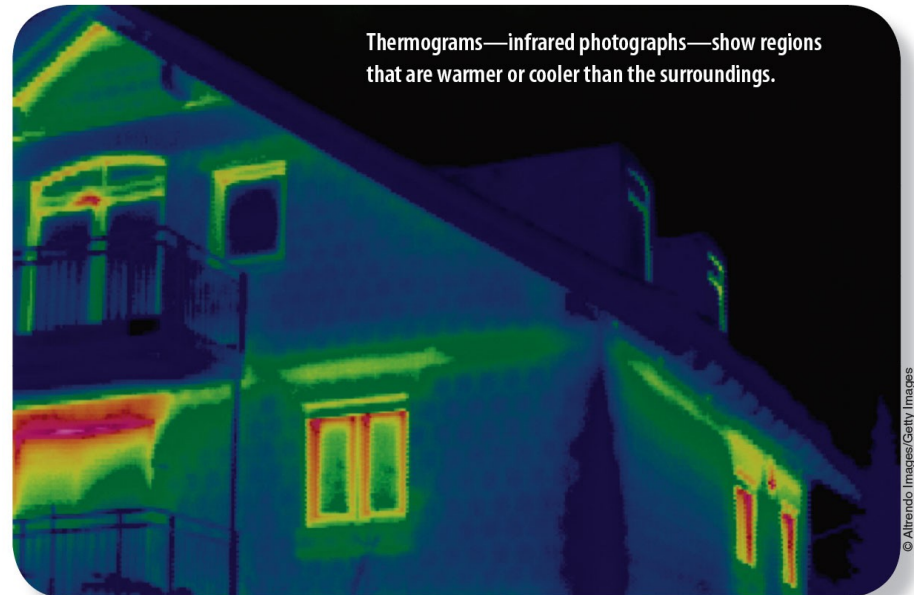
Blackbody radiation=

radiation emitted by thermal emitter (any thing with temperature) .

Power (energy/second) proportional to temperature raised to the 4th power.

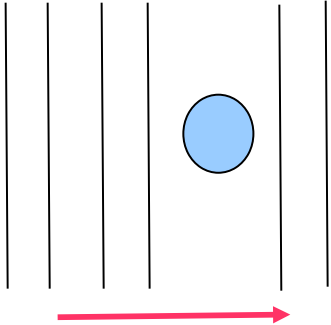
This means we can determine the temperature of an object by measuring the EM radiation it emits.

A *thermogram* is a “picture” of the IR radiation emitted by an object.

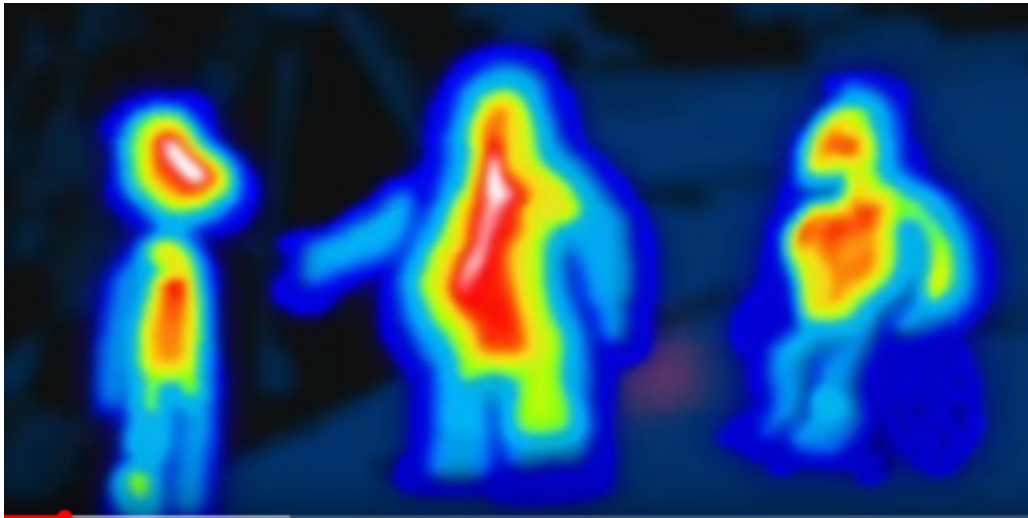


FIRE fighter can use INFRARED CAMERAS to see through the smoke.
Long wavelength diffract more than small wavelength so IR
Diffract around dust and smoke particles

<http://www.infraredcamerasinc.com/infrared-thermography-firefighting.html>



For more information please contact us at: <http://www.infraredcamerasinc.com> 1(409)861-0788

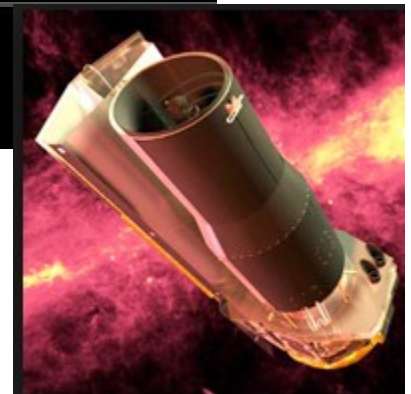


Family guy
As seen by the predator

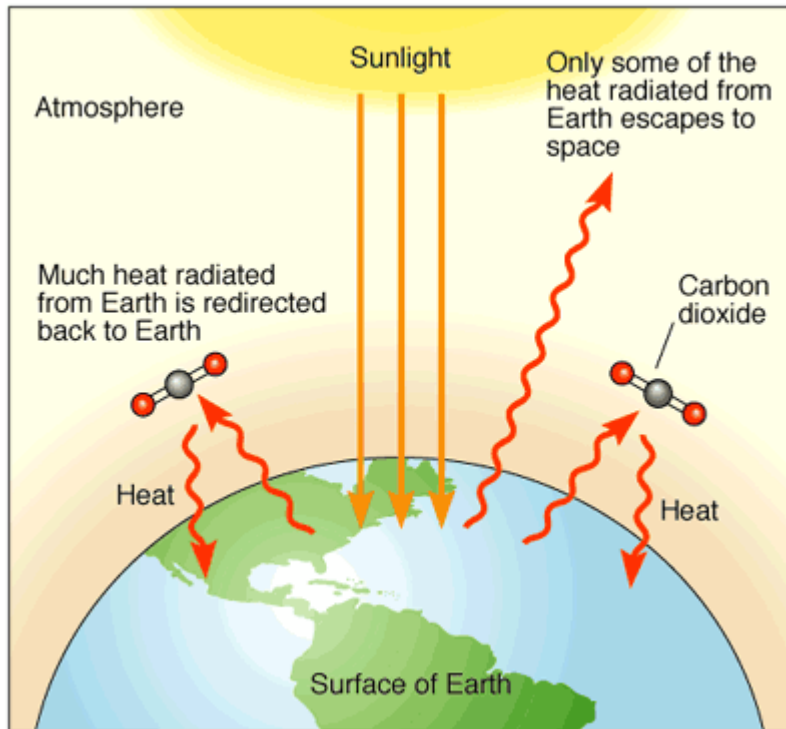
In Astronomy we can pick into nebulae To observe stars



Sptizer telescope



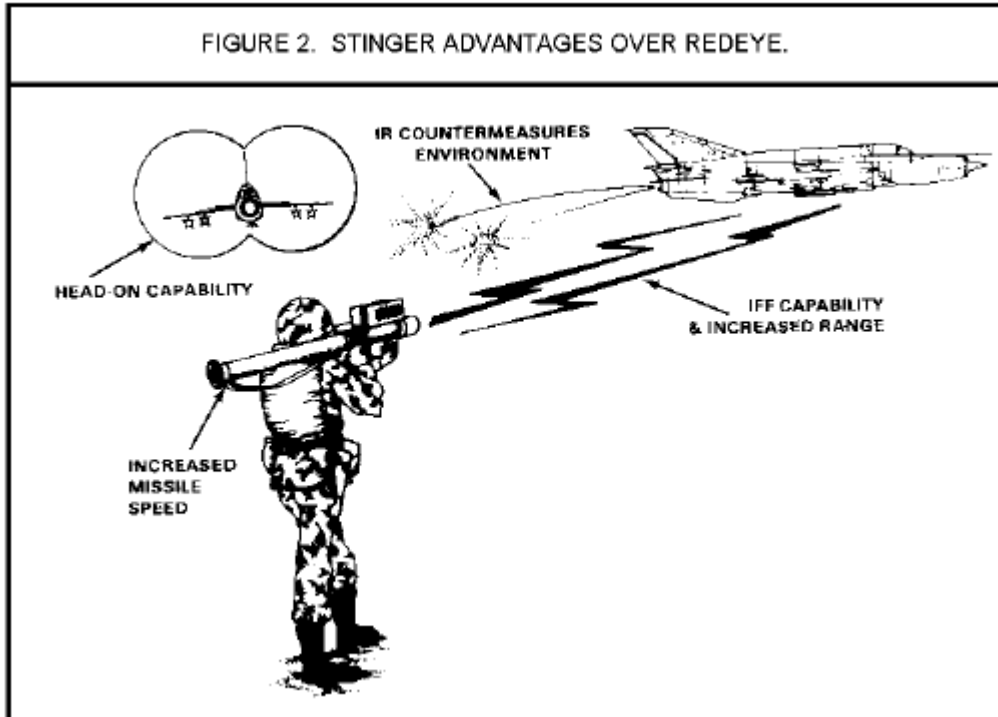
GREENHOUSE EFFECT



H₂O, CO₂ and O₃
Absorb IR radiation.

IR radiations are trapped warming the Earth.

IR can be sensed by Pit Vipers, mosquitoes and stingers !!



Stingers are high tech missiles That sense the IR radiation From a warm source.

Stingers were given to Taliban when they were US friends. Taliban sold them to Rwanda Rebels. One of these missiles Hit the plane to the president. That was the beginning of the genocide.

Stingers are only 35 pounds about And reaches 10,000 ft.



Mosquitoes (female) have IR sensors to find you !!

We can use this ability to get rid of them, !!! see links below.



Mosquito Mania / The New York Times

<http://uniquedistributors.com/armksmokisy.html>

http://www.youtube.com/watch?v=8GXOByRq4Uk&feature=player_embedded#!

HEAT VISION



They operate at a wavelength of about 10 micrometers to see human. (our range See below to compute that).

The goggles can also emit an IR light so they can see each other. This is the Infra red transmission Technology. Wavelength is 2 micrometers.

Application with IR

Also:

Video about IR

How IR radiation was discovered by William Herschel

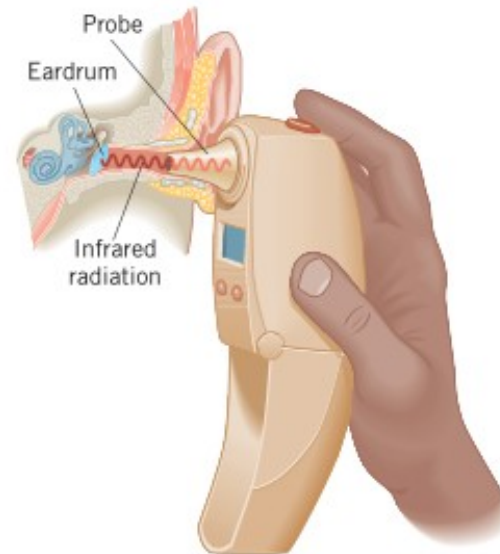


Figure 24.11 A pyroelectric thermometer measures body temperature by determining the amount of infrared radiation emitted by the eardrum and surrounding tissue.

MICROWAVE

- upper limit of radio waves. Range from 0.3m to 0.3mm (so not micrometer).
Resonant wavelengths for molecules of water. The water molecules are Bipolar (+ and -) and feel the electric field of the waves. They start vibrating and heat the Food in the process. (torque is applied to molecules).

This is because the temperature

Is proportional to the

Kinetic energy of

The molecules.

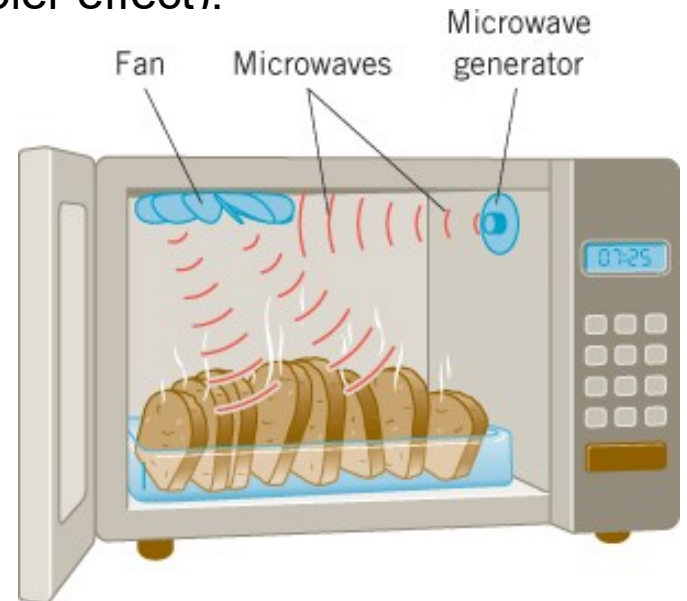
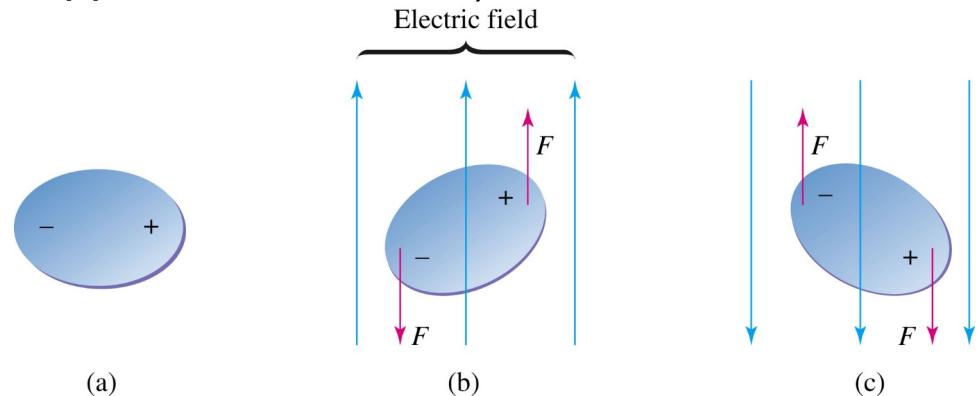
2,450 MHz is used in most ovens

Used in communication. Blue-tooth,

Wi-Fi signals. Used for radars (echolocation + Doppler effect).

Can map surface of planets or Earth.

<http://phet.colorado.edu/en/simulation/microwaves>



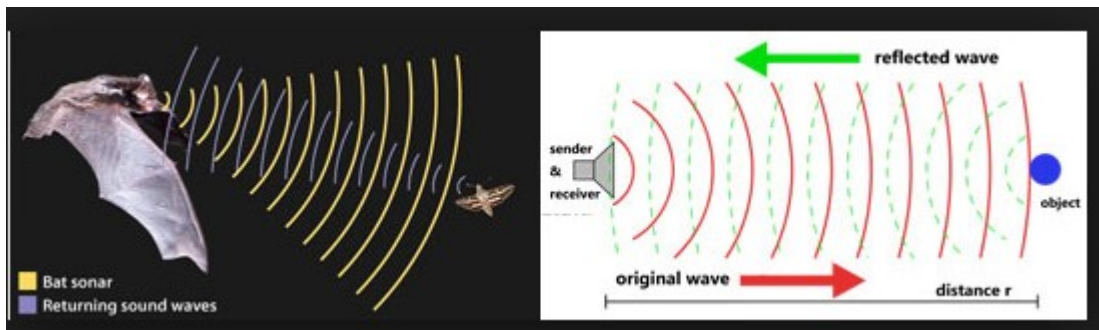


Microwaves spread less than
Radio waves because they have
Shorter wavelengths.
It's easier to focus to use for RADAR.

Because the wavelength is large enough,
It goes through clouds.

MICROWAVES = used in radar and cell phone. Small wavelengths so diffract less.

Microwave dish acts like a lens. Can be aimed at some distant target.

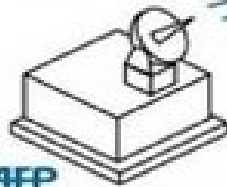


Invisible aircraft

Military stealth aircraft avoid or interfere with radar detection, allowing covert raids into enemy territory

Fooling radar:

- Surface materials that absorb signals, convert them into heat
- Smooth surfaces with seamless joints to prevent radar reflection
- Wing edges moulded to scatter incoming waves
- Engines hidden in body of wing
- Exhaust ducts placed on top of aircraft to reduce heat trail



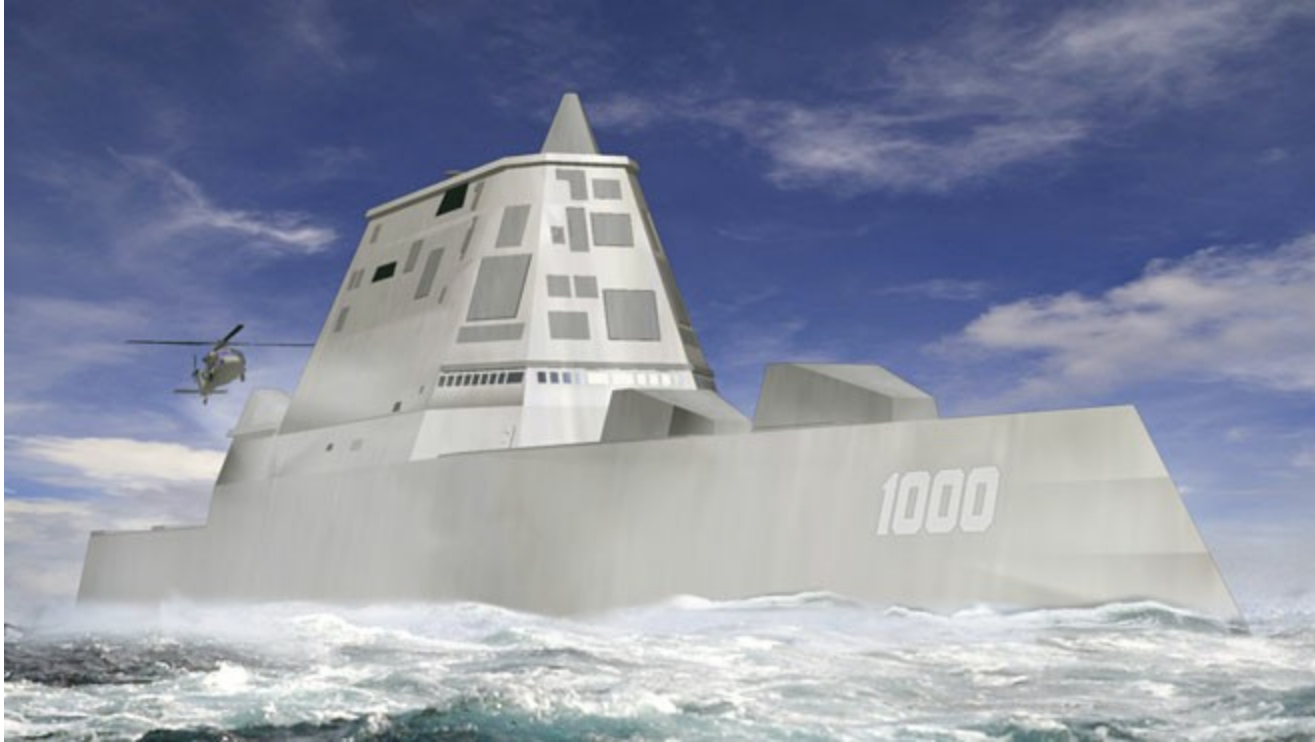
- Electronic counter measures such as jamming and ghost imaging (*images that the camera itself cannot see*)

- Jagged rear edges to scatter signals

AFP

Source: GlobalSecurity/Engineering.com/TheGadgetBook

STEALTH TECHNOLOGY go go invisible



This file image released by Bath Iron Works shows a rendering of the DDG-1000 Zumwalt, the U.S. Navy's next-generation destroyer, which has been funded to be built at Bath Iron Works in Maine and at Northrop Grumman's shipyard in Pascagoula, Miss. (AP/Bath Iron Works)

<http://www.foxnews.com/politics/2012/06/04/planned-stealth-destroyer-could-underpin-us-navy-china-strategy/>

RADAR can be used for imaging cities / surface of Earth / planets.

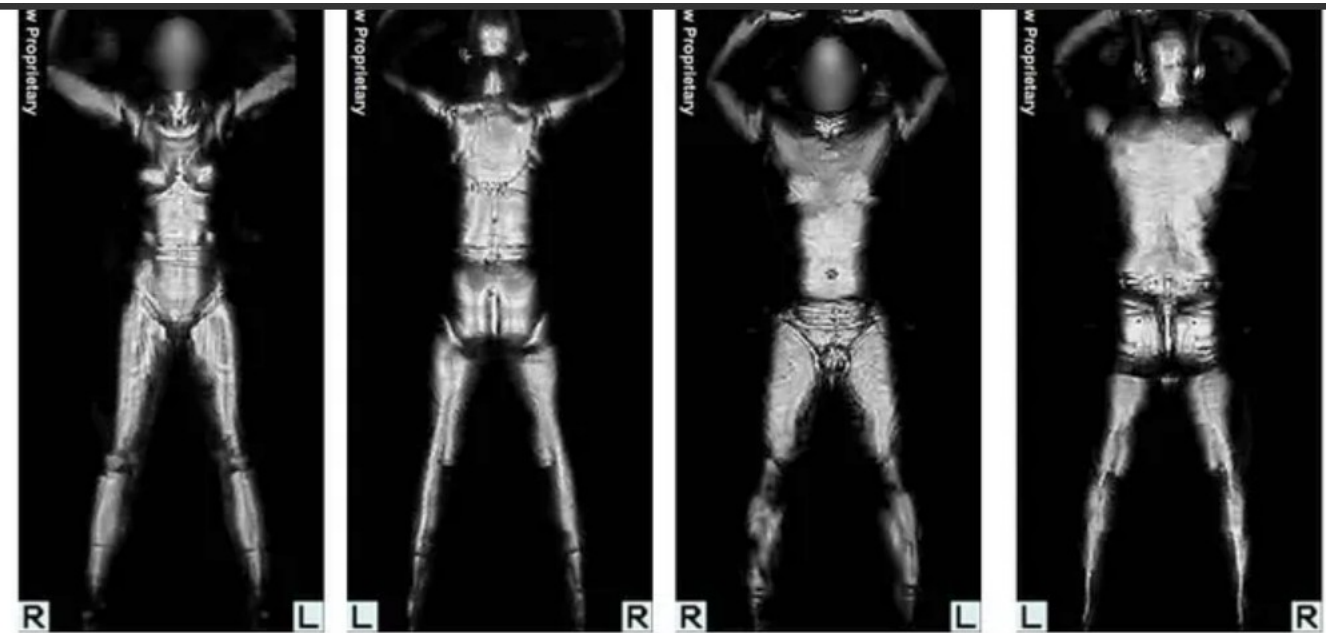
The technology is called **SAR**.

[http://en.wikipedia.org/wiki/
Synthetic_aperture_radar](http://en.wikipedia.org/wiki/Synthetic_aperture_radar)

single beam-forming antenna from which a target scene is repeatedly illuminated with pulses of radio waves at wavelengths anywhere from a meter down to millimeters. The many echo waveforms received successively at the different antenna positions are coherently detected and stored

[http://www.thespacereview.com/article/
790/1](http://www.thespacereview.com/article/790/1)





The TSA has spent \$120 million on the full-body millimeter wave scanners that we have to shuffle through in our sock feet, a figure that works out to about \$150,000 per unit. But not only is that a tremendous amount of money, these machines – and the agents who operate them – have proven to be increasingly inept at, you know, actually *scanning* for weapons or other threats to passenger safety. The agency previously spent \$40 million on the backscatter X-ray scanners that created borderline graphic images of passengers' bodies; those machines were replaced by the current scanners.

<https://www.cnet.com/culture/tsa-bored-of-seeing-you-naked-removing-airport-body-scanners/>



Figure 24.8 This wireless capsule endoscope is designed to be swallowed. As it passes through a patient's intestines, it broadcasts video images of the interior of the intestines. (Courtesy Given Imaging, Ltd.)

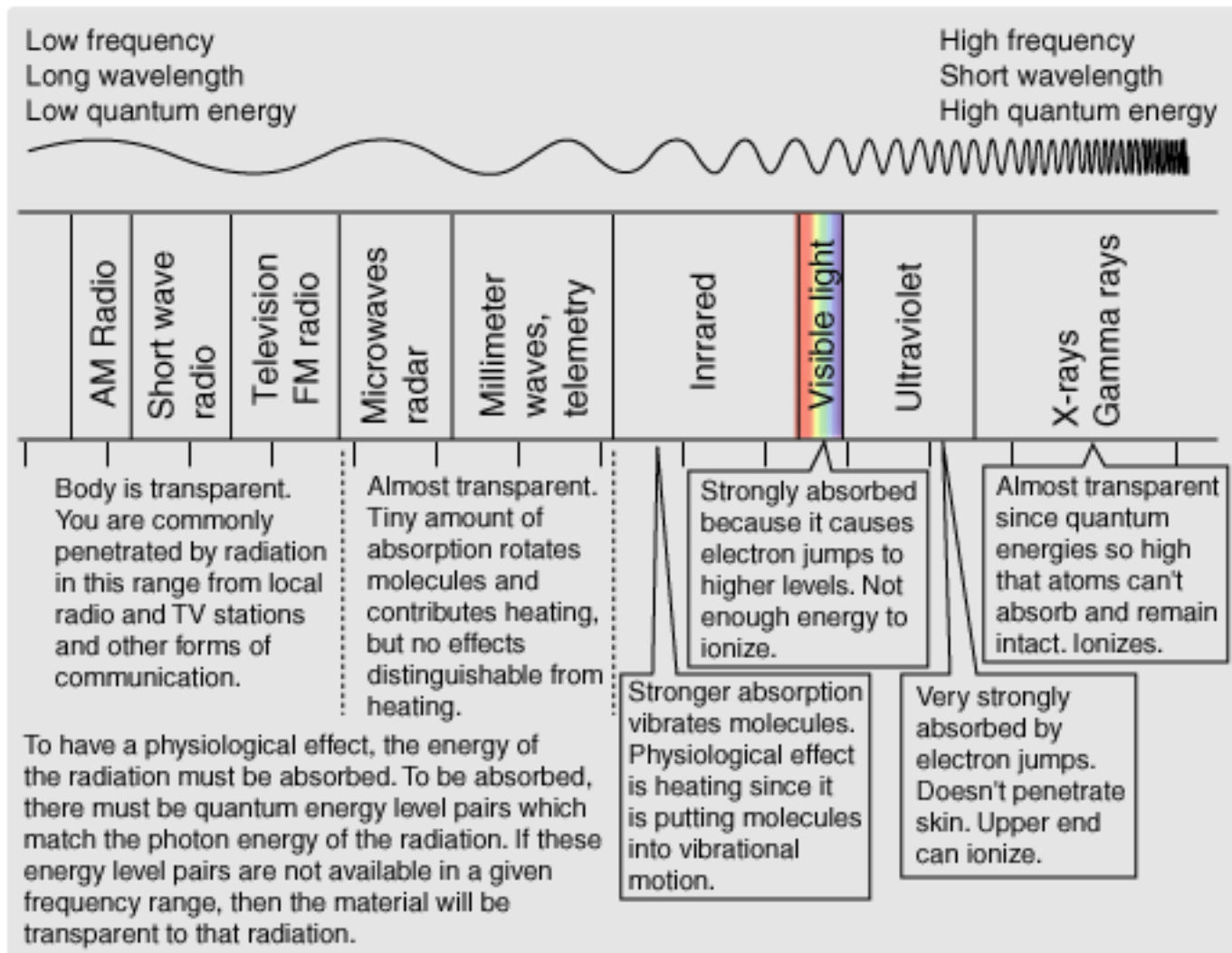
MICROWAVE

RADIO WAVES

- very large range from 100 Hz to 1000 megahertz. Frequencies given in kilohertz and megahertz. Radio waves are produced using AC with the appropriate frequency using an antenna. AM = long wavelength reflected by atmosphere. FM=short wavelength go through the atmosphere. Narrow frequency bands are Assigned for specific purposes. 88 to 108 MHz for commercial FM radio. Some bands are assigned to government and private communication.

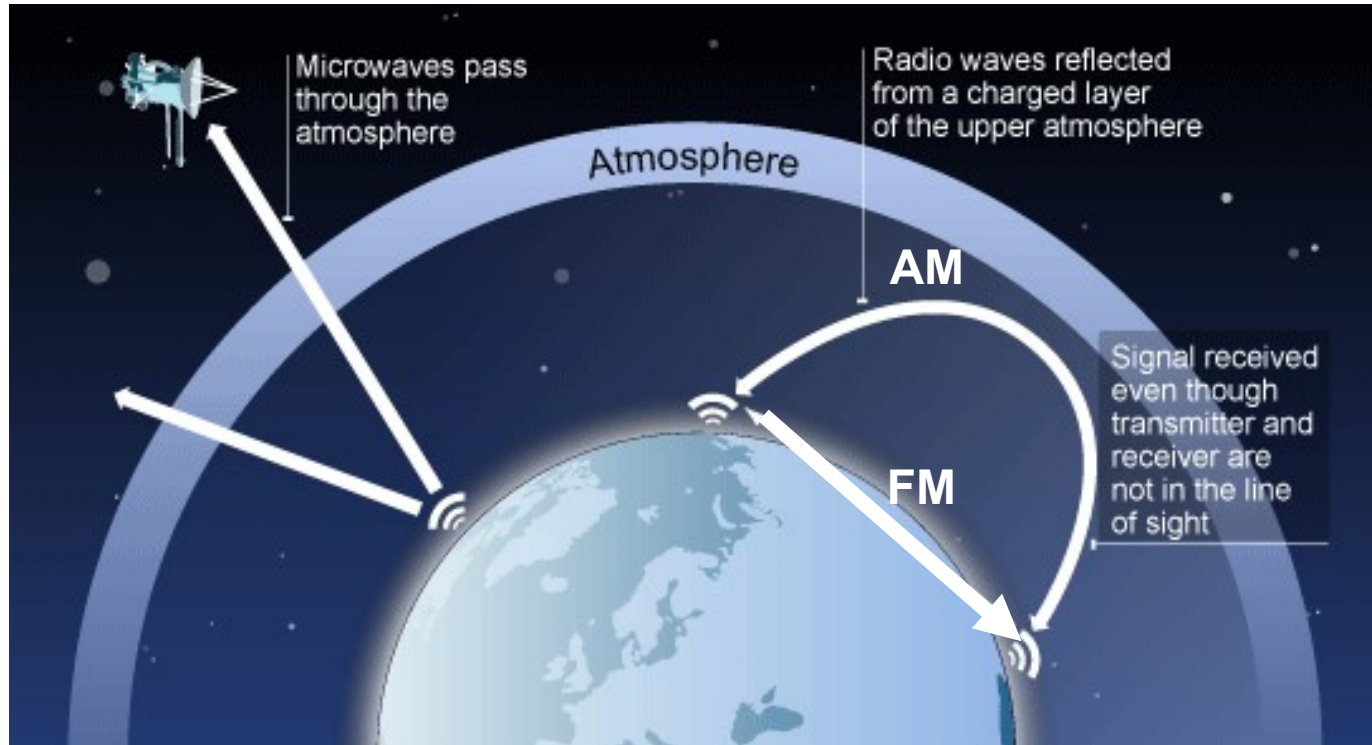
Note: radio waves do go through the body but they do not ionize. Only gamma-rays, X-rays and UV can ionize (and cause cancer or radiation sickness). The Wavelengths are small to the rays behave like photons. Bundles of energy.

Radio waves can have other physiological effects. Electric field act upon our ions. A wave is both an electric field and a magnetic field shaking.



<https://www.quora.com/How-is-it-that-radio-waves-can-go-through-concrete-and-x-rays-through-muscle-What-determines-whether-photons-will-pass-through-without-interacting>

RADIO WAVES



FM + TV = high frequency, better resolution = better music because more info fit in the signal but smaller wavelength means less diffraction. So a building will shade the signal.

AM = larger distance, even larger at night so they decrease the power. Hard to listen at night.

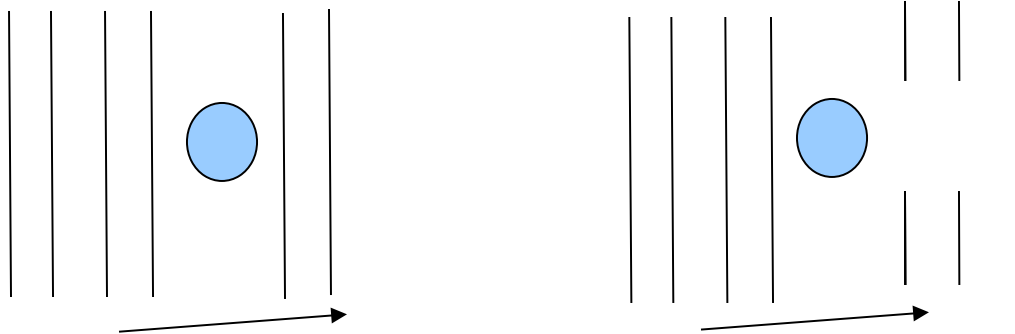
Conceptual Example 2 The Diffraction of AM and FM Radio Waves

Diffraction is the ability of a wave to bend around an obstacle or the edges of an opening. Would you expect AM or FM radio waves to bend more readily around an obstacle such as a building?

AM has longer wavelength (frequency about 100 kHz)

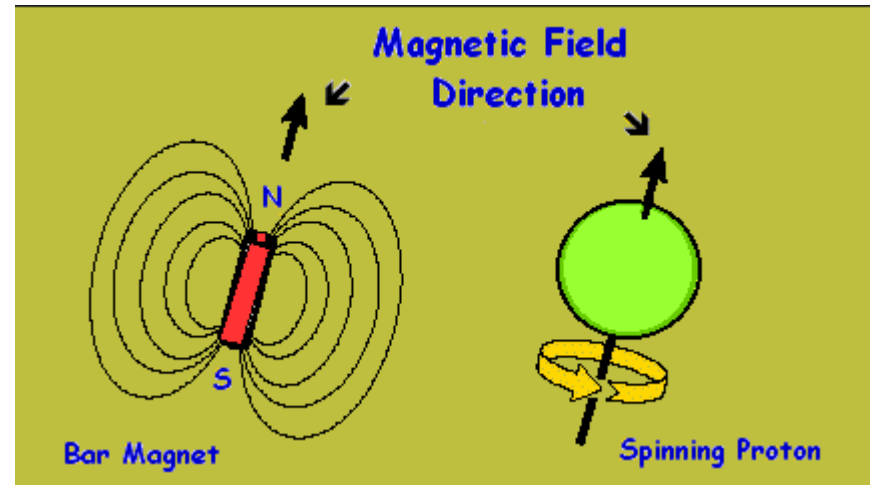
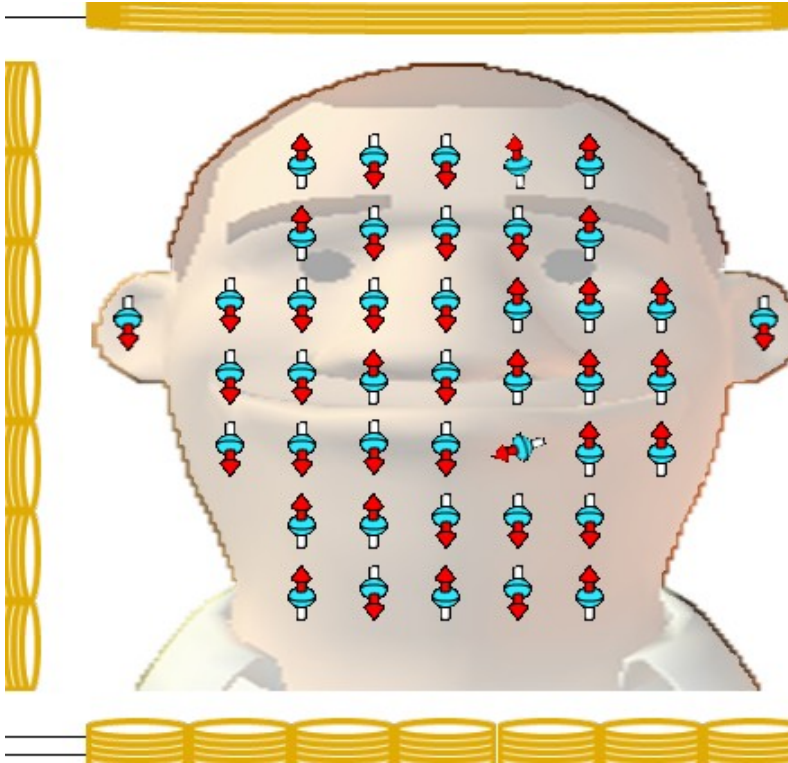
FM has shorter wavelength (frequency about 100 MHz)

Which one is which ?



MRI

1 T $\leftrightarrow$ 40MHz radio wave.



The protons behave like small magnets and are aligned in A VERY strong magnetic field. If you excite them with a Radio wave with the right frequency they jump to a higher level Of energy (they flip) and when they "relax" they burp out Radio waves used to illuminate the brain from inside out.

<http://phet.colorado.edu/en/simulation/mri>

<http://www.schoolphysics.co.uk/age16-19/Atomic%20physics/Atomic%20structure%20and%20ions/text/MRI/index.html?PHPSESSID=4efb7b210aababb8e14bd5e7c386b2a1>

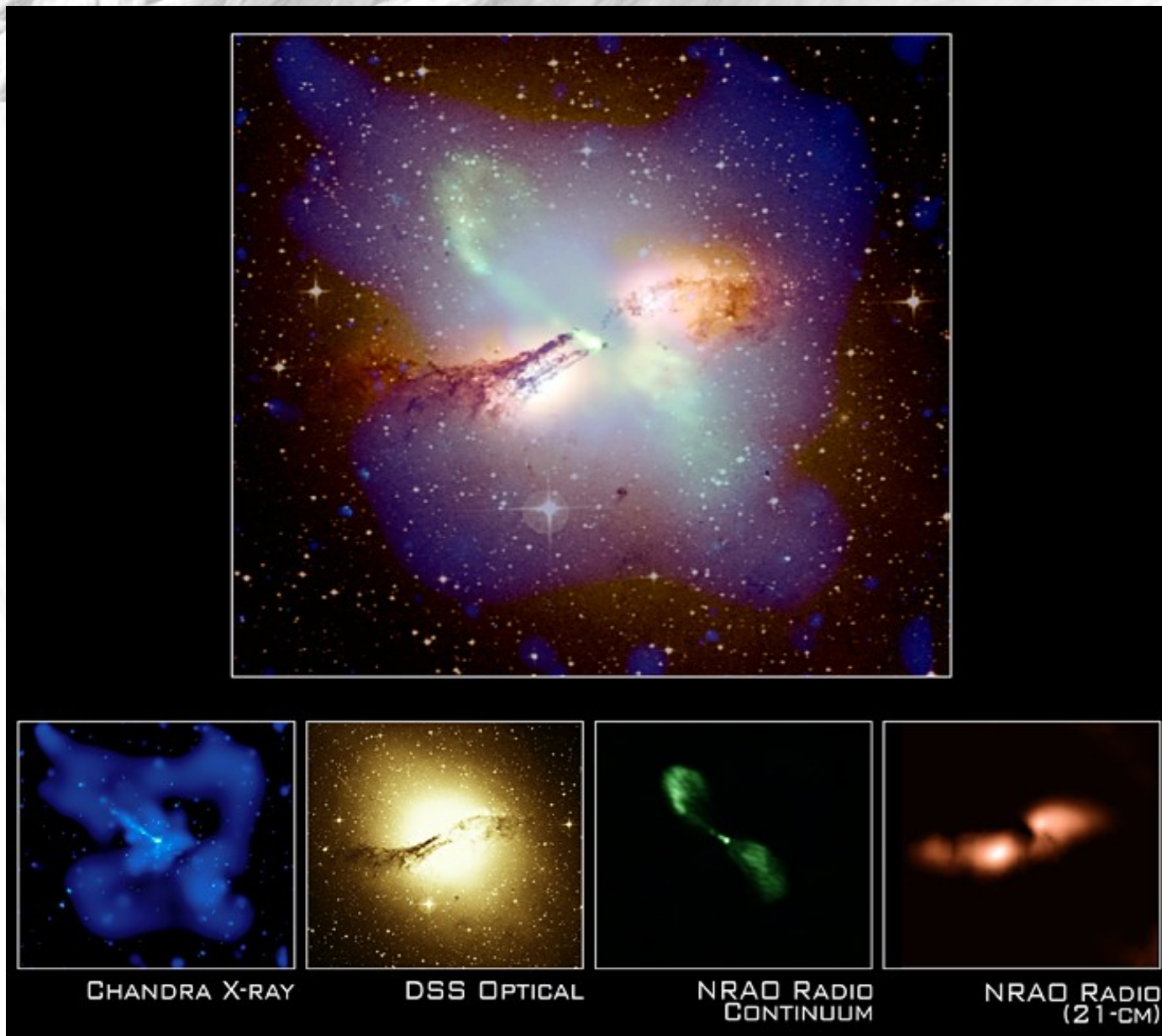
How to turn the MRI into a rail gun

[https://www.youtube.com/watch?](https://www.youtube.com/watch?v=plvIEf7JsKo&nohtml5=False)

[v=plvIEf7JsKo&nohtml5=False](https://www.youtube.com/watch?v=plvIEf7JsKo&nohtml5=False)

With interferometry, radio astronomers can **combine the signals from many antennas, and even many telescopes**. It allows them to create an image that is much brighter and sharper than what is possible from a single antenna dish.

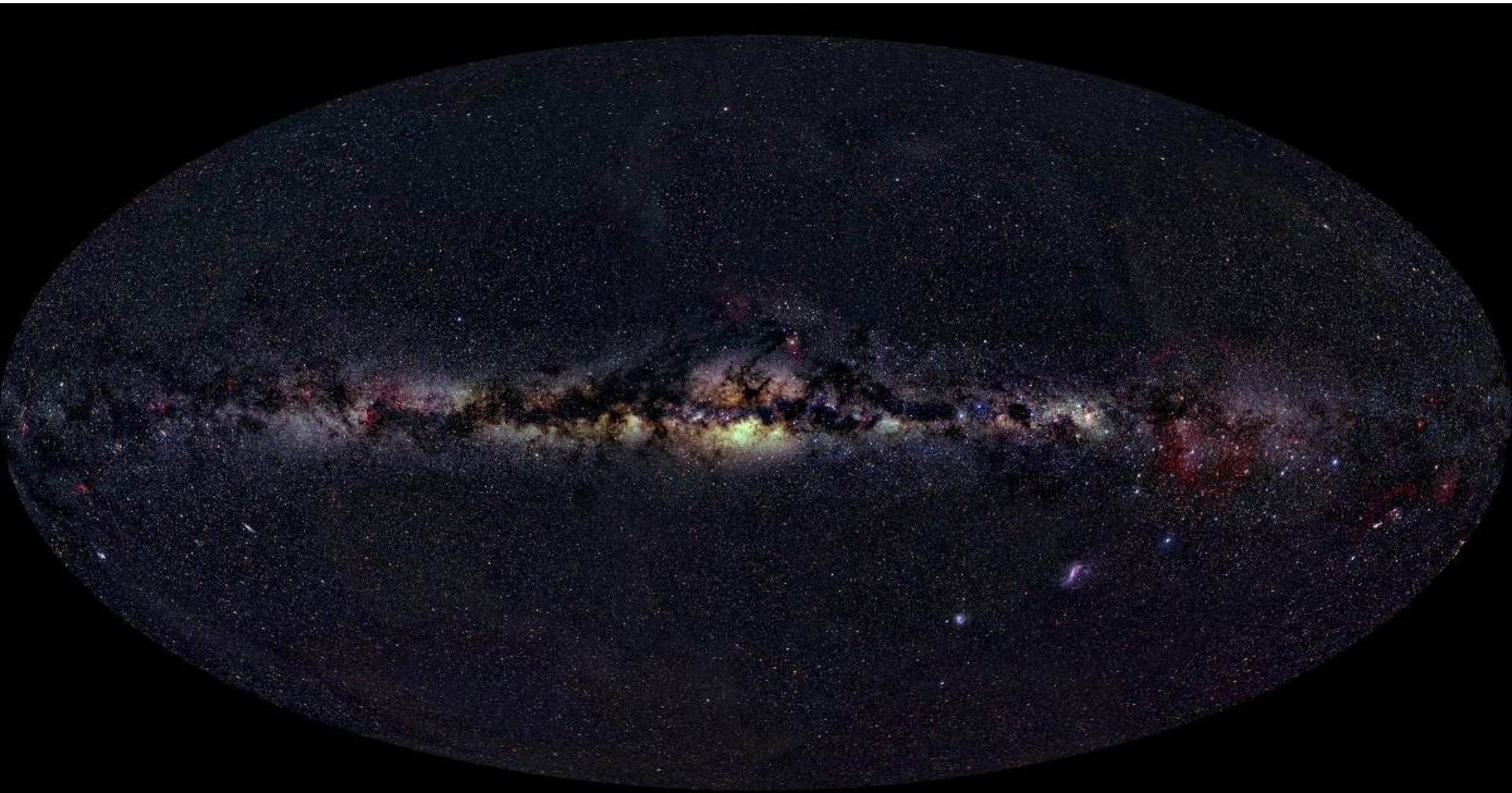




A composite X-ray (blue), radio (pink and green), and optical (orange and yellow) image of the galaxy Centaurus A presents a stunning tableau of a galaxy in turmoil.

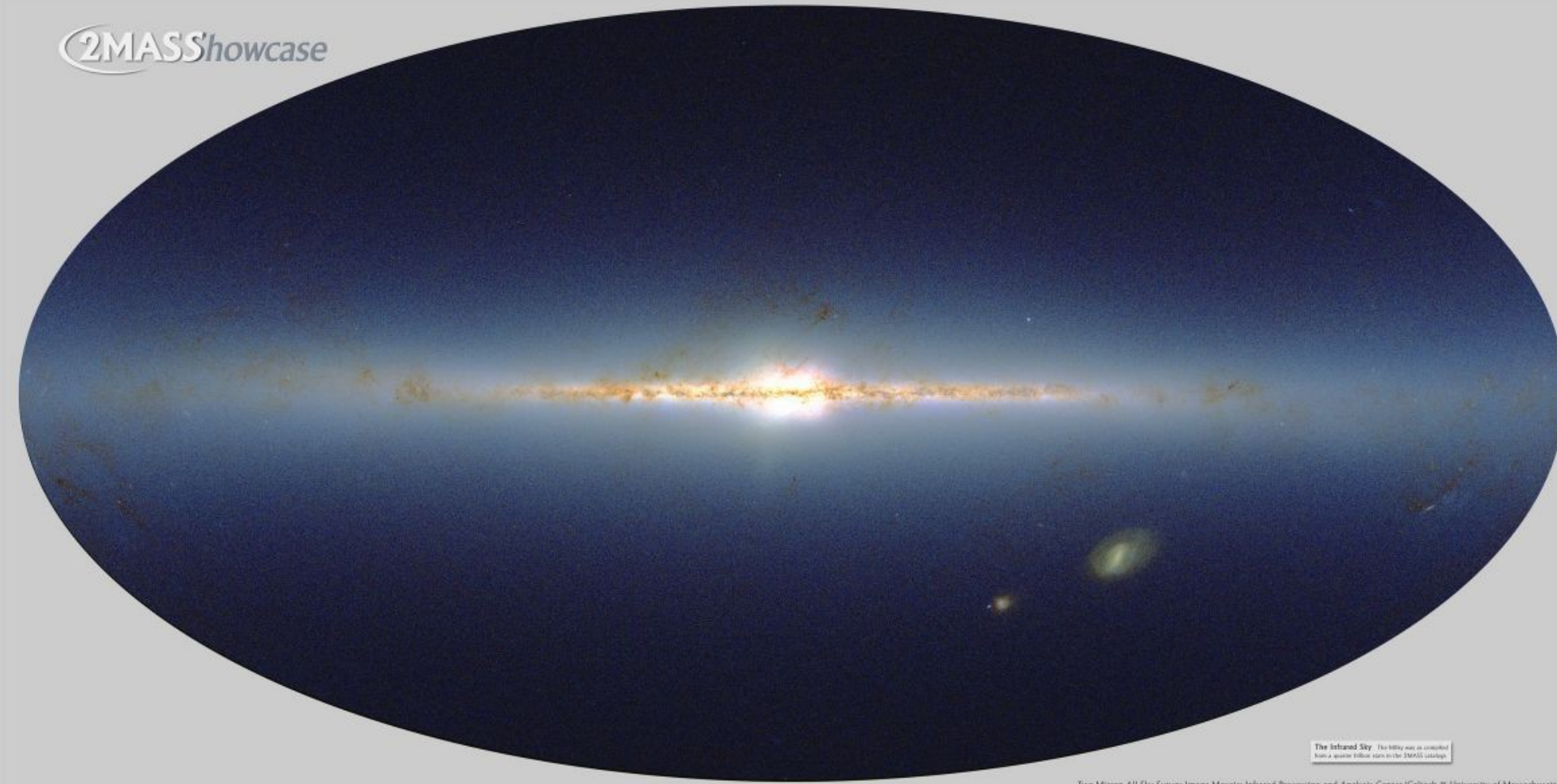
<http://chandra.harvard.edu/photo/2002/0157/>

Optical Sky



Near-infrared sky

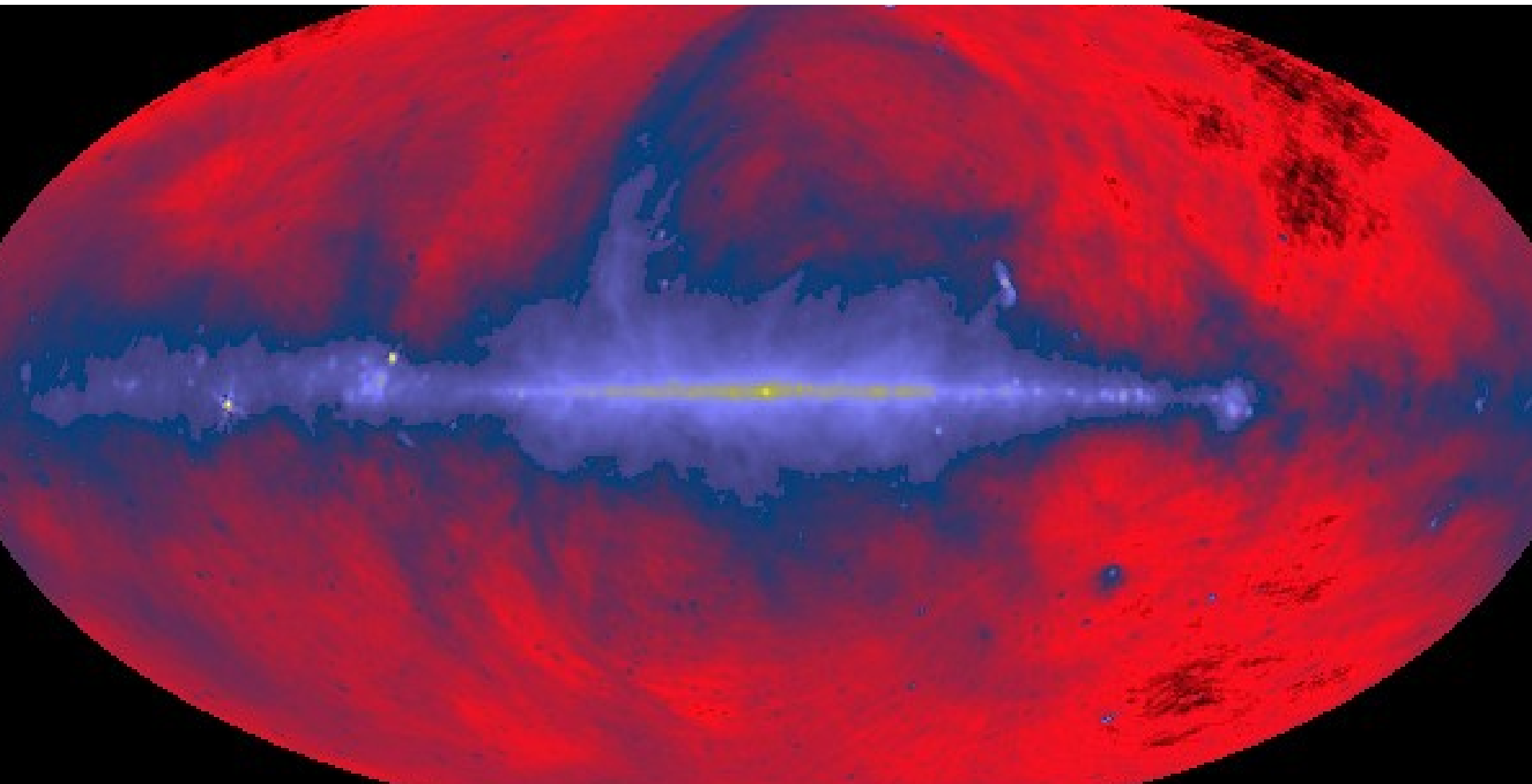
2MASS*showcase*



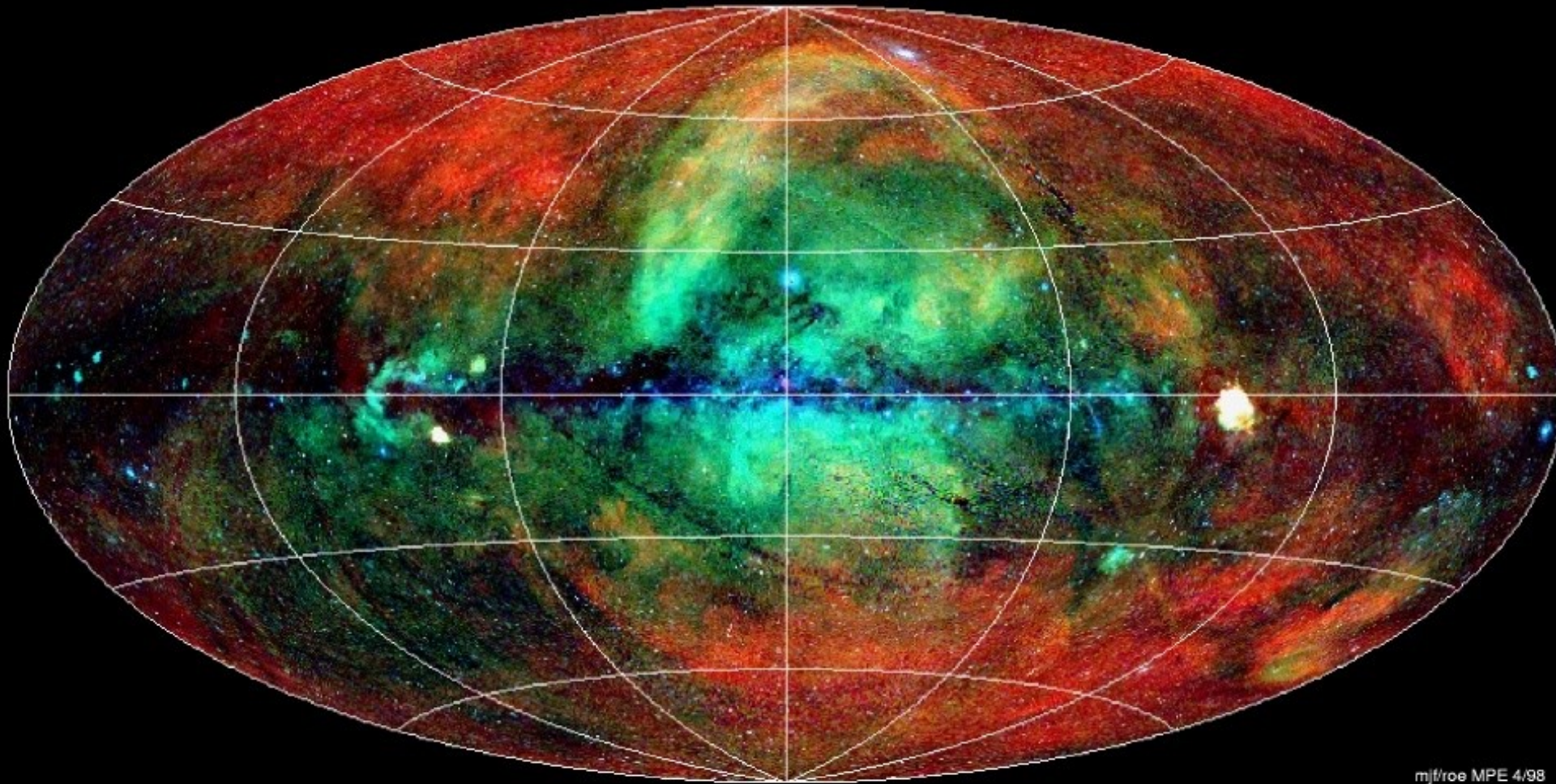
The Infrared Sky This image was compiled from a quarter million stars in the 2MASS catalog.

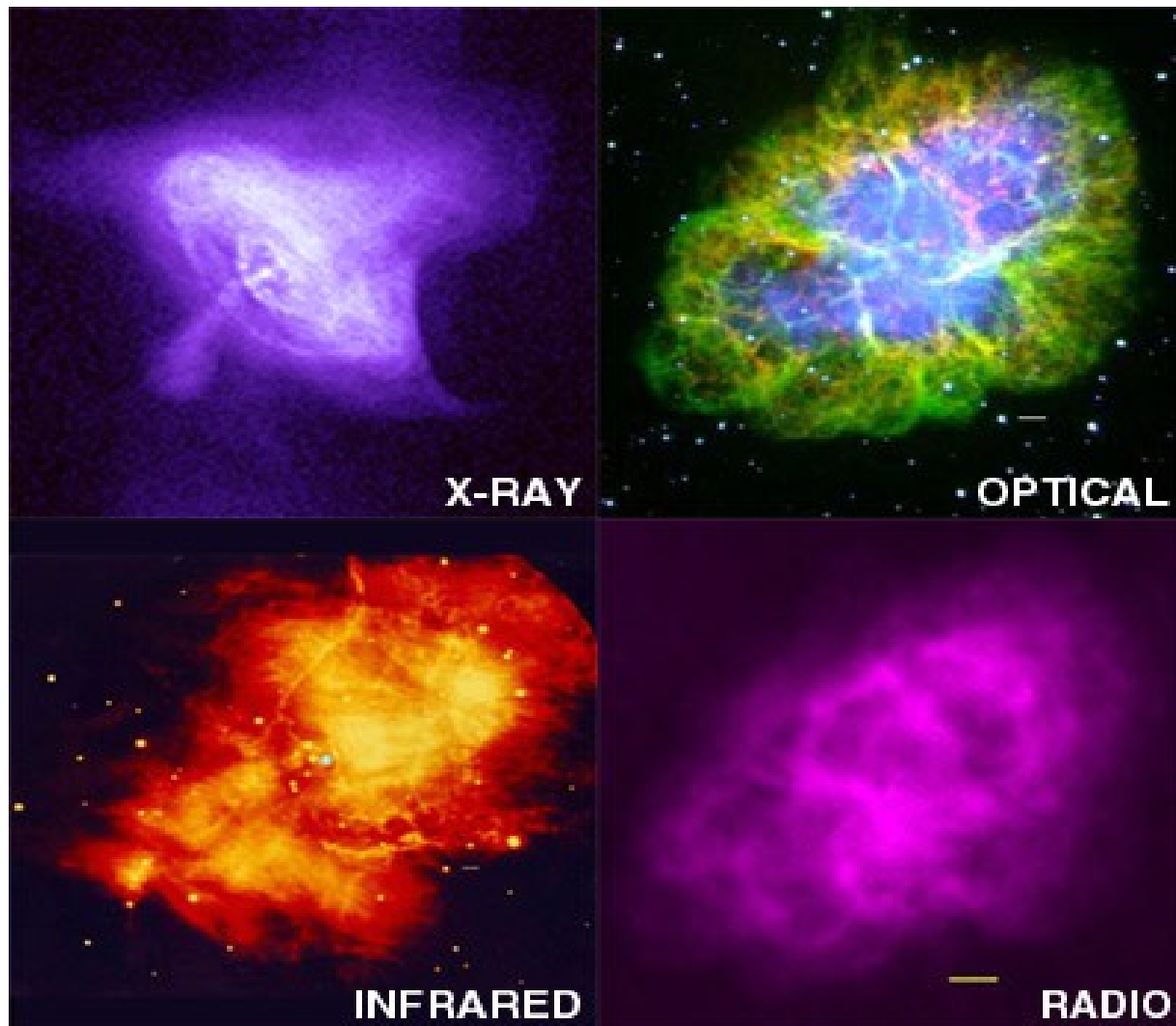
Two Micron All Sky Survey Image Mosaic: Infrared Processing and Analysis Center/Caltech & University of Massachusetts

Radio Sky



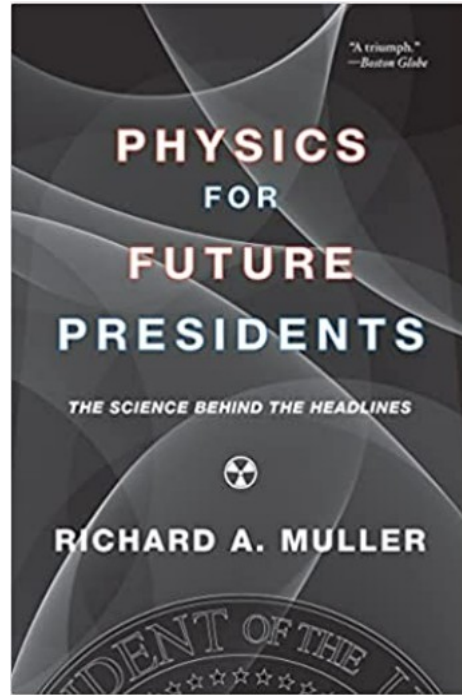
Soft X-ray Sky





Crab nebula

Reference for my slides

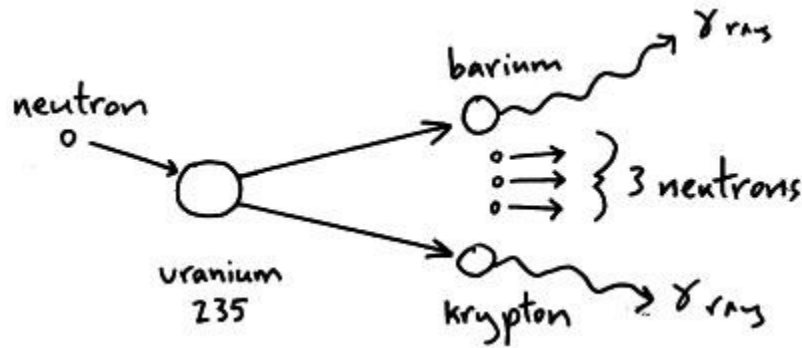


Physics for Future Presidents: The Science
Behind the Headlines Reprint Edition

by [Richard A. Muller](#) (Author)

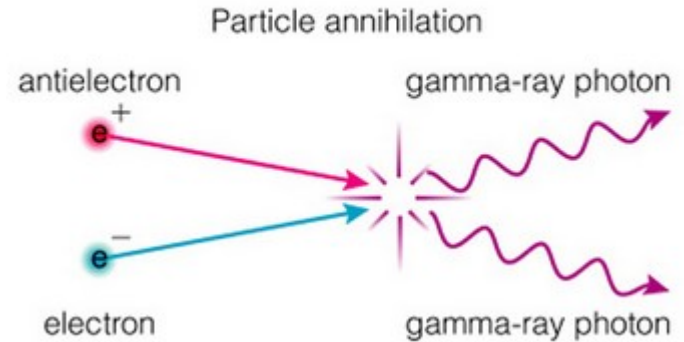
GAMMA RAYS:

Highest frequency range from 3×10^{19} Hz to 10^{23} Hertz. Wavelength same size of nuclei (10^{-12} m and smaller) Emitted in nuclear processes; radioactive decay, Fission, fusion .. damaging. Ionizing radiation. Can cause radiation sickness and Death or induce cancer. But used for imaging.



Gamma rays can be emitted when a nucleus decays. Here is reaction of fission used for the atomic bomb

Explosive and violent event in space
Can produce gamma ray bursts.
It involves neutron stars and black holes.



Gamma rays can be produced when an Electron and an anti-electron (positron) collide.

Matter+ anti_matter = energy

PET SCAN

Using gamma rays

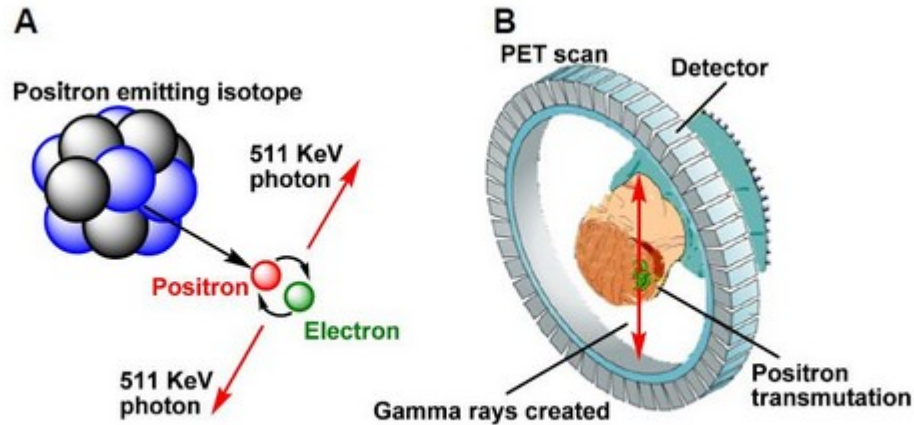
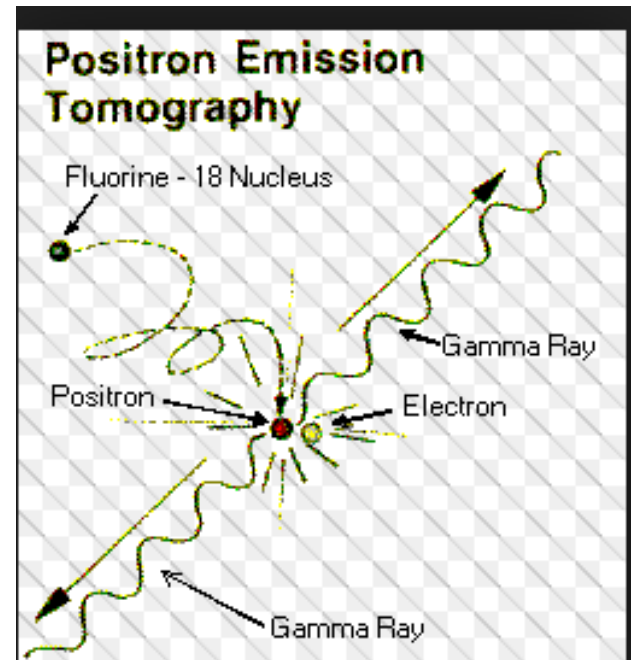
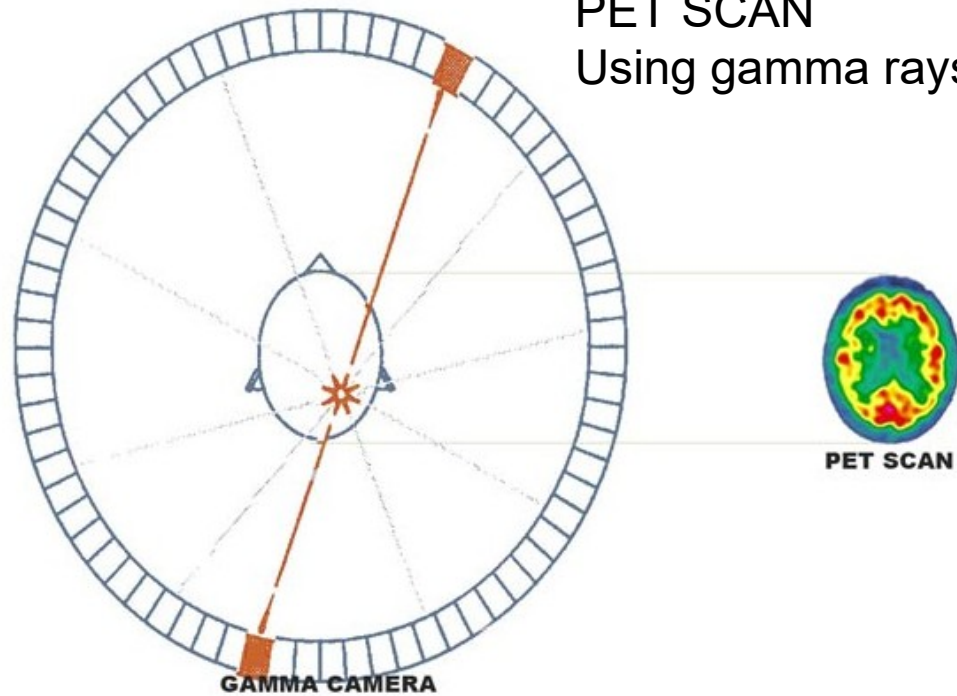
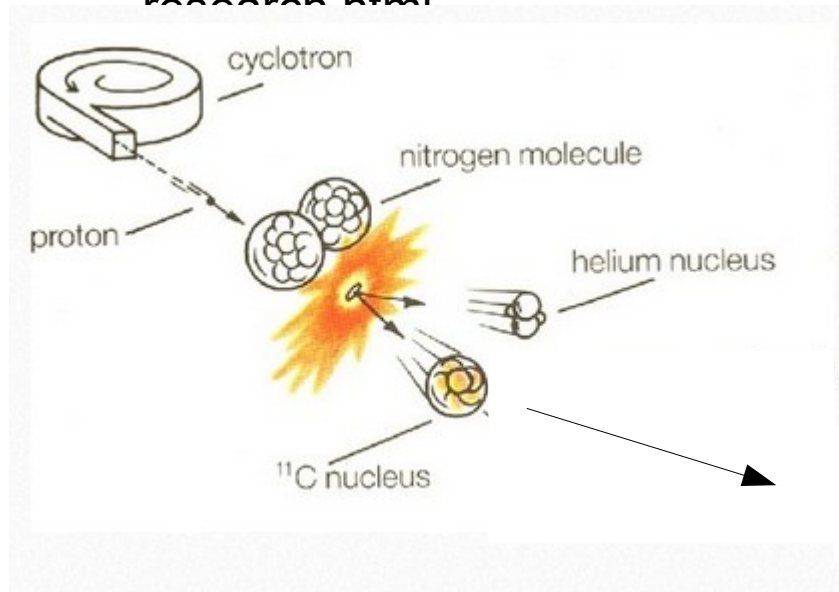


Fig. 1. (A) A positron and an electron annihilate producing two 511 keV photons travelling in opposite directions. (B) The 511 keV photon was registered by the circular gamma ray detector array in the PET camera.

Radionuclide	Half-life	Chemical Form	Nuclear Reaction(s)
^{11}C	20 min	$^{11}\text{CO}_2$ / $^{11}\text{CH}_4$	$^{14}\text{N}(\text{p},\alpha)^{11}\text{C}$
^{13}N	10 min	$^{13}\text{NH}_4^+$ / $^{13}\text{NO}_x$	$^{16}\text{O}(\text{p},\alpha)^{13}\text{N}$
^{15}O	2 min	$^{15}\text{O}_2$	$^{15}\text{N}(\text{p},\text{n})^{15}\text{O}$ $^{14}\text{N}(\text{d},\text{n})^{15}\text{O}$
^{18}F	110 min	$^{18}\text{F}^-$ or $^{18}\text{F}_2$	$^{18}\text{O}(\text{p},\text{n})^{18}\text{F}$ $^{20}\text{Ne}(\text{d},\alpha)^{18}\text{F}$

<http://www.imperial.ac.uk/people/a.gee/research.html>



The radioisotopes used for PET scan have to be made in the hospitals using a cyclotron. (remember a magnetic field is used to trap protons around a circular path. An electric field is used to speed them up.)



Here is a list of some positron emitters :

Carbon-11 (20 min)

Nitrogen -13 (9 min)

Oxygen-15 (2min)

Fluorine-18 (110 min)

Iodine-124 (4.2 days)



The radioisotopes have a very short life. They have to be “ made” in the hospital Before the session. This is expensive because it requires accelerators.

In beta plus decay, a proton is converted, via the weak force, to a neutron, a positron (also known as the "beta plus particle", the antimatter counterpart of an electron), and a neutrino.

Iodine-124 for example, goes to the thyroid. Doctors can detect if there is blank spots or hyperactivity by detecting gamma radiation.

O-15 can flow to the brain for probing the brain.

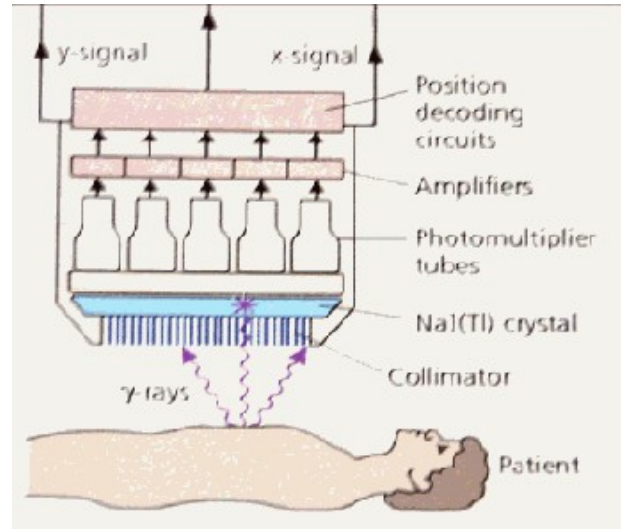
C-11 can flow to biological compounds.

http://www.isotopeworld.com/filestore/EIR_Medical%20Isotopes%20in%20the%2021st%20Century.pdf

http://www.search.com/reference/Positron_emission

SPECT (single photon emission tomography)

Like PET but cheaper:



SPECT is similar to PET in its use of radioactive tracer material and detection of gamma rays. In contrast with PET, however, the tracer used in SPECT emits gamma radiation that is measured directly, whereas PET tracer emits positrons that annihilate with electrons up to a few millimeters away, causing two gamma photons to be emitted in opposite directions. A PET scanner detects these emissions "coincident" in time, which provides more radiation event localization information and, thus, higher resolution images than SPECT (which has about 1 cm resolution). SPECT scans, however, are significantly less expensive than PET scans, in part because they are able to use longer-lived more easily-obtained radioisotopes than PET

http://www.fas.org/irp/imint/docs/rst/Intro/Part2_26d.html

<http://www.slideshare.net/Panduekoyudho/spect-medicine>

Radioisotopes for this are imported from Canada.

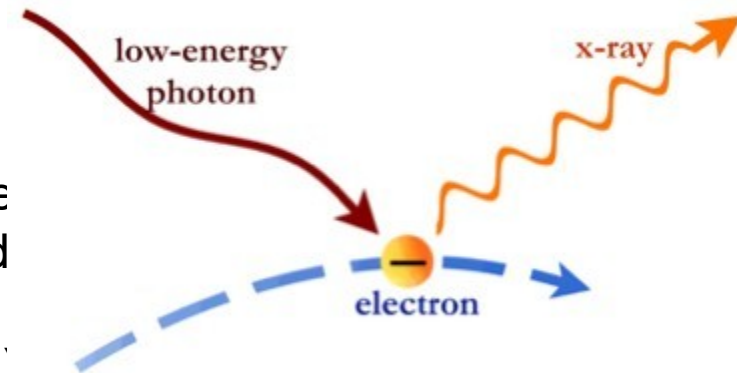
http://en.wikipedia.org/wiki/Single-photon_emission_computed_tomography

http://www.umsl.edu/~tsytsarev/tsytsarev_files/Lecture18.htm



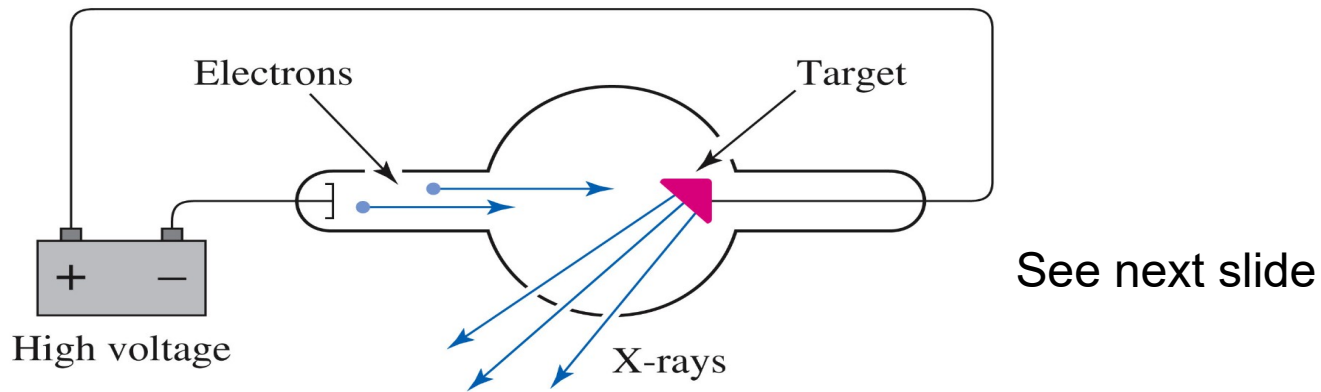
Other ways gamma Rays are emitted.

Crab nebula, electrons accelerate so much because Strong magnetic field, they emit flares of X-rays and Gamma rays. Or high energy electrons can boost Low energy photons to high energy photons (x-rays,



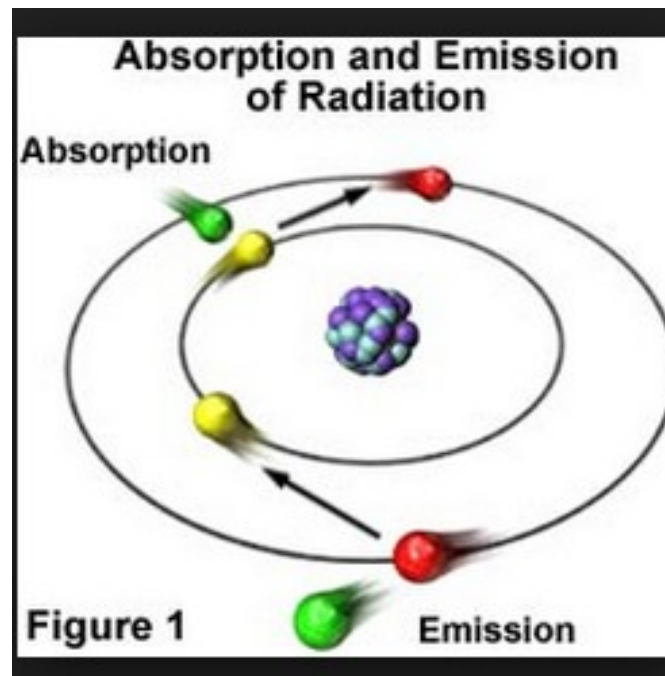
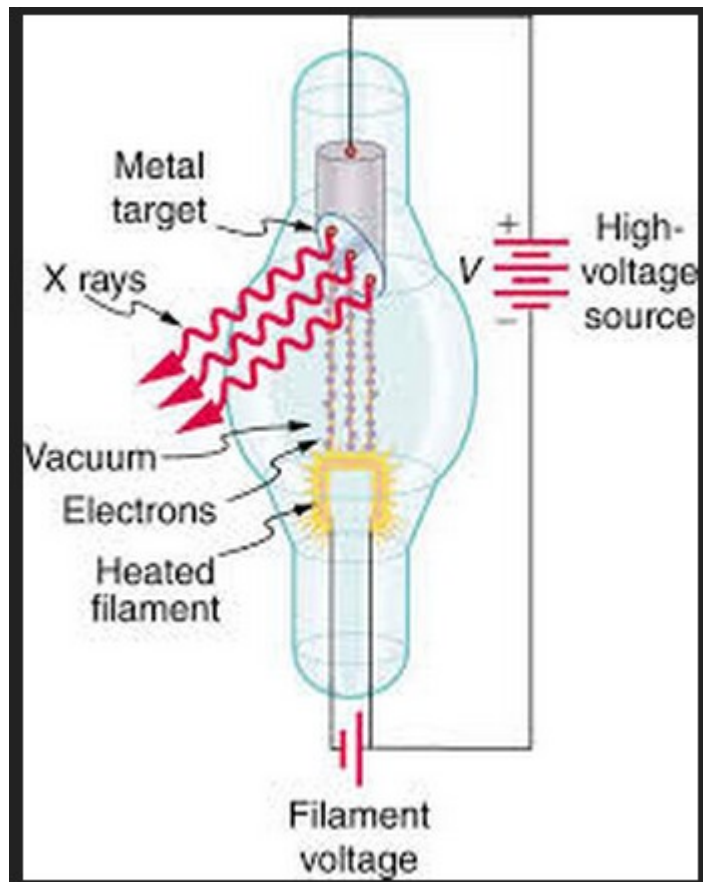
X-RAYS

- penetrating and ionizing too. Can cause damage to cell too.
Size of atoms and molecules (10^{-10}m). Can be used to probe crystals and molecules.
X-rays are produced by smashing high-speed electrons into a “target” made of Tungsten or other metal. This is how x-rays are produced in hospitals.



The electron lose their kinetic energy when decelerating as radiation.
Plus they kick out some electrons in the metals.
Other electrons (from higher energy Levels) take their Place
Emitting X-rays.

<http://www.youtube.com/watch?v=IRBKN4h7u80>



X-RAYS

X-rays pass through the body but they are absorbed by large atoms like calcium. Carbon and hydrogen let them go through. Lead atoms stop x-rays.

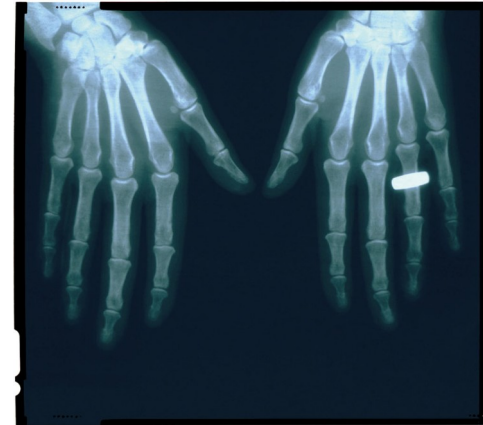
Discovered by 1895 by Wilhelm Conrad Röntgen, who received the first Nobel Prize in Physics in 1901.

He used a cathode ray tube and didn't know what these radiations were. He called them X-rays !!

Very quickly used in the medical field.

First, damaging. Now very safe.

<http://onlinephys.com/giants2.html>



X-RAY EXPOSURE
GIVE ME A DOSE
OH MY POOR HEAD
I CAN'T STAND IT!
IT'S QUICK IT'S SAFE IT'S SURE
OF KOHLER'S ANTIDOTE

KOHLER'S ANTIDOTE

FOR HEADACHE HAS RELIEVED THOUSANDS

WHY NOT YOU? IT WILL CURE THE WORST KIND OF HEADACHE. WHETHER CAUSED BY Sick Stomach, Excess of Spirituous Liquors or Neuralgia

GIVES RELIEF IN 15 MINUTES

8 DOSES Mailed to any address in U. S. 25 CENTS. Post paid, on receipt of price.

KOHLER MFG. CO., BALTIMORE, MD.

When you write, please mention "The Cosmopolitan."



The 8-year-old from Ivory Coast was inside this suitcase when authorities discovered him. (AP)

In the 50s, they used Xrays in shoe stores.

1/5 → cancer

Remember app

See links on website.



Radiation was 30 rem. Find the probability to get cancer in %
1 rem u have a change of 1/2500 to get cancer
If 100,000 people use the machine. How many cancer ? (1200)

CT SCAN or CAT SCAN



The disadvantage is that it uses x-rays.
(you increase the probability of cancer).

CT scans are made by taking x-rays from
Different directions. That's the tomography part.



Less expensive then PET but more invasive
The radiation received is more important.
Not efficient at probing soft tissue.

<http://health.learninginfo.org/pneumothorax.htm>

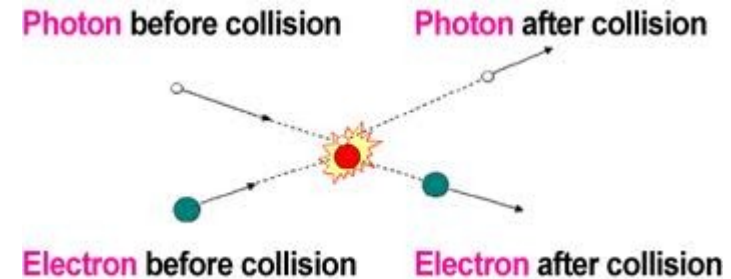
<http://www.ctlab.geo.utexas.edu/overview/index.html>

<http://www.physicscentral.com/explore/action/scans-1.cfm>



XRAY backscatter

Remember : X-ray behaves more like particle than wave because of The high energy of the photons (frequency). The X-ray can either go through The object but they can also bounce off electrons. The amount of back scatter Is roughly determined by the density of the material.



http://articles.cnn.com/2010-05-06/travel/tsa.scanner.assault_1_full-body-scanning-tsa-screener?_s=PM:TRAVEL

**Illegal immigrants
Attempting to enter
Southern Mexico from
Guatemala in truckload of
Bananas !**



UV radiation

Just above the violet. Still energy high enough to induce skin cancer. Trigger a Chemical process responsible for tanning. Not penetrating like x-rays and gamma rays. Emitted by very hot objects. The UV radiation from the Sun is blocked by the Ozone layer. Application : Indian-born physicist Dr. Ashok Gadgil developed a cheap Way to kill the germ in water using UV radiation. Can treat 15 liters per minute.

<http://www.wipo.int/ipadvantage/en/details.jsp?id=2564>

Used to sterilize tools and workplaces. UV laser are used in dermatology and metallurgy(engraving), forensic (to detect body fluid).

Also in fluorescent lamps.

Atoms of mercury are excited by electric discharge.

Emit UV. UV absorbed by a powder coating the lamp.

Fluorescent materials. Re emit visible light.



UV light

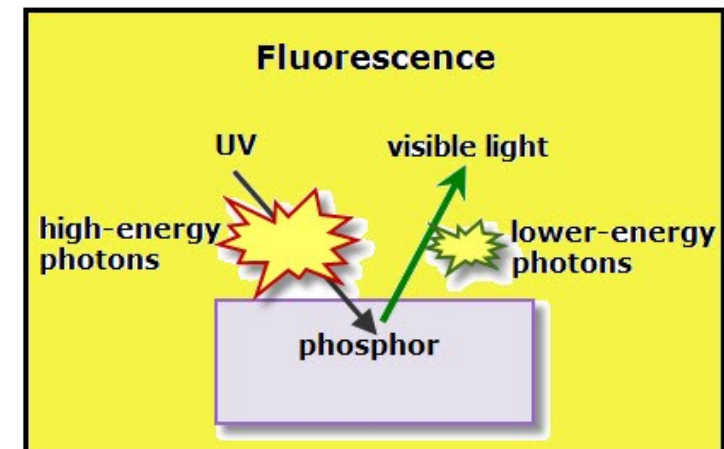
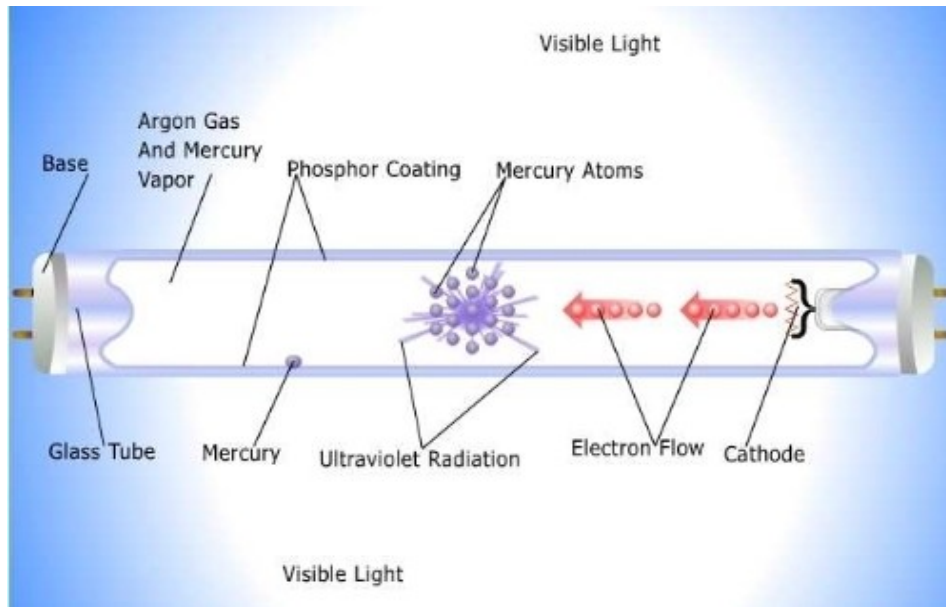
Bug zappers work by emitting a **UV light** that attracts bugs to the center of the device, where they're electrocuted, usually between two metal grids. Because of the irresistible lure of their light, bug zappers are incredibly effective at killing bugs. Jul 27, 2022

 The New York Times



Also in **fluorescent lamps**.

Atoms of mercury are excited by electric discharge. Mercury changes from liquid to gas. The electrons of mercury are bumped to higher levels of energy by collisions (with the Argon gas). When the electrons go back to lower energy level they emit UV photons. The UV photons excite in turn the electrons of the phosphor coating the inside of the lamp. When they go back, visible light is emitted.



<http://www.myclimatechange.net/default.aspx?cat=3&subjectId=51>

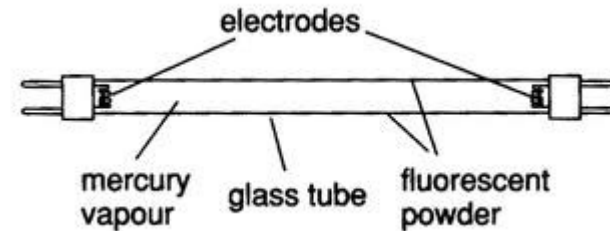
WHITER THAN WHITE

UV light is absorbed by phosphor or fluorescent chemical (for making the jump between energy gaps) and re emit light.



We use this idea in fluorescent light.

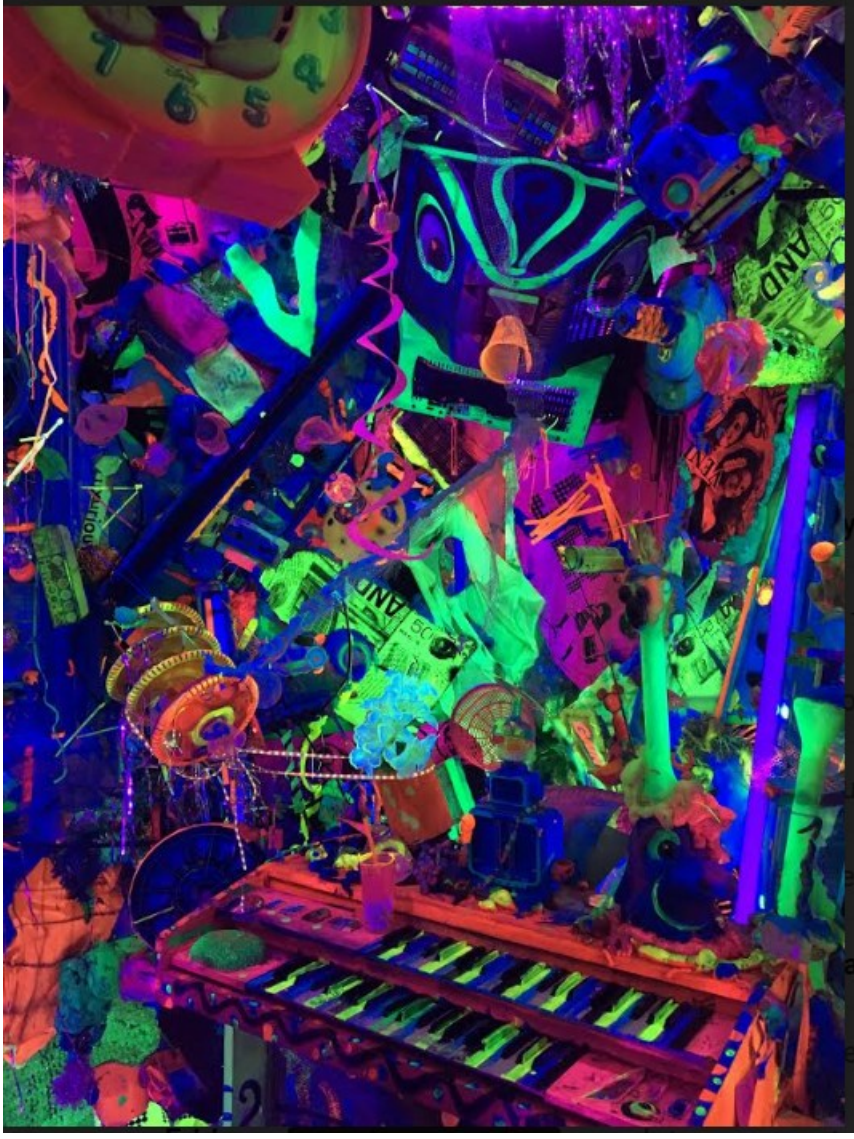
Fluorescent minerals and **laundry soap**.



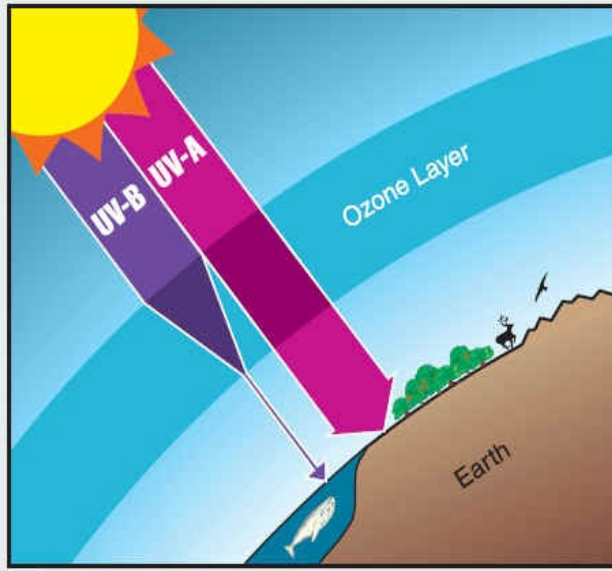
cloth washed with fluorescent chemicals will appear Whiter under the Sun.

UV radiations are absorbed and visible Light emitted Making white brighter.

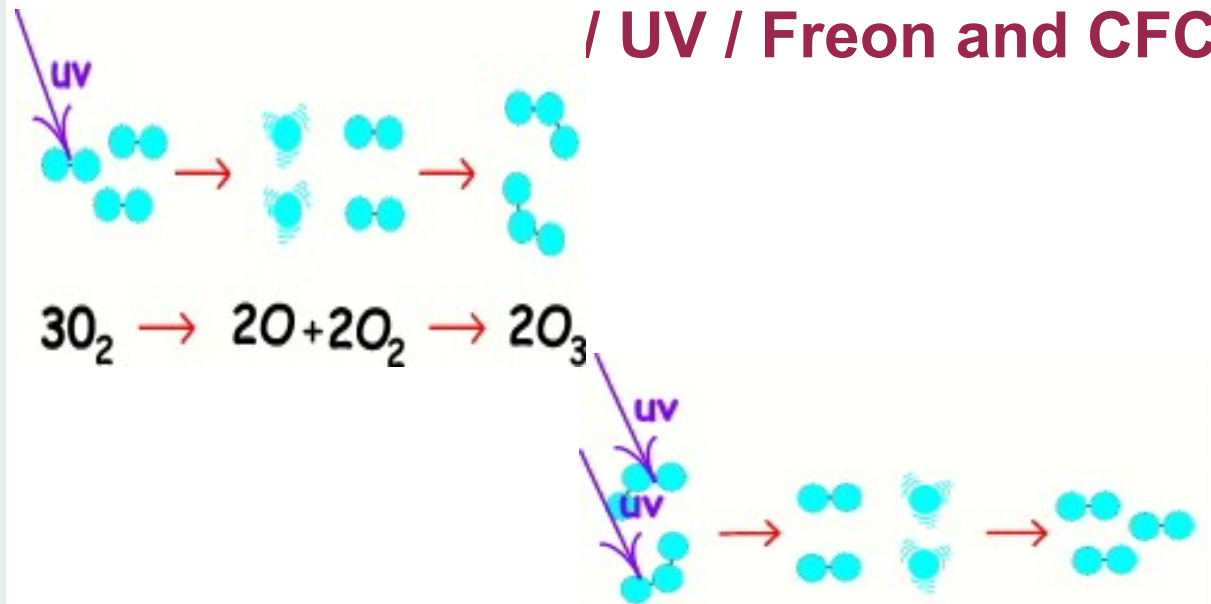
<http://blogs.discovermagazine.com/notrocketscience/2011/12/23/why-do-scorpions-glow-in-the-dark-and-could-their-whole-bodies-be-one-big-eye/#.WsfpMpW5vIV>



UV Protection by the Ozone Layer



/ UV / Freon and CFCs



<https://www.youtube.com/watch?v=IniJx-vRHG0>

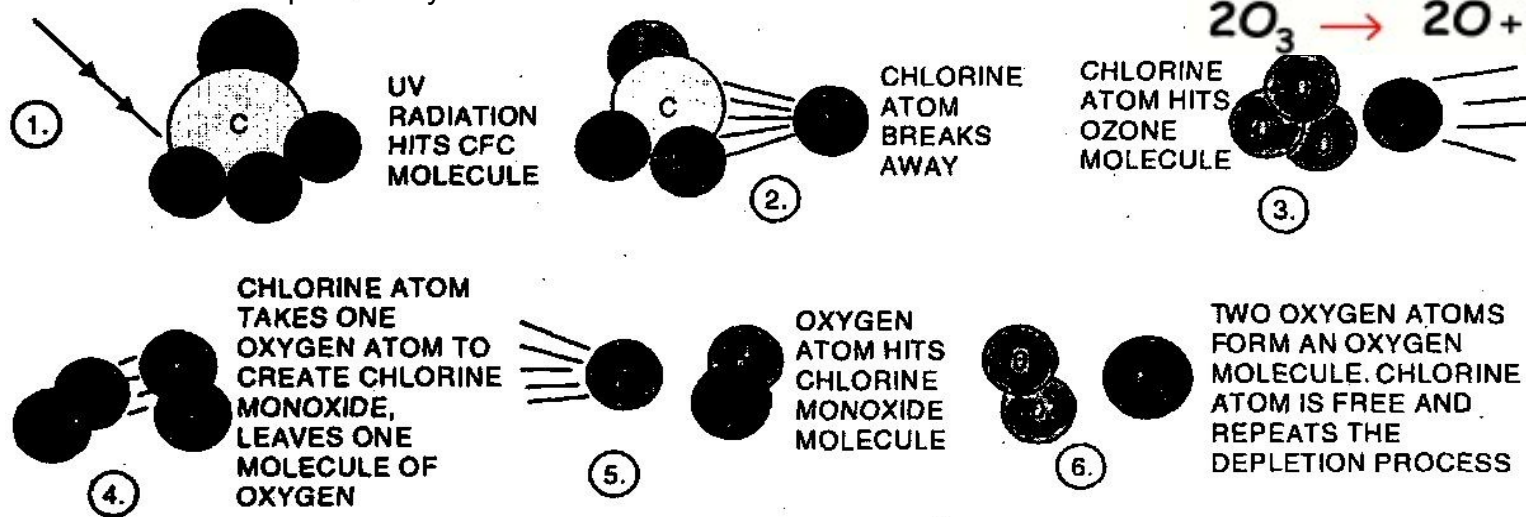
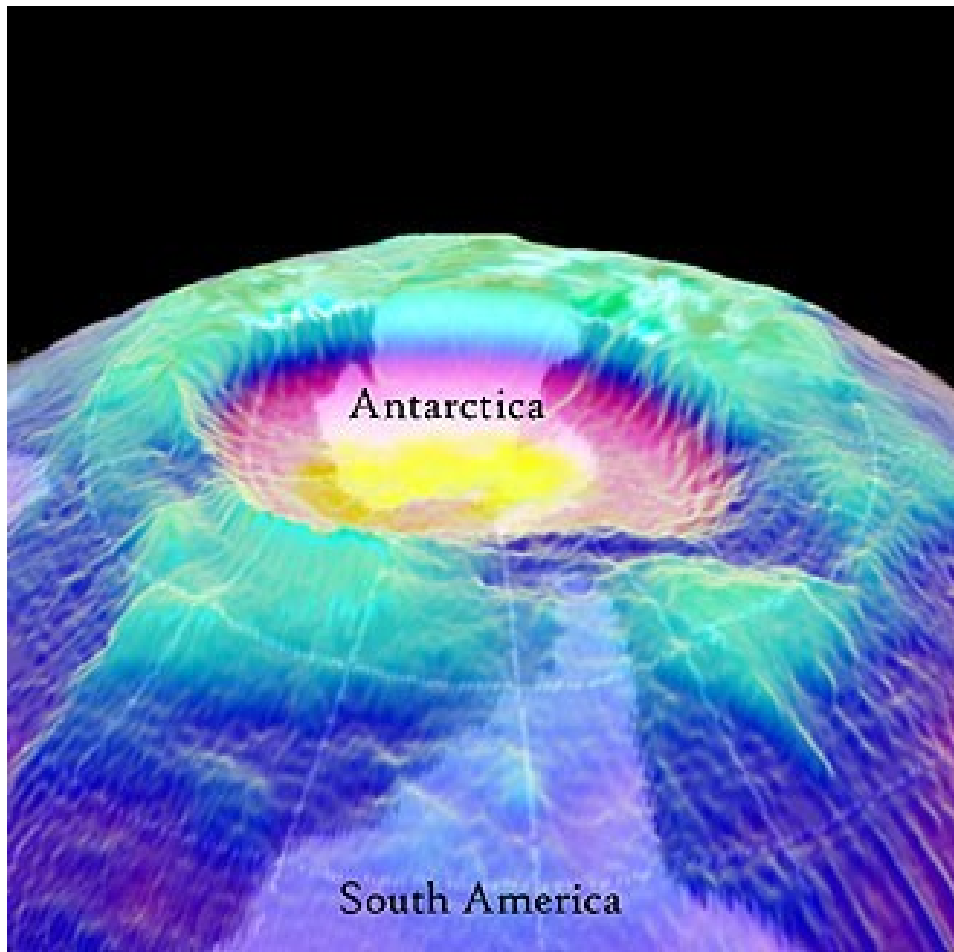


Fig. 1.9. Interaction of CFCs and UV radiation damaging the ozone layer.

<http://resources.yesican-science.ca/trek/scisat/final/grade9/ozone1.html>



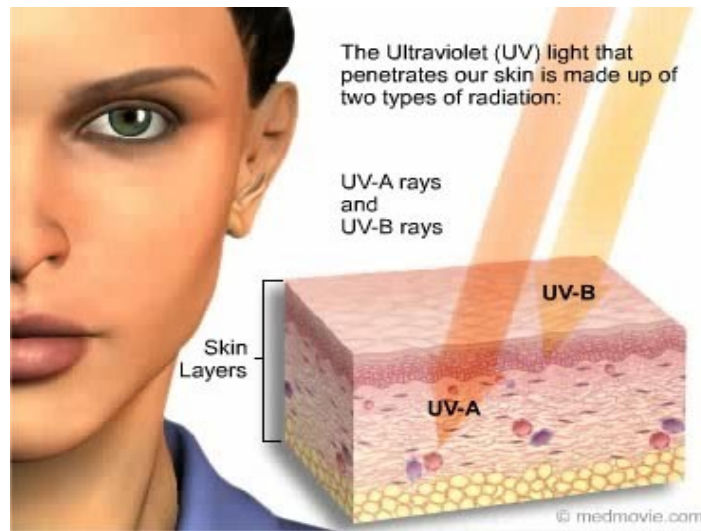
Ozone layer between 40,000 ft to 60,000 ft

Nobody knew why the hole
Was above Antarctica and not
Some where else. Scientists
Realized that certain crystals
Of nitric acid formed there
In the early spring, and on the
Surface of those crystals,
The chlorine and fluorine
Was far more effective at
Destroying the ozone.

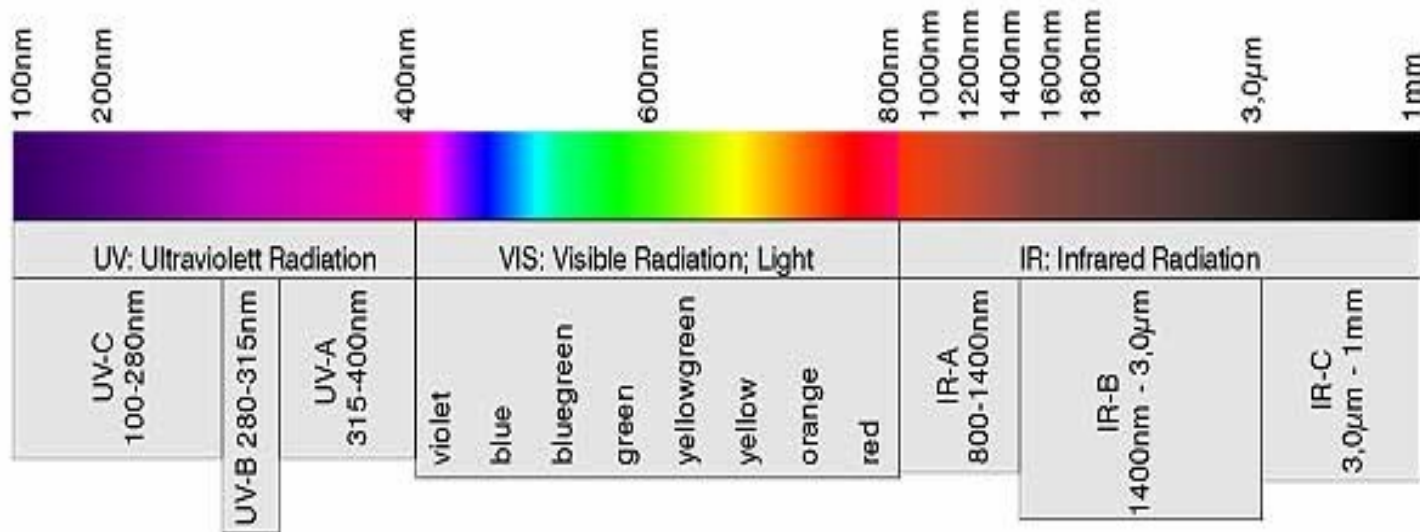
Freon was used in AC,
Refrigerator and propellants. Now,
It's illegal but the CFCs stick for
A long time.

<http://www.freedomsphoenix.com/News/049633-2009-05-01-ozone-hole-purportedly-modifies-antarctic-winds.htm?EdNo=001&From=>

Today, nitrous oxide is used as a propellant



Notice:
You have near IR and
Far IR

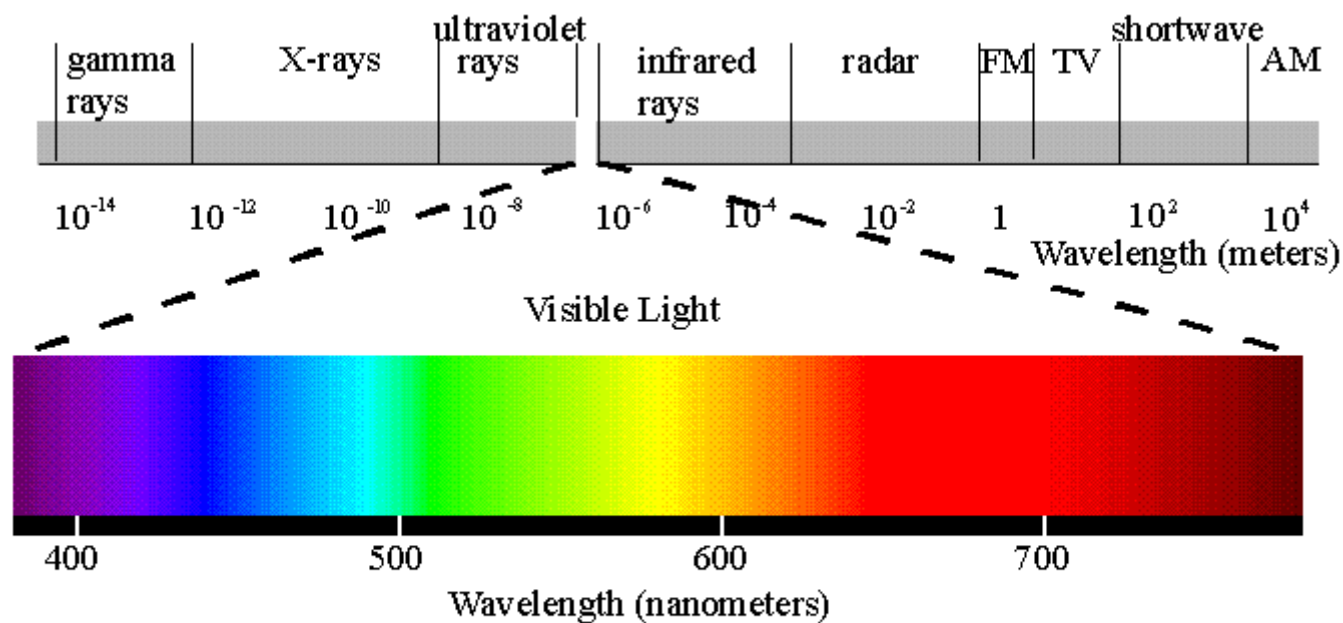


The form of UV that is most potent for burning and cancer has a wavelength of about 300nm. This kind of radiation is absorbed by ordinary window glass.

Normal glass blocks all of UVB but allows UVA to come through.

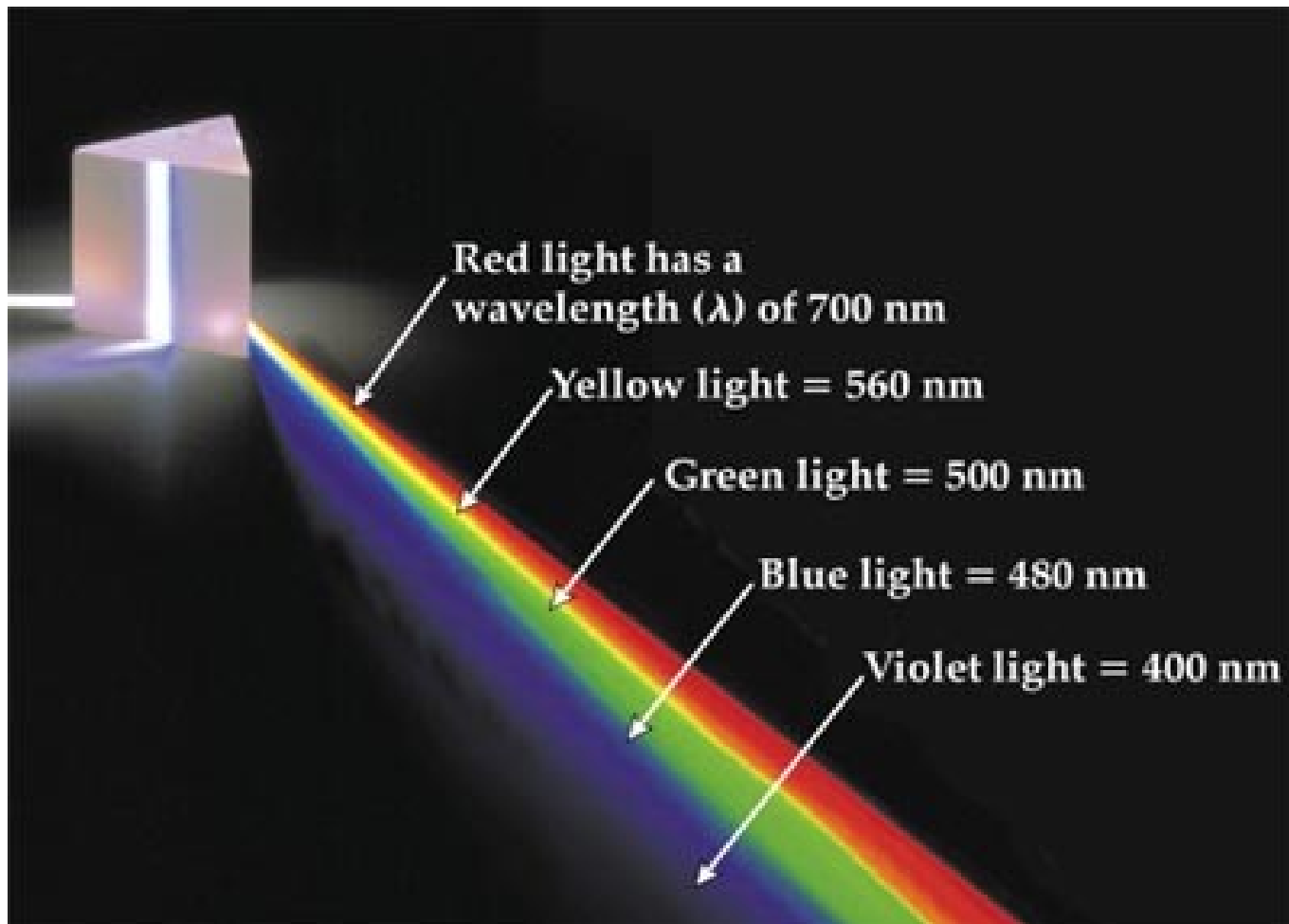
Example 1 The Wavelength of Visible Light

Find the range in wavelengths for visible light in the frequency range between $4.0 \times 10^{14} \text{ Hz}$ and $7.9 \times 10^{14} \text{ Hz}$.



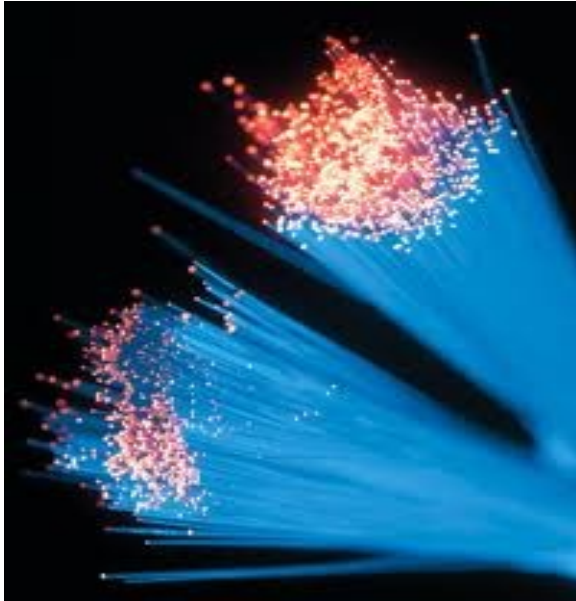
[Table of Contents](#)

[Visual Stimulus](#)



fiber optics: visible light can transport more data

The optical fibers have become very efficient to transport information.



Because of the high frequency of visible light,
You can transport more data per second.
The bits per second is called the baud.
The number of bits of second you can transport
Is the frequency of the wave. (about).

Indeed, information is coded are series of 0 and 1.
So you turn off and on the wave carrying the
Information. You can not send signal faster than the
Frequency. The number of on or off per second =
Frequency of the wave at most.

So you can transport about 10^{14} bits per second. Higher is the frequency, more information can Be transmitted per second.

So visible light can transport more info than Radio wave (smaller frequency).

This rule was found in 1940 by Claude Shannon,

INFRA RED radiations

Heat waves. Produced by any warm body.
(any thing with temperature).

Emitted by the molecules jiggling around.

Used in low energy lasers. Remote

Control. Objects in space emit in the IR.

Used by military to detect tank, planes ...

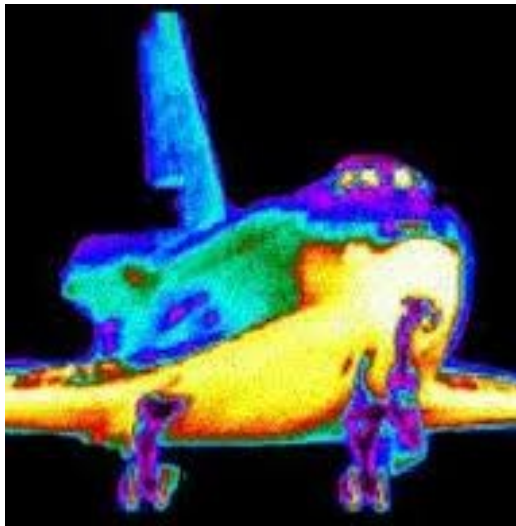
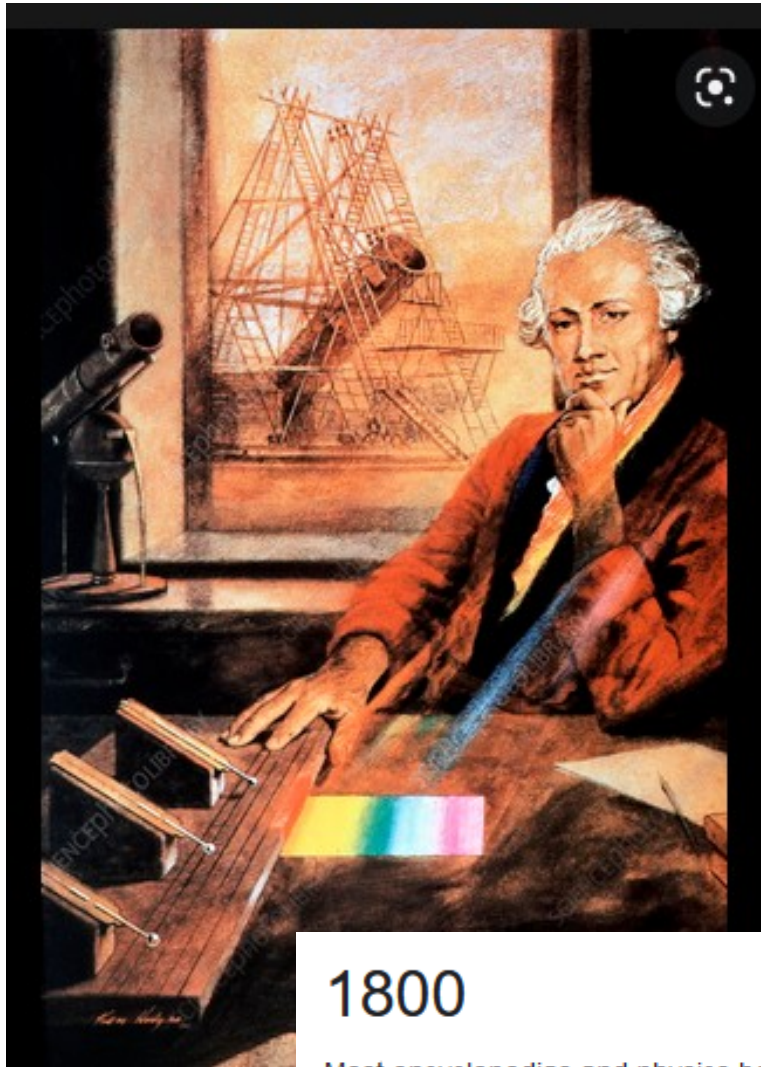


Table 8.1 Approximate Frequencies and Wavelengths of Different Colors

Color	Frequency Range ($\times 10^{14}$ Hz)	Wavelength Range ($\times 10^{-7}$ m)
Red	4.0–4.8	7.5–6.3
Orange	4.8–5.1	6.3–5.9
Yellow	5.1–5.4	5.9–5.6
Green	5.4–6.1	5.6–4.9
Blue	6.1–6.7	4.9–4.5
Violet	6.7–7.5	4.5–4.0



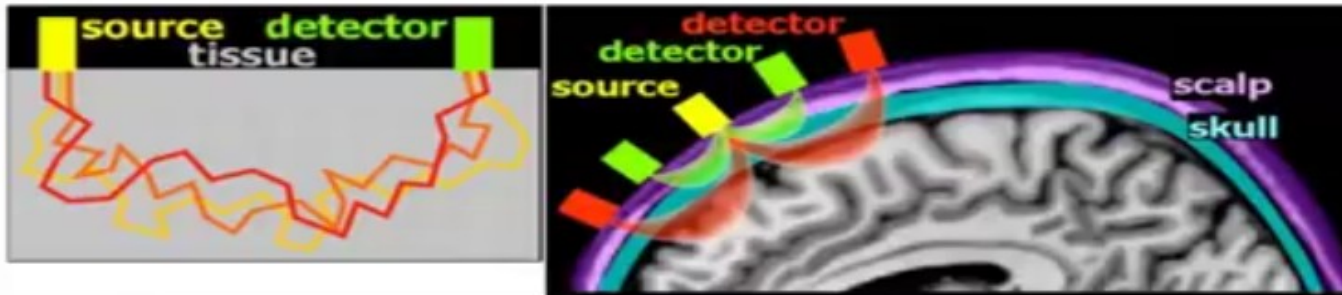
1800

Most encyclopedias and physics books credit the great British astronomer Sir William Herschel with the discovery of infrared radiation **in 1800**.

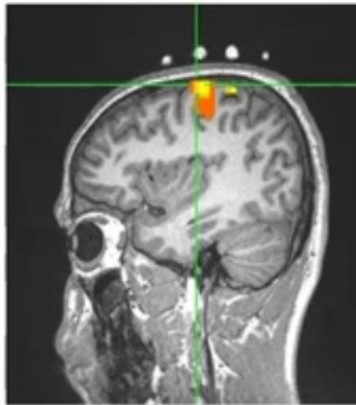
He discovered Uranus

Functional Near-Infrared Spectroscopy (fNIRS)

- Each source-detector pair probes a 'banana-shaped' region
- fNIRS can only image the surface of the brain (cerebral cortex)
- Multiple source-detector pairs are used simultaneously to map neuronal activity on the brain's surface



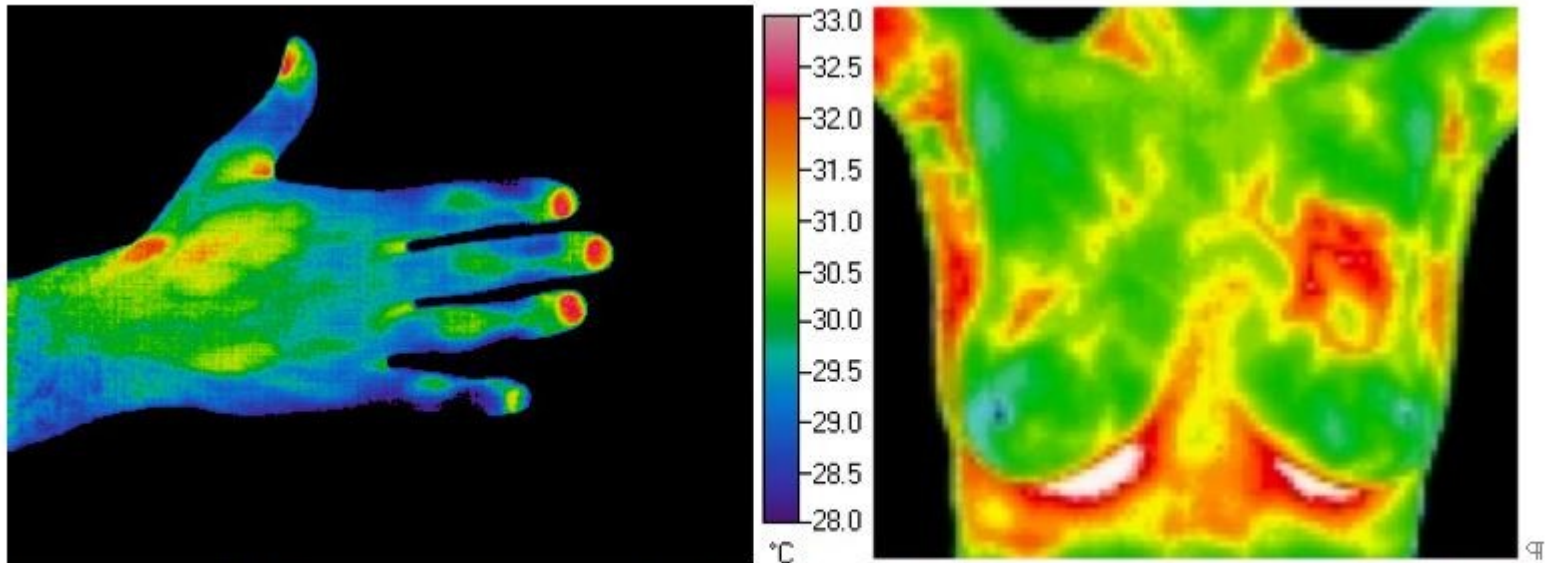
Using NIR to probe the surface of the brain.



https://en.wikipedia.org/wiki/Near-infrared_spectroscopy

A thermogram can also depict abnormal heating indicating infection

In the image below, the right foot of the patient is noticeably cooler, suggesting reduced circulation related to nerve damage



Thermograph image of a human hand, with an infection on the wrist (near the thumb), and thermograph showing breast cancer (on upper breast on right). ©

(Left photo from NASA; right photo copyright Meditherm.) ©

<https://link.springer.com/article/10.1007/s13755-022-00194-8>

[http://www.fas.org/irp/imint/docs/rst/Intro/
Part2_26d.html](http://www.fas.org/irp/imint/docs/rst/Intro/Part2_26d.html)

Note: thermography can also be done by painting the skin with a chemical (known as a liquid Crystal) that shows visible color change with temperature

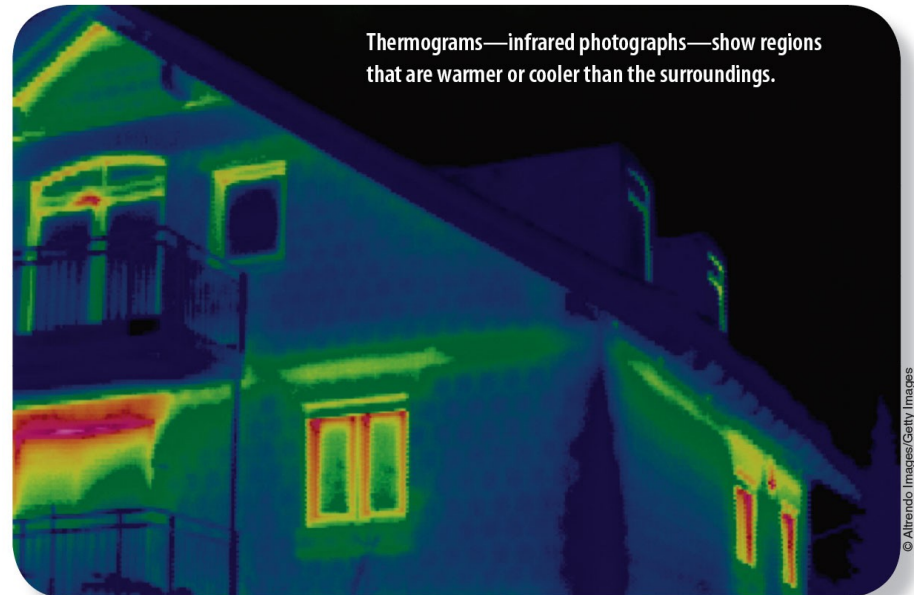
Blackbody radiation=

radiation emitted by thermal emitter (any thing with temperature) .

Power (energy/second) proportional to temperature raised to the 4th power.

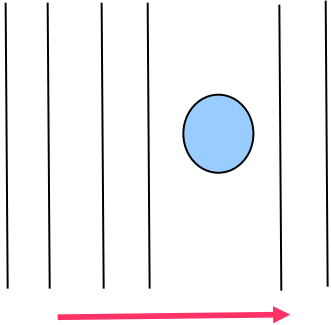
This means we can determine the temperature of an object by measuring the EM radiation it emits.

A thermogram is a “picture” of the IR radiation emitted by an object.

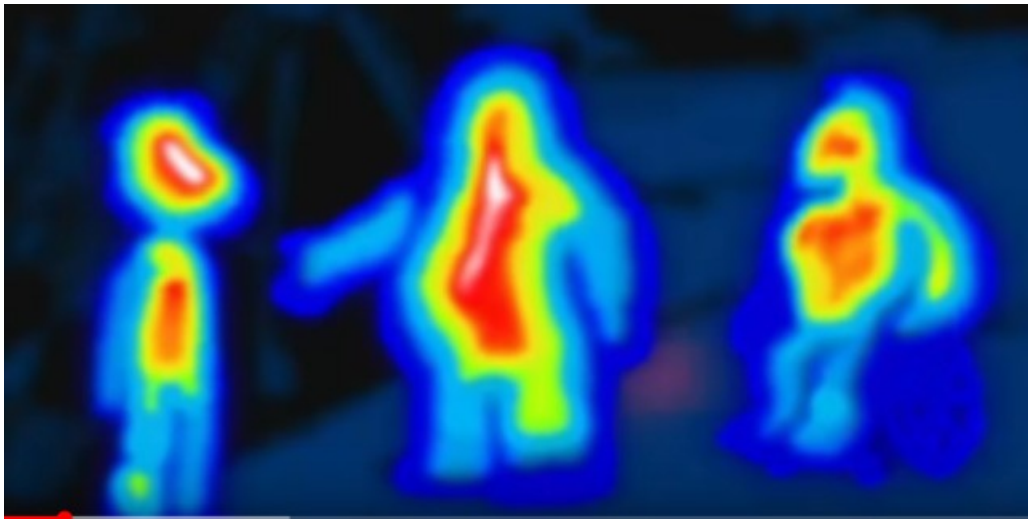


FIRE fighter can use INFRARED CAMERAS to see through the smoke.
Long wavelength diffract more than small wavelength so IR
Diffract around dust and smoke particles

<http://www.infraredcamerasinc.com/infrared-thermography-firefighting.html>

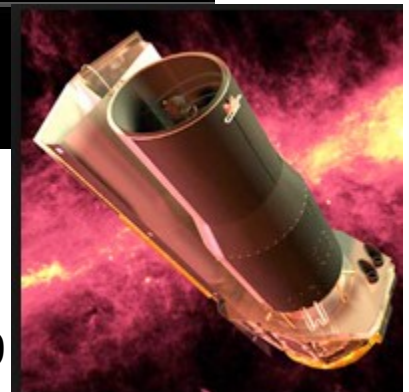


For more information please contact us at: <http://www.infraredcamerasinc.com> 1(409)861-0788



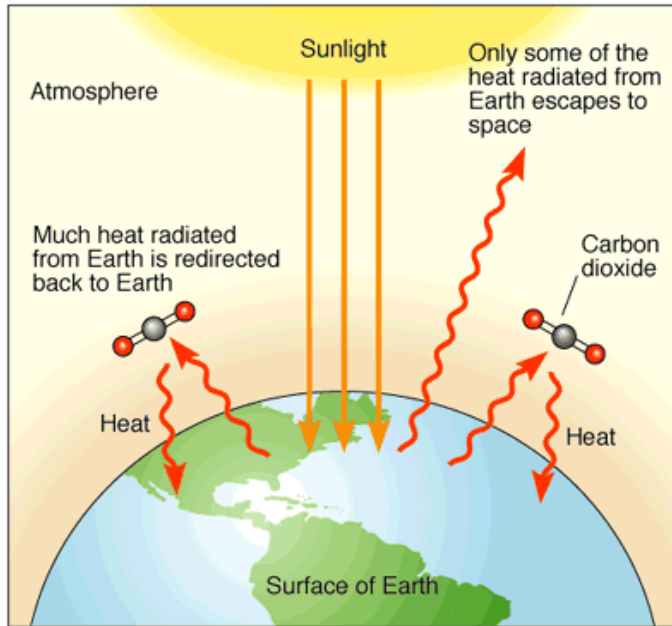
Family guy
As seen by the predator

In Astronomy we can pick into nebulae To observe stars



Spitzer telescope

GREENHOUSE EFFECT



H₂O, CO₂ and O₃

Absorb IR radiation.

IR radiations are trapped warming the Earth
→ made life possible on Earth (15°C average)

→ more deaths because of cold vs heat

→ The most potent greenhouse gas
Is not CO₂ (don't blame the cows)



Sulfur hexafluoride (SF₆): This colorless, odorless, and non-toxic gas is incredibly potent at trapping heat in the atmosphere. It's estimated to be over 20,000 times more effective at trapping heat CO₂ over a 100-year period.

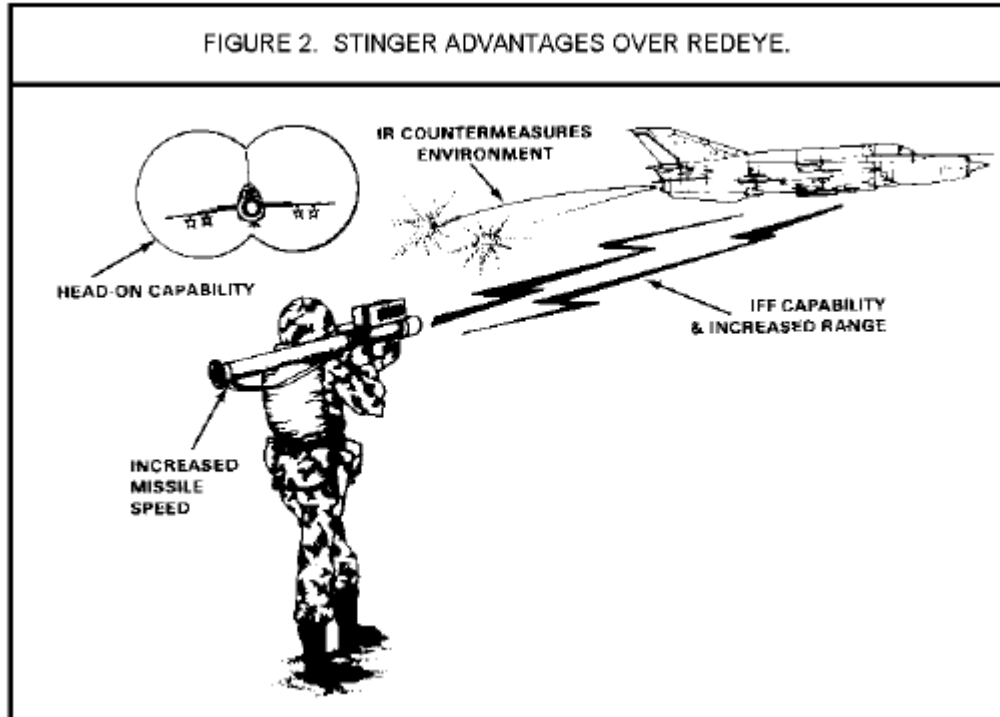
Electrical equipment insulation in high voltage transformers and switchgear.

Semiconductor manufacturing.

Magnets for medical imaging machines (MRI)

Nitrous oxide (N₂O), another potent greenhouse gas (around 300 times more potent than CO₂) used in some fertilizers.

IR can be sensed by Pit Vipers, mosquitoes and stingers !!



Stingers are high tech missiles That sense the IR radiation From a warm source.

Stingers were given to Taliban when they were US friends. Taliban sold them to Rwanda Rebels. One of these missiles Hit the plane to the president. That was the beginning of the genocide.

Stingers are only 35 pounds about And reaches 10,000 ft.



Mosquitoes (female) have IR sensors to find you !!

We can use this ability to get rid of them, !!! see links below.



Mosquito Mania / The New York Times

<https://www.electronicproducts.com/bug-zapper-tracks-and-kills-bugs-using-infrared-and-laser-beams/>

http://www.youtube.com/watch?v=8GXOByRq4Uk&feature=player_embedded#!

HEAT VISION



They operate at a wavelength of about 10 micrometers to see human. (our range See below to compute that).

The goggles can also emit an IR light so they can see each other. This is the Infra red transmission Technology. Wavelength is 2 micrometers.

Application with IR

Also:

Video about IR

How IR radiation was discovered by William Herschel

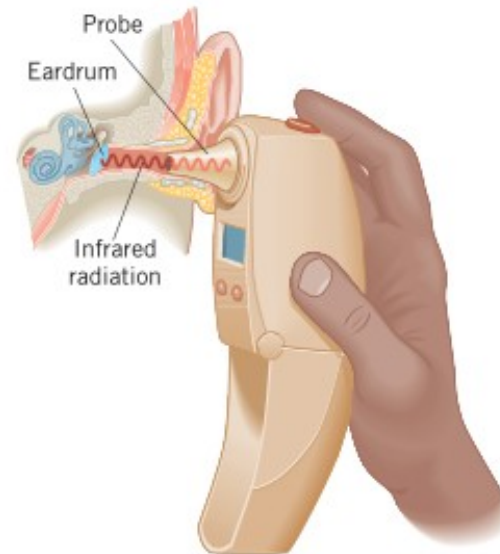


Figure 24.11 A pyroelectric thermometer measures body temperature by determining the amount of infrared radiation emitted by the eardrum and surrounding tissue.

MICROWAVE

- upper limit of radio waves. Range from 0.3m to 0.3mm (so not micrometer).
Resonant wavelengths for molecules of water. The water molecules are Bipolar (+ and -) and feel the electric field of the waves. They start vibrating and heat the Food in the process. (torque is applied to molecules).

This is because the temperature

Is proportional to the

Kinetic energy of

The molecules.

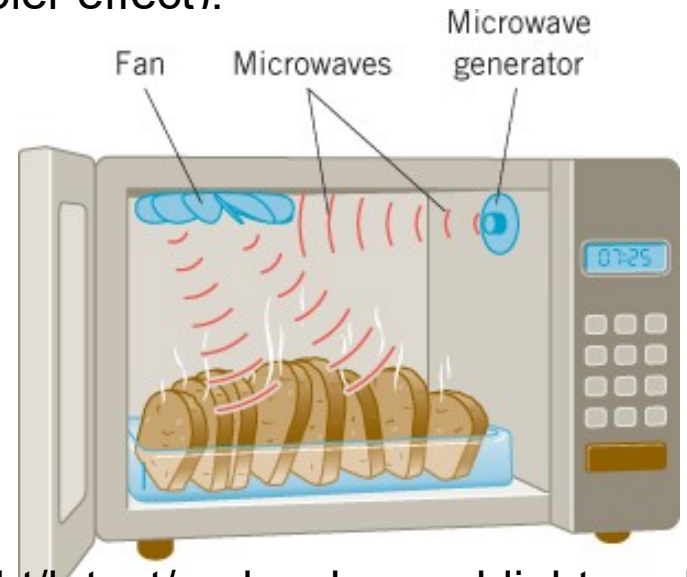
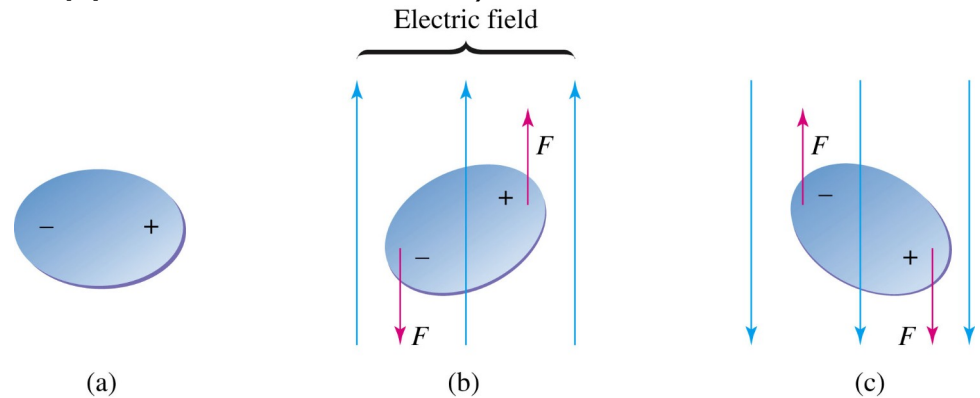
2,450 MHz is used in most ovens

Used in communication. Blue-tooth,

Wi-Fi signals. Used for radars (echolocation + Doppler effect).

Can map surface of planets or Earth.

<http://phet.colorado.edu/en/simulation/microwaves>



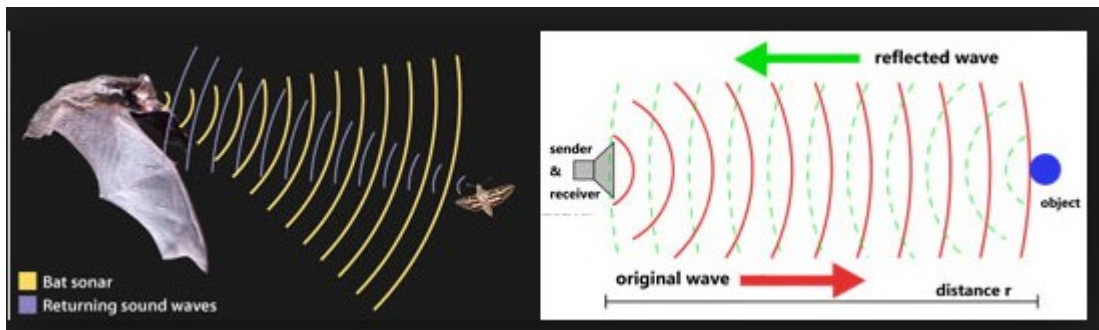


Microwaves spread less than
Radio waves because they have
Shorter wavelengths.
It's easier to focus to use for RADAR.

Because the wavelength is large enough,
It goes through clouds.

MICROWAVES = used in radar and cell phone. Small wavelengths so diffract less.

Microwave dish acts like a lens. Can be aimed at some distant target.

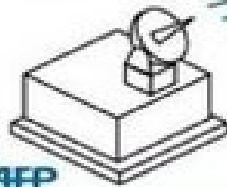


Invisible aircraft

Military stealth aircraft avoid or interfere with radar detection, allowing covert raids into enemy territory

Fooling radar:

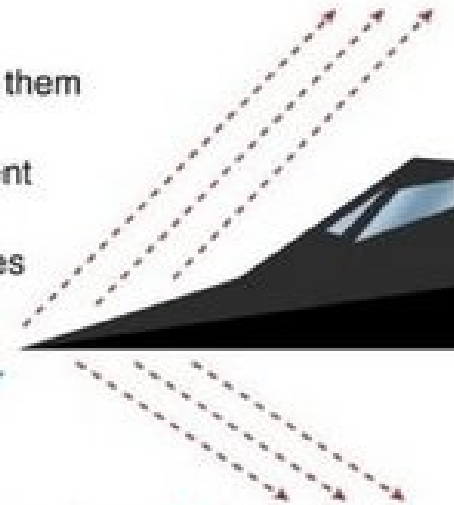
- Surface materials that absorb signals, convert them into heat
- Smooth surfaces with seamless joints to prevent radar reflection
- Wing edges moulded to scatter incoming waves
- Engines hidden in body of wing
- Exhaust ducts placed on top of aircraft to reduce heat trail



AFP

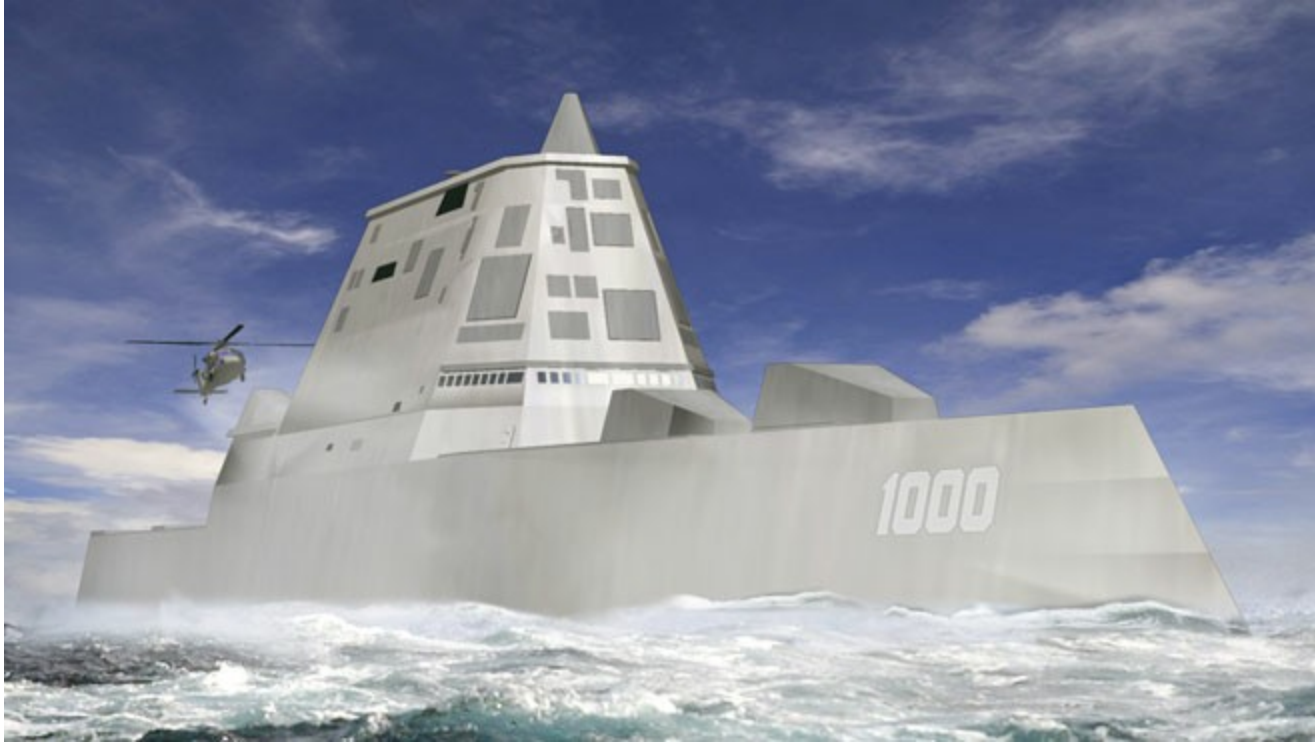
- Electronic counter measures such as jamming and ghost imaging (*images that the camera itself cannot see*)

- Jagged rear edges to scatter signals



Source: GlobalSecurity/Engineering.com/TheGadgetBook

STEALTH TECHNOLOGY go go invisible



This file image released by Bath Iron Works shows a rendering of the DDG-1000 Zumwalt, the U.S. Navy's next-generation destroyer, which has been funded to be built at Bath Iron Works in Maine and at Northrop Grumman's shipyard in Pascagoula, Miss. (AP/Bath Iron Works)

<http://www.foxnews.com/politics/2012/06/04/planned-stealth-destroyer-could-underpin-us-navy-china-strategy/>

RADAR can be used for imaging cities / surface of Earth / planets.

The technology is called **SAR**.

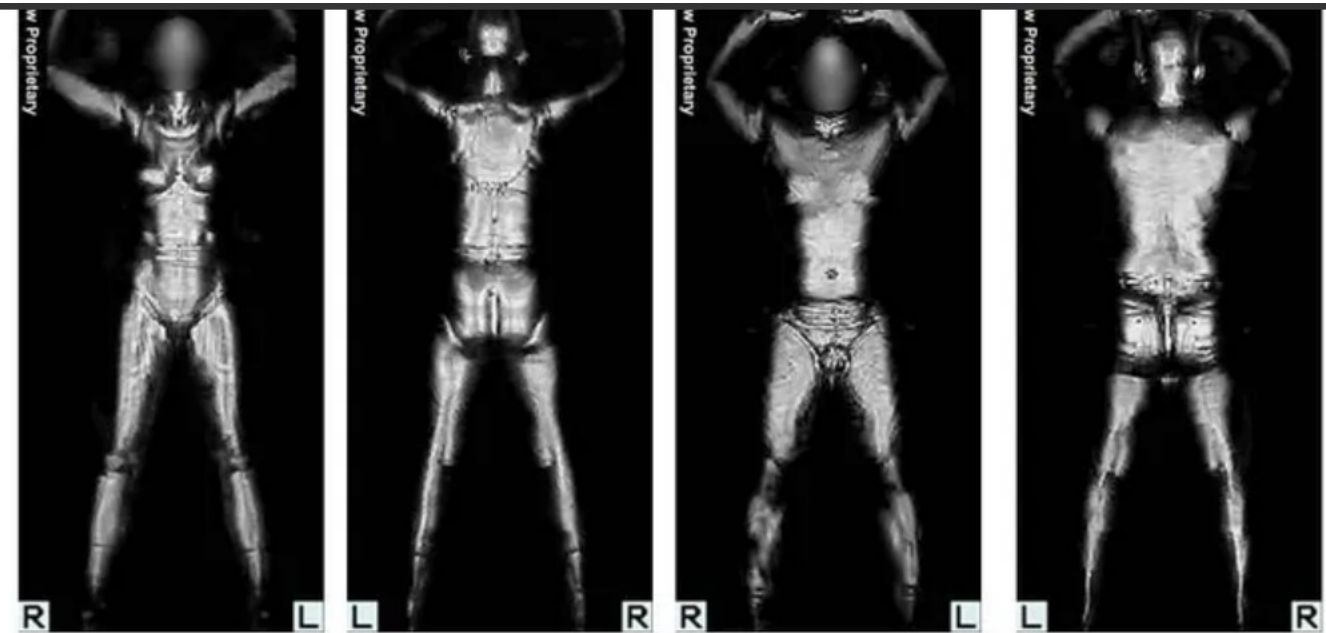
[http://en.wikipedia.org/wiki/
Synthetic_aperture_radar](http://en.wikipedia.org/wiki/Synthetic_aperture_radar)

single beam-forming antenna from which a target scene is repeatedly illuminated with pulses of radio waves at wavelengths anywhere from a meter down to millimeters. The many echo waveforms received successively at the different antenna positions are coherently detected and stored

[http://www.thespacereview.com/article/
790/1](http://www.thespacereview.com/article/790/1)

[https://www.ncbi.nlm.nih.gov/pmc/
articles/PMC9229681/#:~:text=With
%20the%20emergence%20of
%20Millimeter,%2C%20concealed
%20threat%20detection%2C%20etc.](https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9229681/#:~:text=With%20the%20emergence%20of%20Millimeter,%2C%20concealed%20threat%20detection%2C%20etc.)





The TSA has spent \$120 million on the full-body millimeter wave scanners that we have to shuffle through in our sock feet, a figure that works out to about \$150,000 per unit. But not only is that a tremendous amount of money, these machines – and the agents who operate them – have proven to be increasingly inept at, you know, actually *scanning* for weapons or other threats to passenger safety. The agency previously spent \$40 million on the backscatter X-ray scanners that created borderline graphic images of passengers' bodies; those machines were replaced by the current scanners.

<https://www.cnet.com/culture/tsa-bored-of-seeing-you-naked-removing-airport-body-scanners/>



Figure 24.8 This wireless capsule endoscope is designed to be swallowed. As it passes through a patient's intestines, it broadcasts video images of the interior of the intestines. (Courtesy Given Imaging, Ltd.)

Before discussing AM vs FM, let's take a tangent and talk about diffraction
A wave property of light. (like interference and the double slit experiment).

When a wave goes through an opening (lens of microscope, telescope ..)
It will diffract. When a wave "meet" an obstacle, it will bend around the corners.

This will limit the resolution for microscope or telescope. (see picture resolution).

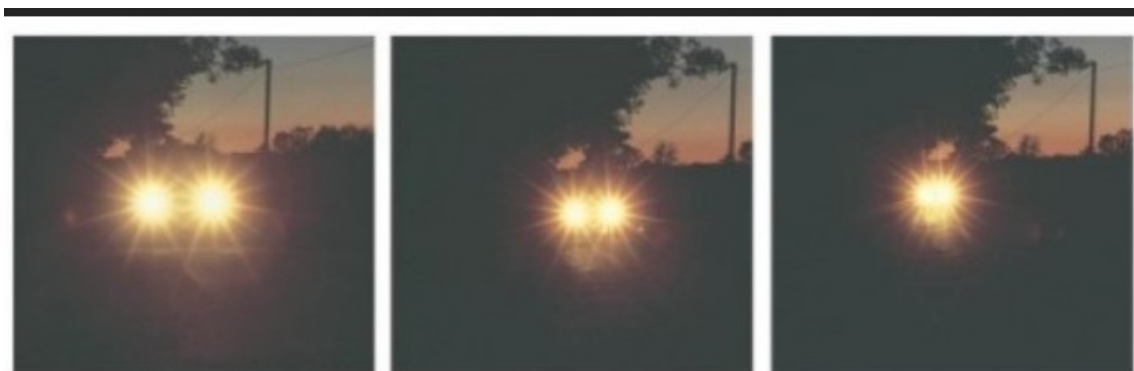
The amount of "bending" depends on the size of the wavelength.

If wavelength $<$ opening or obstacle $\rightarrow$ little spreading.

If wavelength $>$ opening or obstacle $\rightarrow$ large spreading.

We can not see things smaller than the wavelength we are using.

See pictures in folder.



MICROWAVE

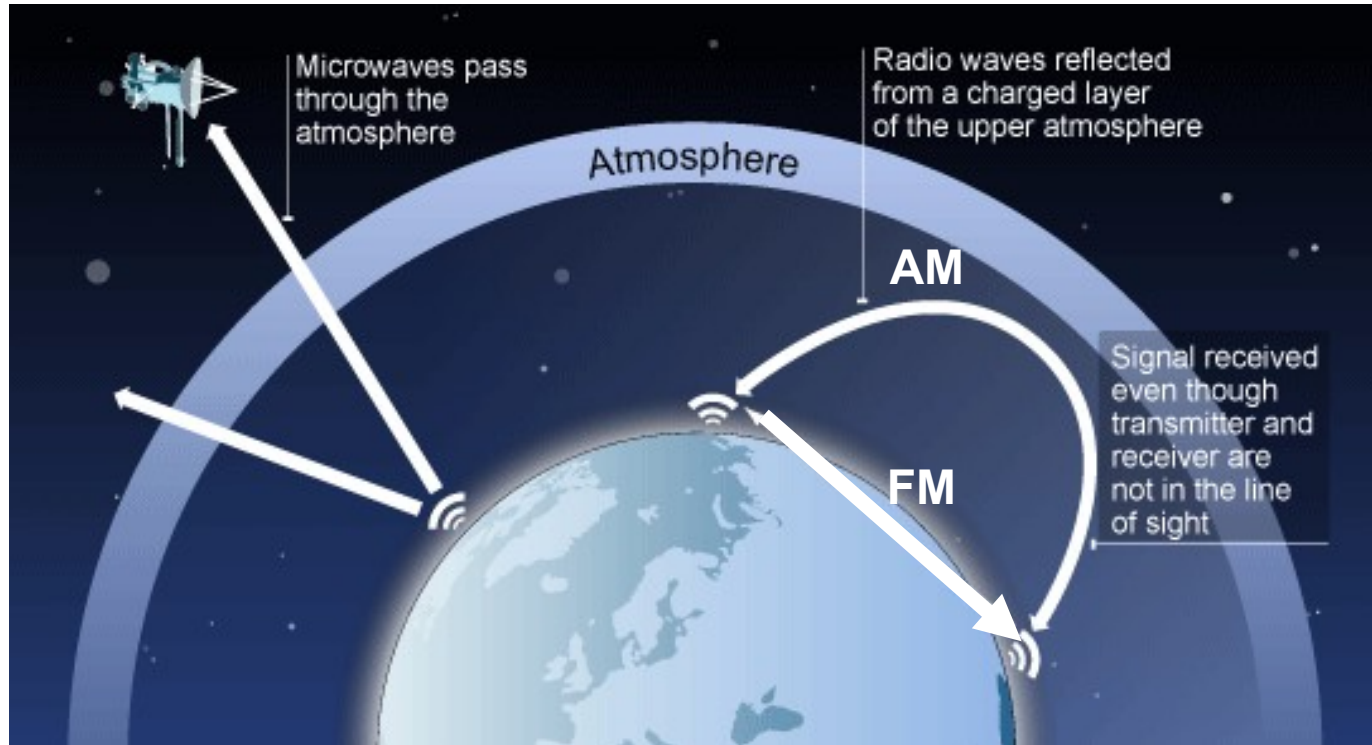
RADIO WAVES

- very large range from 100 Hz to 1000 megahertz. Frequencies given in kilohertz and megahertz. Radio waves are produced using AC with the appropriate frequency using an antenna. AM = long wavelength reflected by atmosphere. FM=short wavelength go through the atmosphere. Narrow frequency bands are Assigned for specific purposes. 88 to 108 MHz for commercial FM radio. Some bands are assigned to government and private communication.

Note: some radio waves do go through the body (if wavelegnth smaller than us) but they do not ionize. Only gamma-rays,X-rays and UV can ionize (and cause cancer or radiation sickness). Those behave like bullets (photons)

Radio waves can have other physiological effects. Depending on their energy And wavelength. (strong magnetic field produced by transformers can be unhealthy) A wave is both an electric field and a magnetic field shaking.

RADIO WAVES



FM + TV = high frequency, better resolution = better music because more info fit in the signal but smaller wavelength means less diffraction. So a building will shade the signal.

AM = larger distance, even larger at night so they decrease the power. Hard to listen at night.

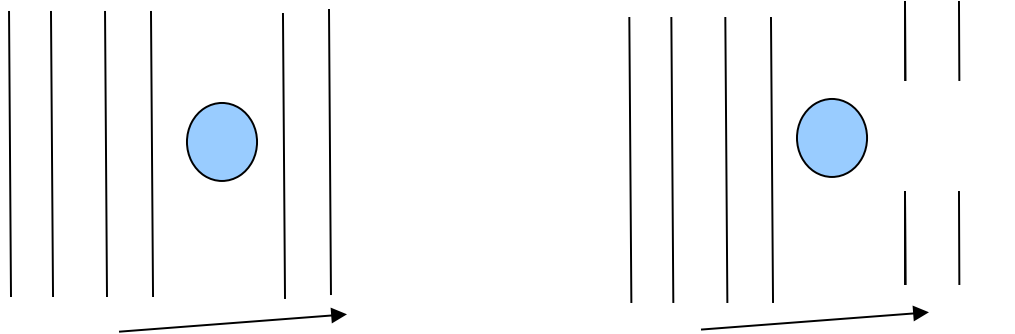
Conceptual Example 2 The Diffraction of AM and FM Radio Waves

Diffraction is the ability of a wave to bend around an obstacle or the edges of an opening. Would you expect AM or FM radio waves to bend more readily around an obstacle such as a building?

AM has longer wavelength (frequency about 100 kHz)

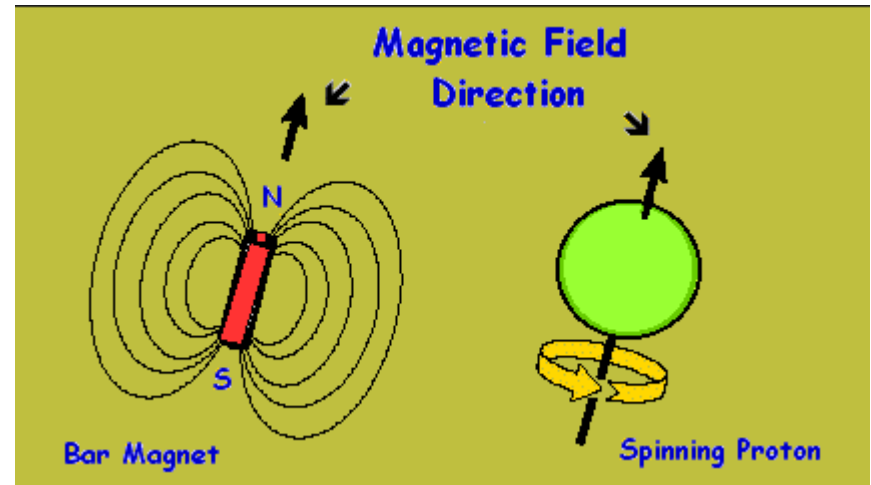
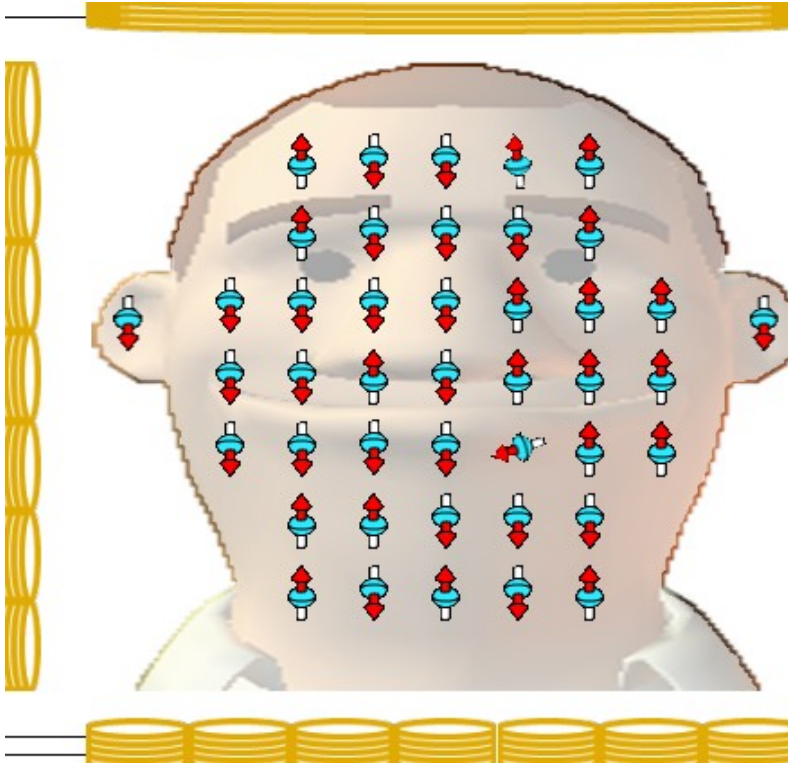
FM has shorter wavelength (frequency about 100 MHz)

Which one is which ?



MRI

1 T $\leftrightarrow$ 40MHz radio wave.



The protons behave like small magnets and are aligned in A VERY strong magnetic field. If you excite them with a Radio wave with the right frequency they jump to a higher level Of energy (they flip) and when they "relax" they burp out Radio waves used to illuminate the brain from inside out.

<http://phet.colorado.edu/en/simulation/mri>

<http://www.schoolphysics.co.uk/age16-19/Atomic%20physics/Atomic%20structure%20and%20ions/text/MRI/index.html?PHPSESSID=4efb7b210aababb8e14bd5e7c386b2a1>

How to turn the MRI into a rail gun

[https://www.youtube.com/watch?](https://www.youtube.com/watch?v=plvIEf7JsKo&nohtml5=False)

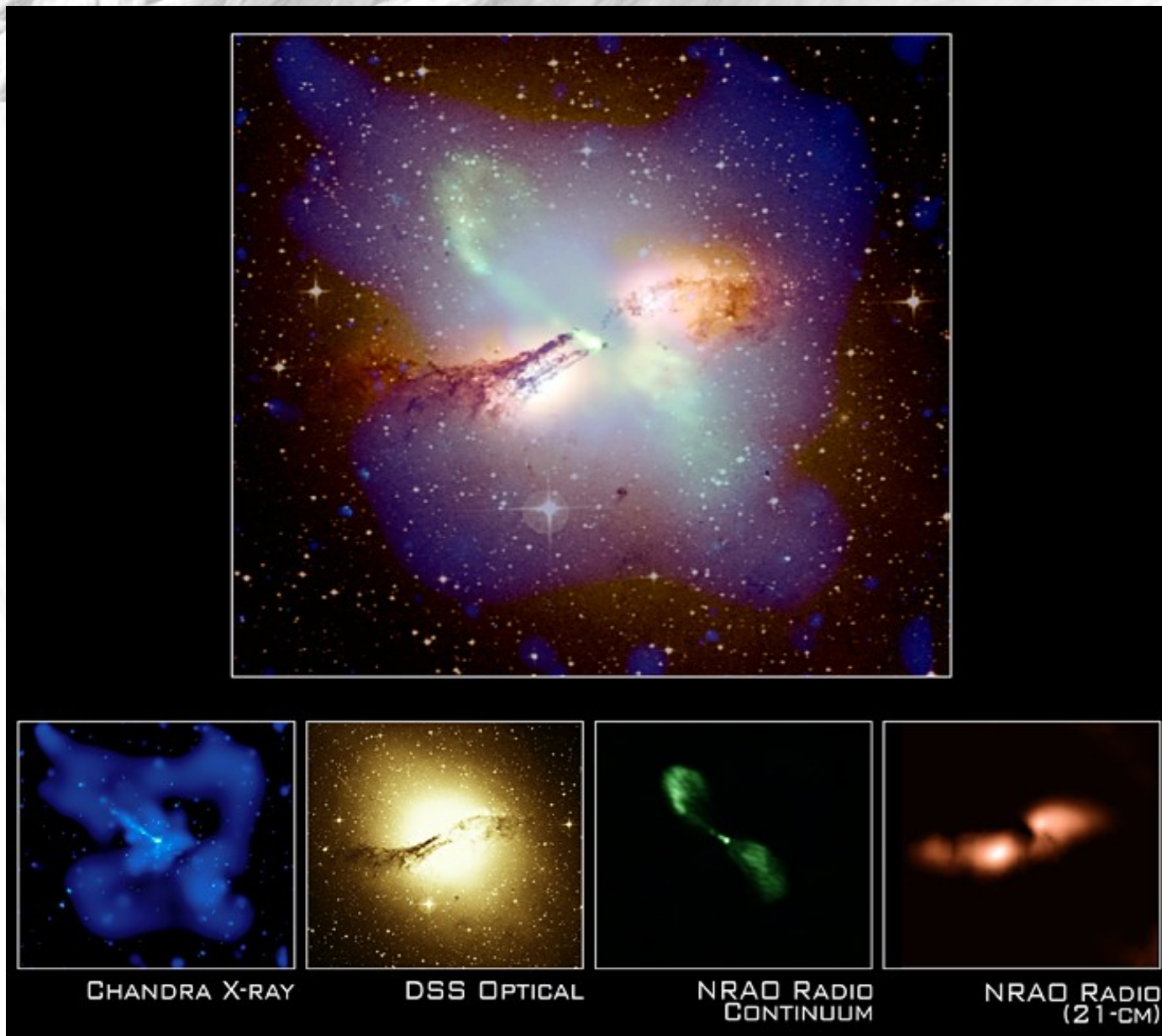
[v=plvIEf7JsKo&nohtml5=False](https://www.youtube.com/watch?v=plvIEf7JsKo&nohtml5=False)

With interferometry, radio astronomers can **combine the signals from many antennas, and even many telescopes**. It allows them to create an image that is much brighter and sharper than what is possible from a single antenna dish.



If you want to use radiowaves coming From space, you need large Opening to get good resolution. So large dishes.

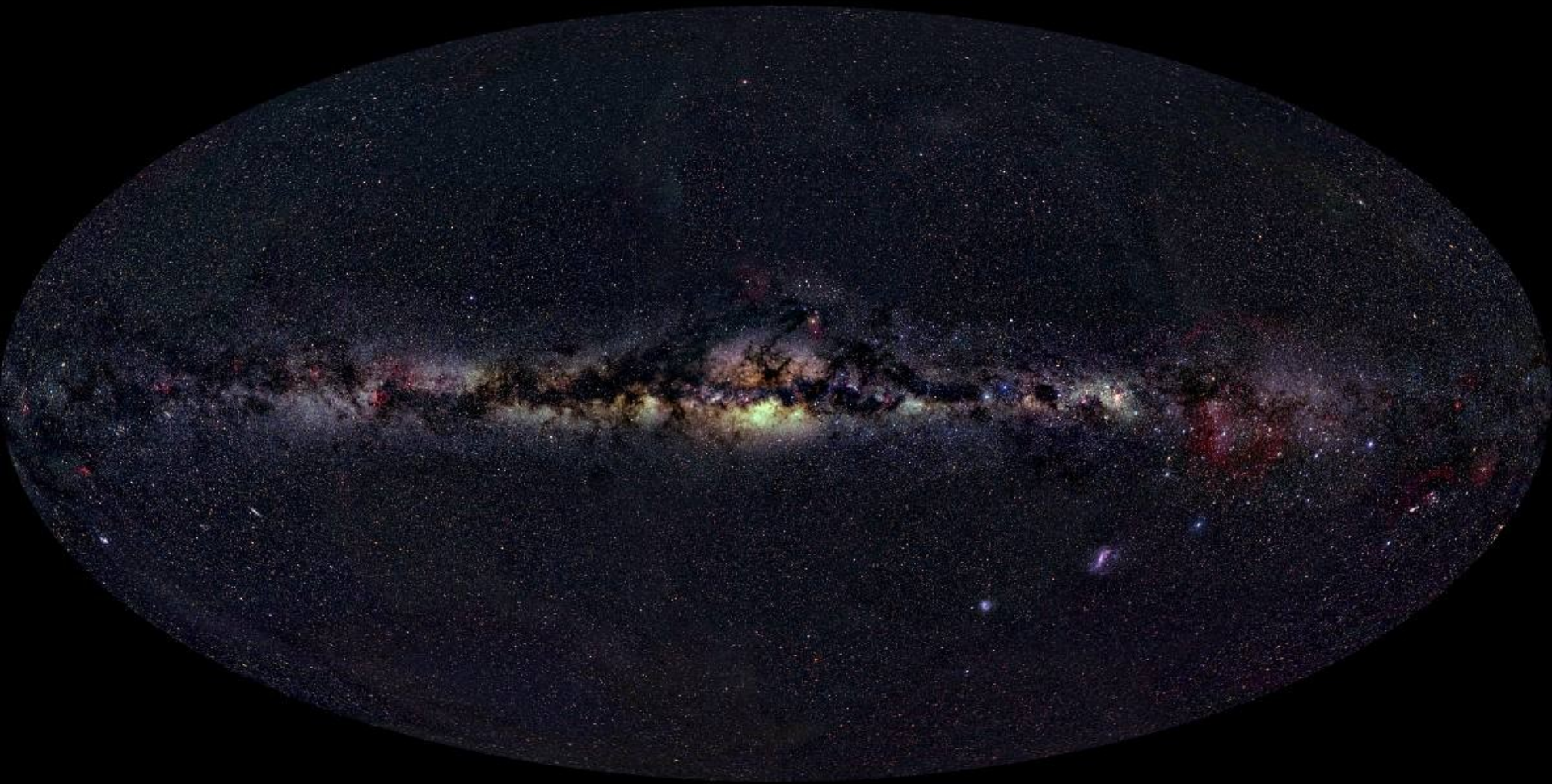




A composite X-ray (blue), radio (pink and green), and optical (orange and yellow) image of the galaxy Centaurus A presents a stunning tableau of a galaxy in turmoil.

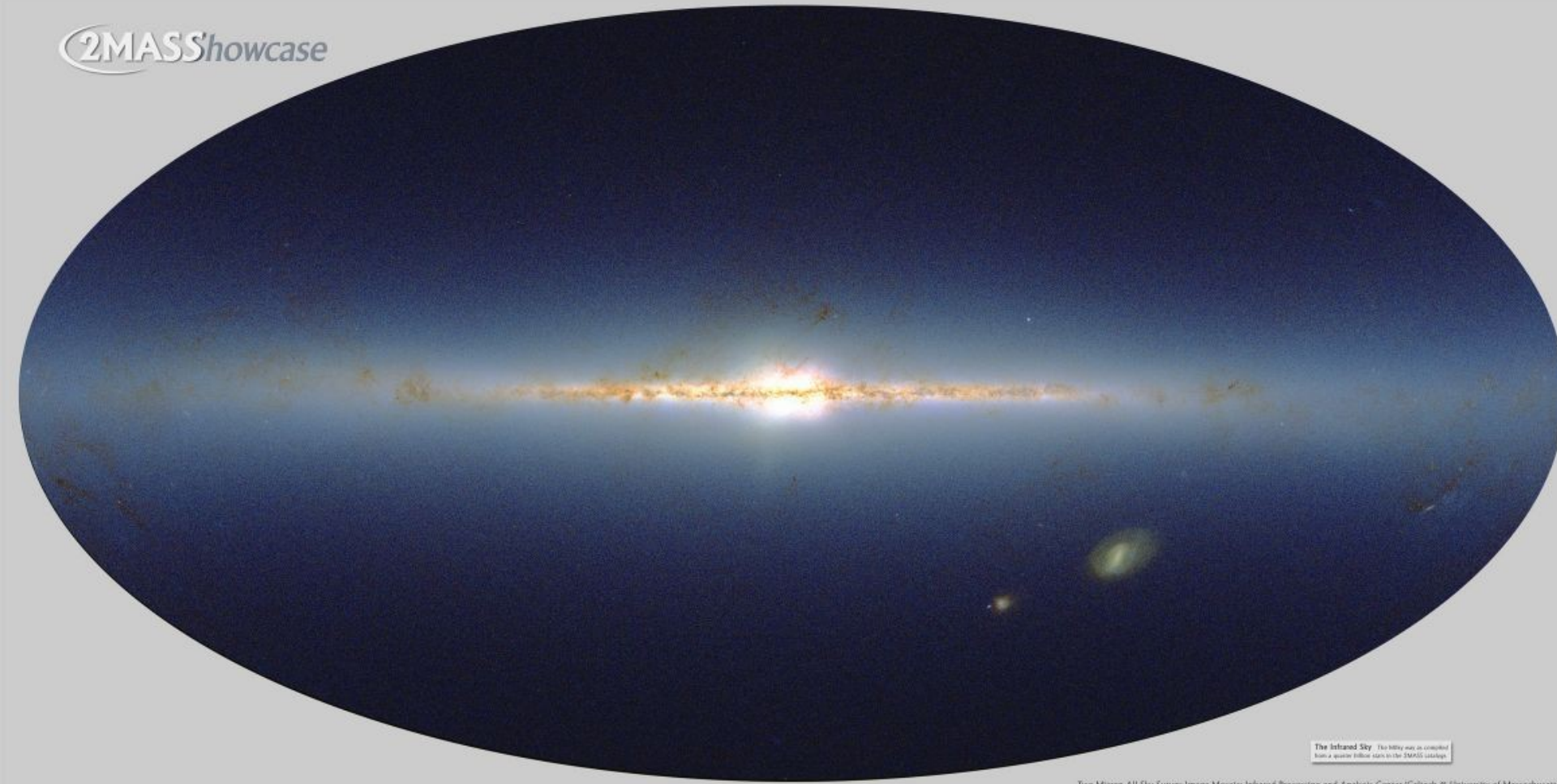
<http://chandra.harvard.edu/photo/2002/0157/>

Optical Sky



Near-infrared sky

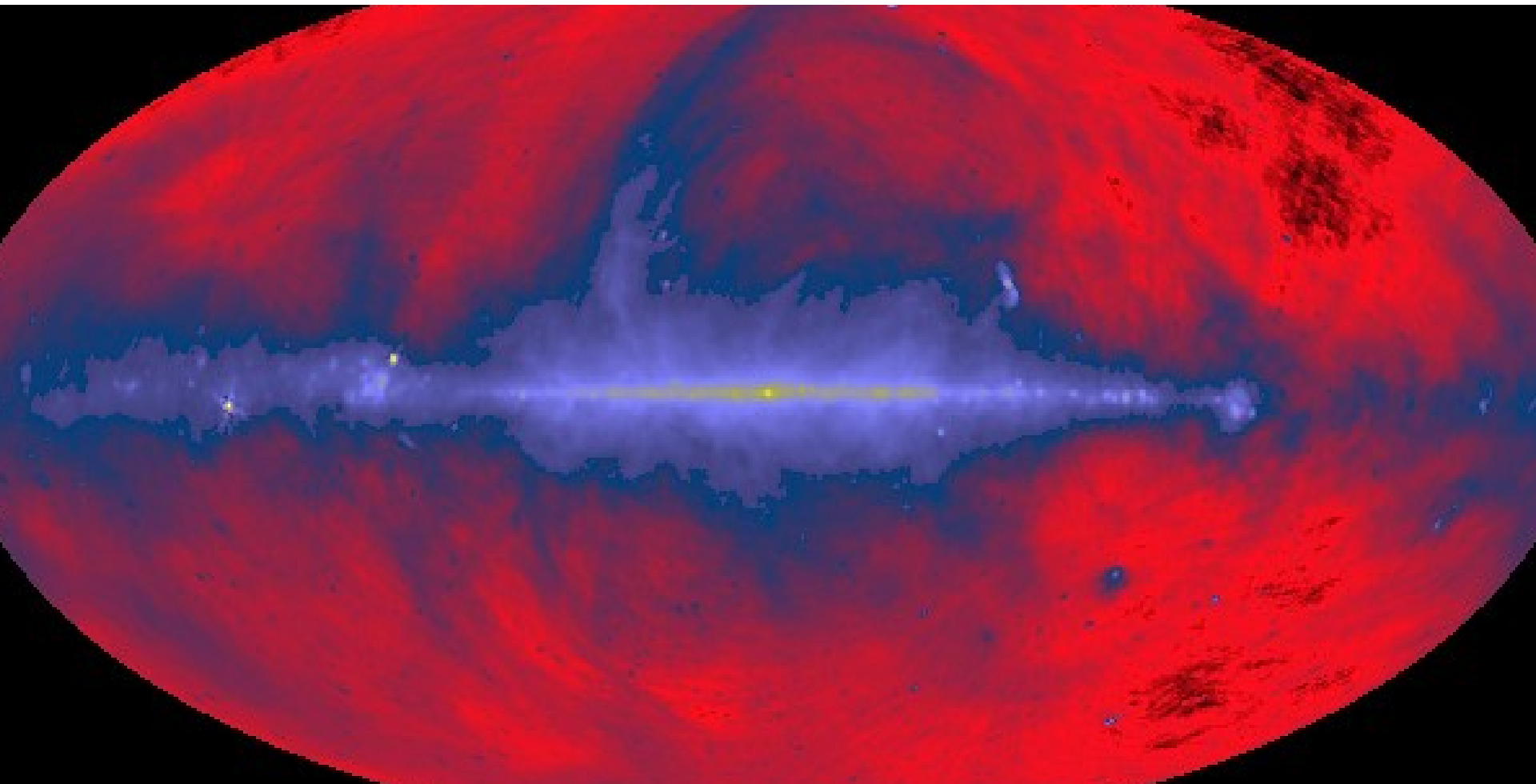
2MASS*showcase*



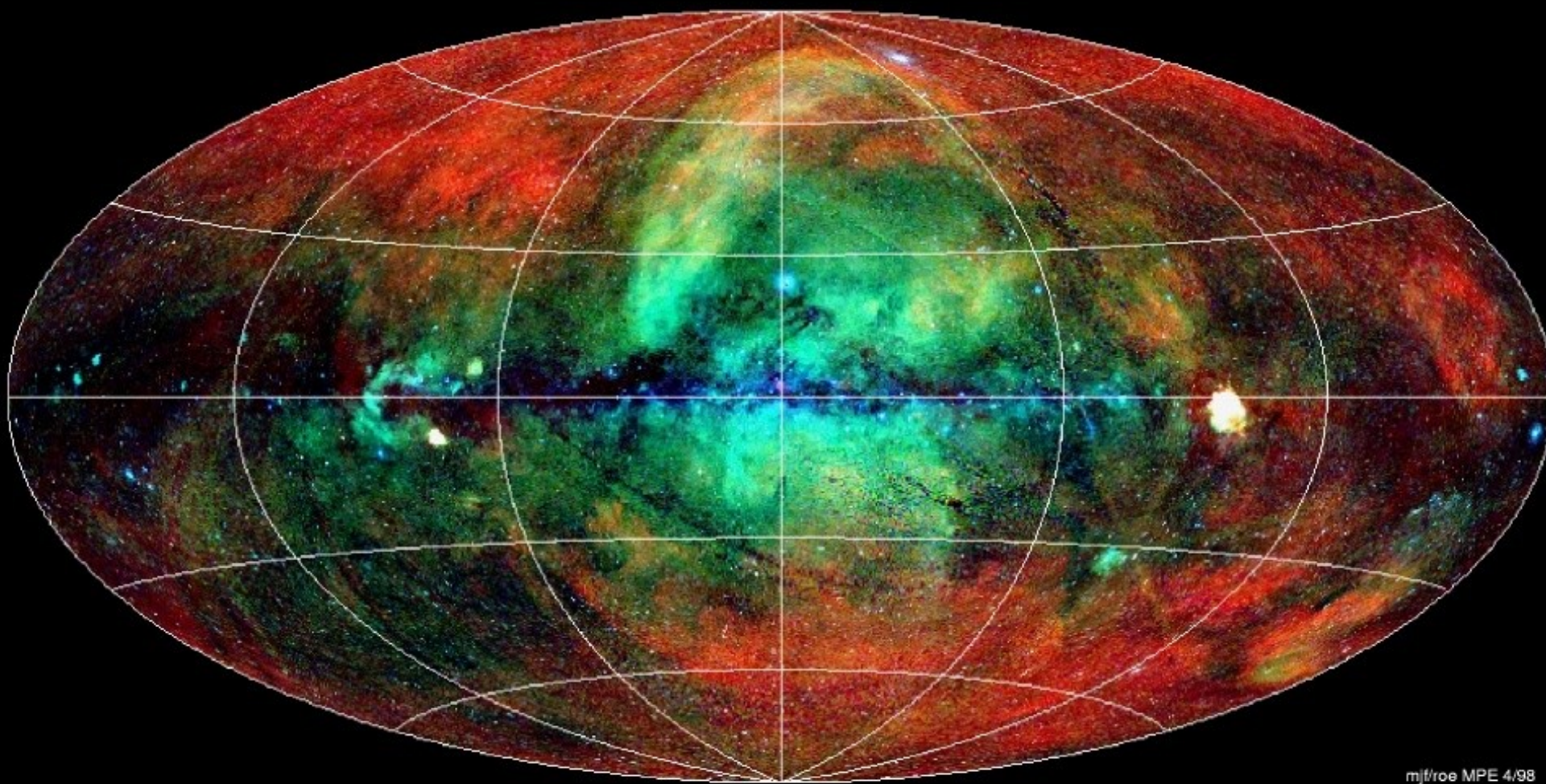
The Infrared Sky This image was compiled from a quarter million stars in the 2MASS catalog.

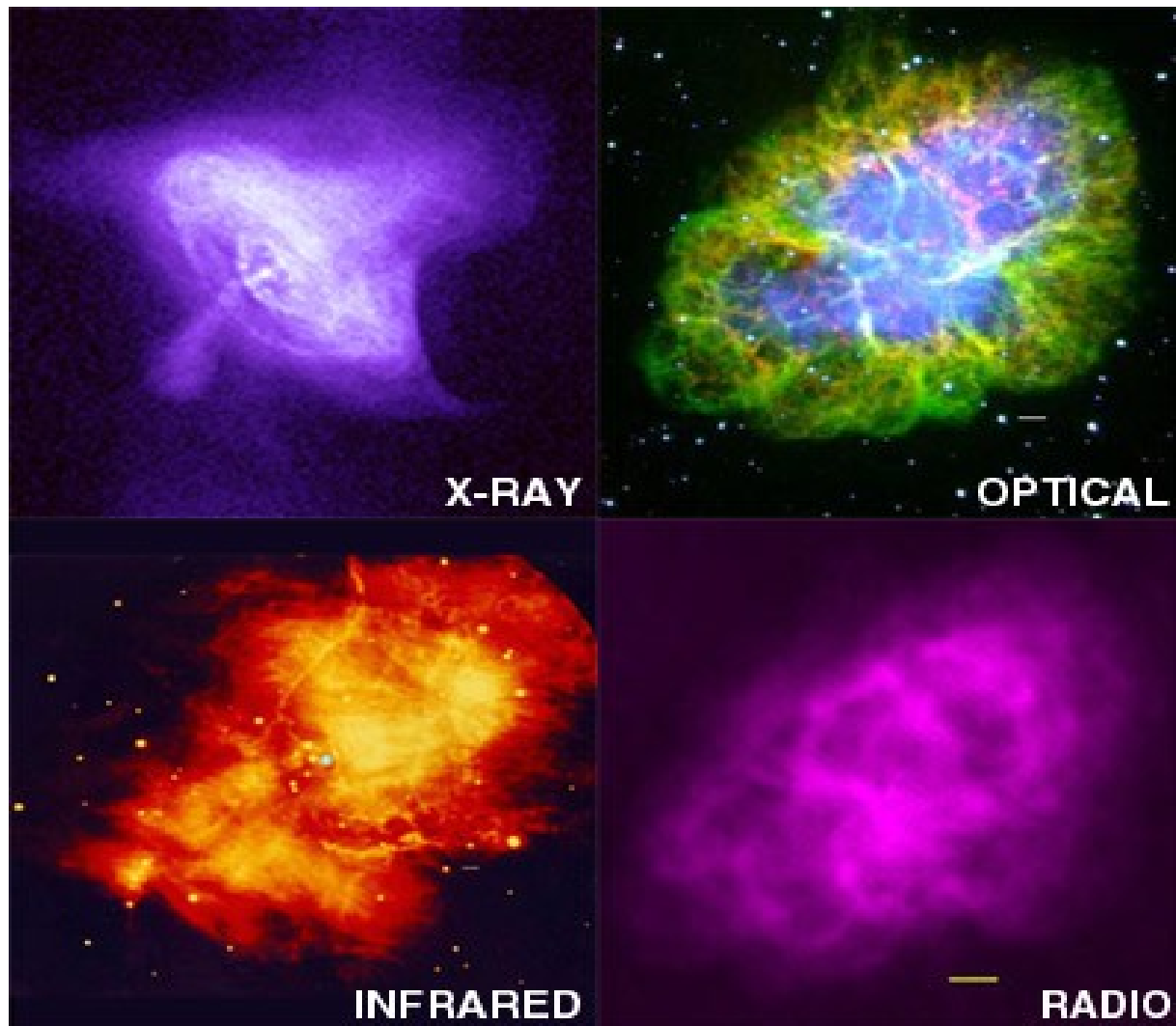
Two Micron All Sky Survey Image Mosaic: Infrared Processing and Analysis Center/Caltech & University of Massachusetts

Radio Sky

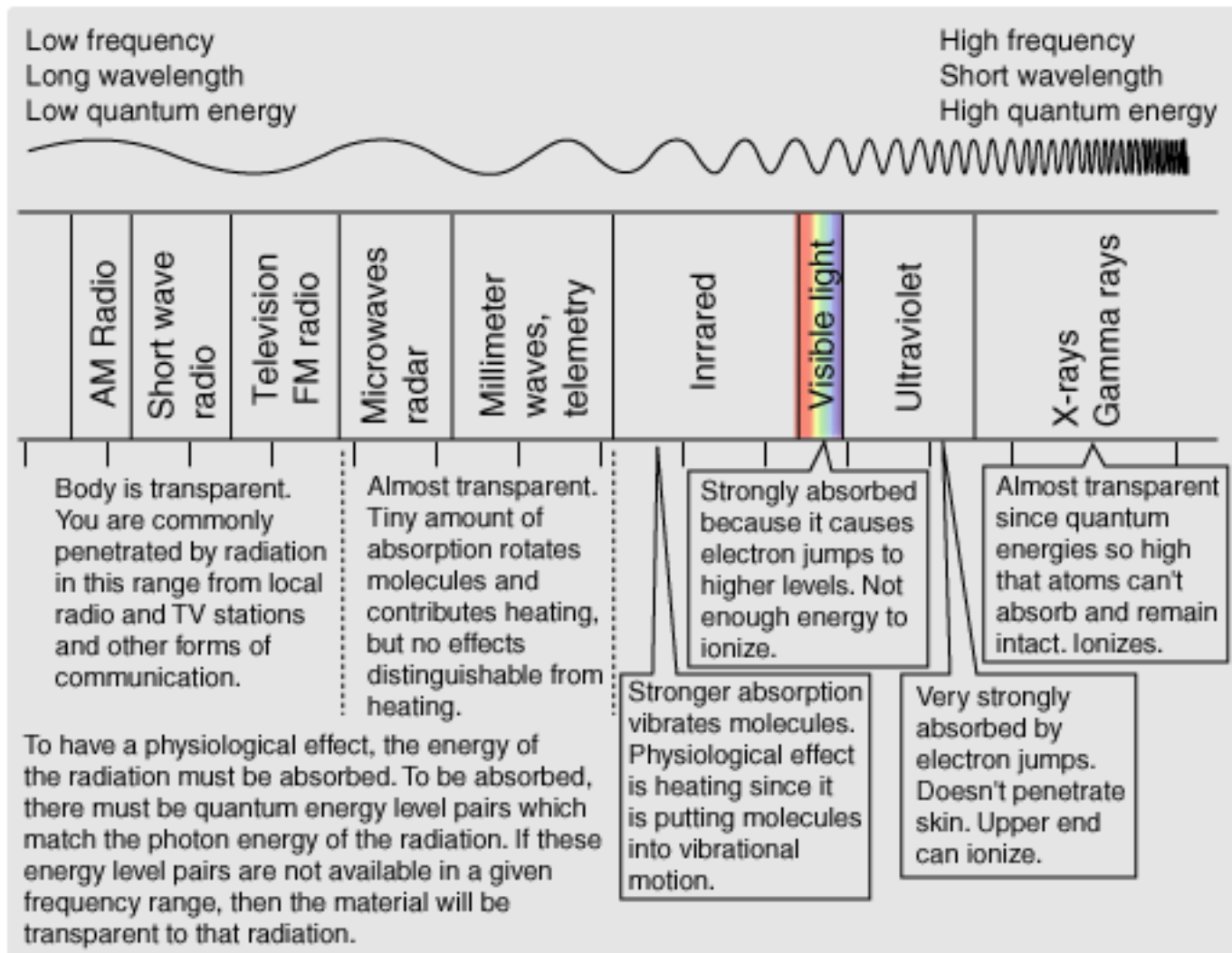


Soft X-ray Sky





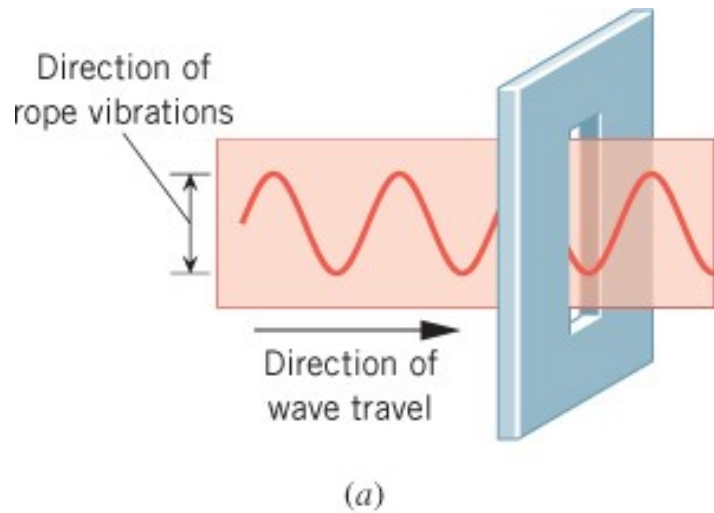
Crab nebula



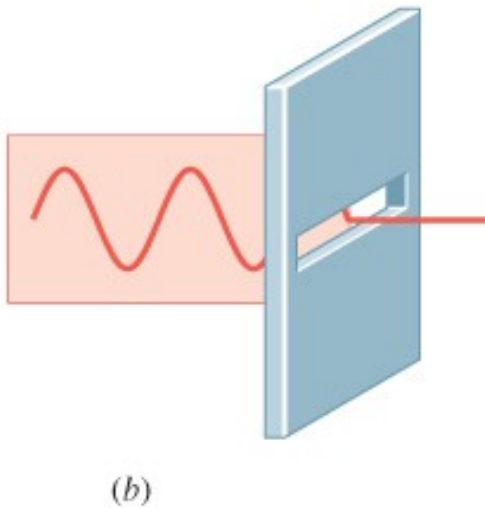
<https://www.quora.com/How-is-it-that-radio-waves-can-go-through-concrete-and-x-rays-through-muscle-What-determines-whether-photons-will-pass-through-without-interacting>

- 1) Compute the wavelength of the carrier of your favorite radio station
- 2) What is the wavelength of the 60,000 Hz radio wave used by “radio-controller” clocks and wristwatches ?
- 3) Compute the frequency of an EM wave with a wavelength of 1 in (0.0254m)
- 3) The wavelength of an electromagnetic wave is measured to be 600m.
 - A) What is the frequency of the wave ?
 - B) What type of EM wave is it ? (check the scale)
- 4) Determine the range of wavelengths in the UV radiation band
- 5) The simplest electromagnetic wave is a sinewave and it is produced by sending alternative current in an Antenna (see below). For best result, the antenna should be half-wavelength long. How long should the antenna be to broadcast at 98MHz (MHz means Mega Hertz or times 1000,000)?

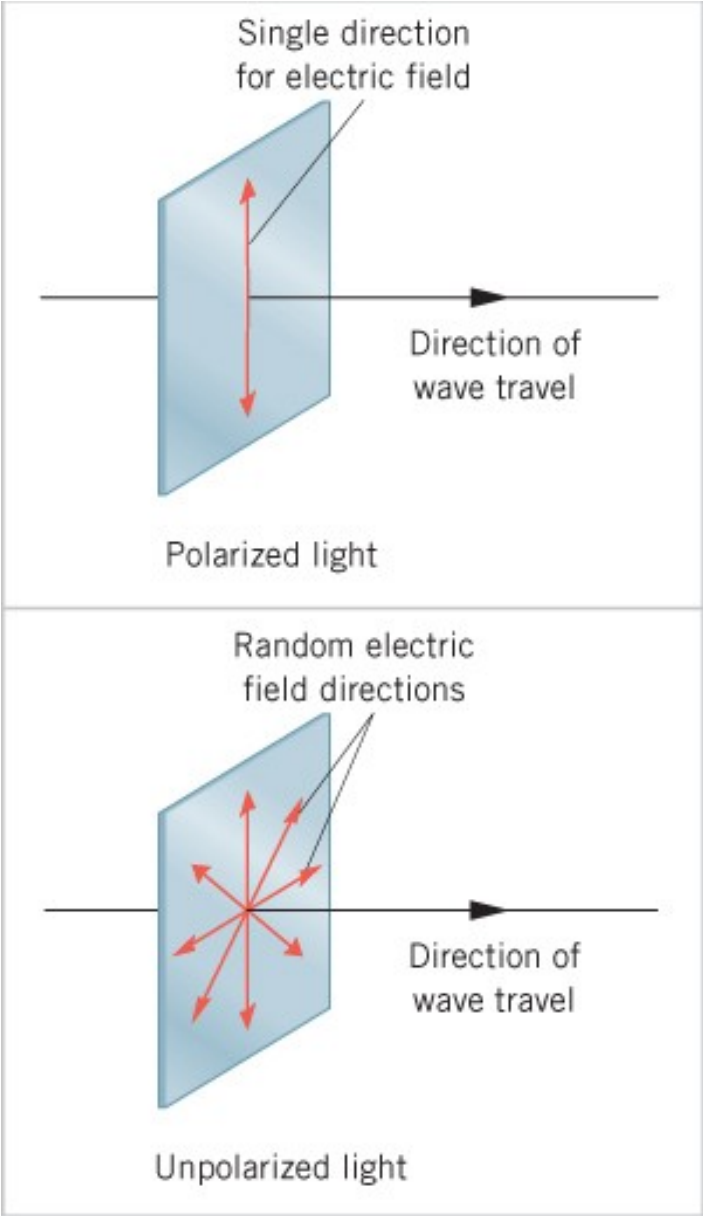
POLARIZED ELECTROMAGNETIC WAVES



Linearly polarized wave
on a rope.

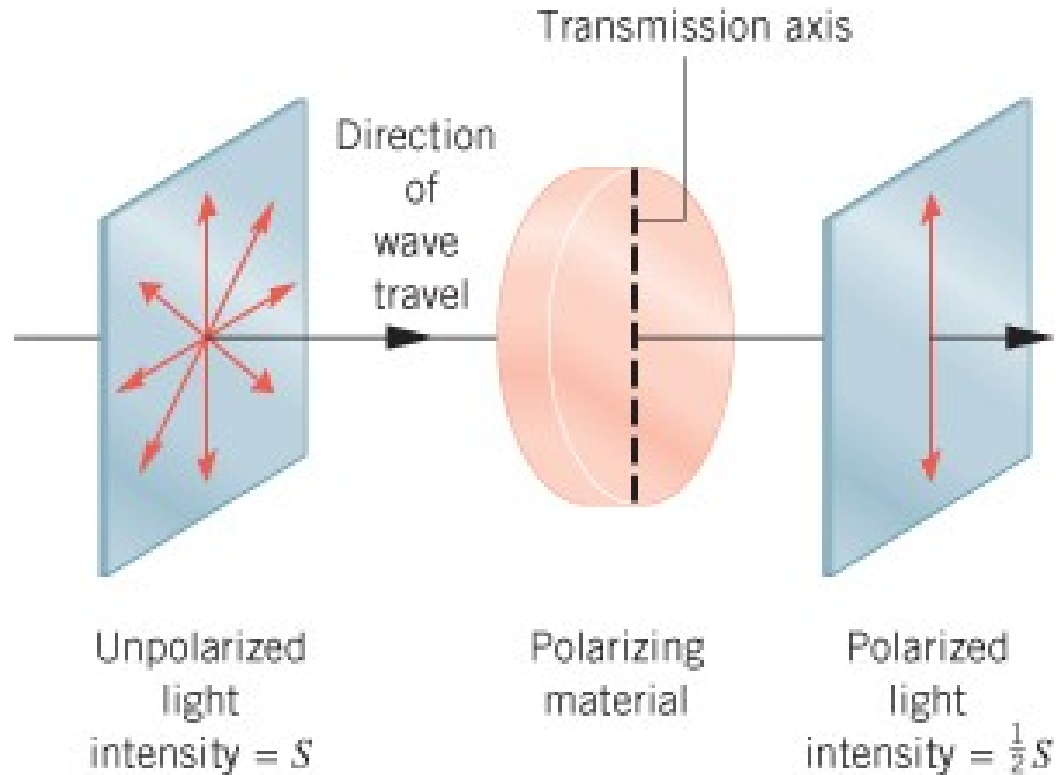


24.6 Polarization



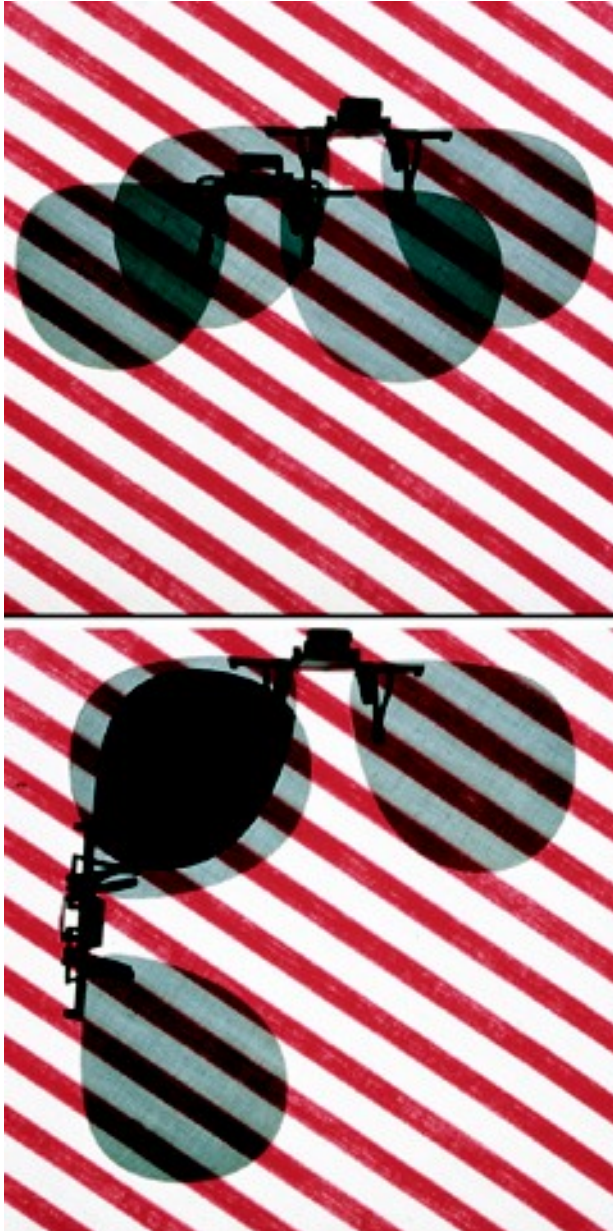
In polarized light, the electric field fluctuates along a single direction.

24.6 Polarization



Polarized light may be produced from unpolarized light with the aid of polarizing material.

24.6 Polarization



When Polaroid sunglasses are crossed, the intensity of the transmitted light is reduced to zero.

